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Probabilistic Number Theory in Additive Arithmetic Semigroups, III

WEN-BIN ZHANG
Department of Mathematics, University of Illinois, 1409 West Green Street, Urbana, Illinois 61801, USA; Department of Mathematics, South China University of Technology, Guangzhou, People's Republic of China

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Abstract. We extend the investigation of quantitative mean-value theorems of completely multiplicative functions on additive arithmetic semigroups given in our previous paper. Then the new and old quantitative mean-value theorems are applied to the investigation of local distribution of values of a special additive function $\Omega^*(a)$. The result is unexpected from the point of view of classical number theory. This reveals the fact that the essential divergence of the theory of additive arithmetic semigroups from classical number theory is not related to the existence of a zero of the zeta function $Z(y)$ at $y = -q^{-1}$.

Key words: quantitative mean-value theorem, completely multiplicative function, density, additive function, additive arithmetic semigroup

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1. Introduction

The investigation of density of additive functions, the so-called local theorems in classical probabilistic number theory, was originated by Hardy and Ramanujan [3] and is continued by Erdös, Halász [2] and others. Especially, Halász applies the quantitative mean-value theorems in this investigation (see, e.g., Elliott's monograph [1]).

In this paper, we shall extend this investigation to additive functions on additive arithmetic semigroups [5]. To this end, we shall first extend our investigation of quantitative mean-value theorems of completely multiplicative functions on such semigroups given in our previous paper. Then we shall apply our new and old quantitative mean-value theorems to the investigation of local theorems [1], for a special additive function $\Omega^*(a)$ (see Theorem 5.1).

The two quantitative mean-value theorems, proved in our previous paper [11], are mainly designed for the investigation of rate of convergence to the normal law of distribution of values of additive functions on additive arithmetic semigroups.

To investigate the local theorems, or density theorems, for additive functions, extension of these two quantitative mean-value theorems to more general multiplicative functions is required.

We recall that an additive arithmetic semigroup $\mathcal{G}$ is, by definition [5], a free commutative semigroup with identity element 1 such that $\mathcal{G}$ has a countable free generating set $\mathcal{P}$ of "primes" p and such that $\mathcal{G}$ admits a "degree" mapping $\partial : \mathcal{G} \to \mathbb{N} \cup \{0\}$ satisfying

(i) $\partial(ab) = \partial(a) + \partial(b)$ for all $a, b \in \mathcal{G}$ and
(ii) the total number $G(n)$ of elements of degree n in $\mathcal{G}$ is finite for each $n \geq 0$.

Then, from (i) and (ii), $\partial(a) = 0$ if and only if $a = 1$ and $G(0) = 1$.

A complex-valued function f defined on an additive arithmetic semigroup $\mathcal{G}$, not identically equal to zero, is said to be multiplicative or completely multiplicative (respectively, additive or completely additive) according as $f(ab) = f(a)f(b)$ (respectively, $f(ab) = f(a) + f(b)$) holds for all coprime $a, b \in \mathcal{G}$ or for all $a, b \in \mathcal{G}$.

We shall consider completely multiplicative functions f satisfying either conditions (1) and (3) or conditions (1) and one of conditions (2) and (2*) as follows. These are

(1) $f(p) = 0$ or

$$0 < \lambda_1 \leq |f(p)| \leq \lambda_2 < q$$

for all $p \in \mathcal{P}$,

(2) there exists a constant $\mu > 0$ such that, when $f(p) \neq 0$,

$$-\frac{\pi}{2} + \mu \leq \theta_p \leq \pi - \mu$$

for a value θ_p of argument of $f(p)$,

(2*) there exists a constant $\mu > 0$ such that, when $f(p) \neq 0$,

$$-\pi + \mu \leq \theta_p \leq \frac{\pi}{2} - \mu,$$

and

(3) when $f(p) \neq 0$,

$$\frac{5\pi}{6} \leq \theta_p \leq \frac{7\pi}{6} \quad \text{or} \quad -\frac{\pi}{6} \leq \theta_p \leq \frac{\pi}{6}$$

for a value θ_p of argument of $f(p)$.

We shall prove, in Section 4, two new quantitative mean-value theorems, i.e., Theorem 4.1, which corresponds to conditions (2) and its conjugate (2*), and Theorem 4.2, which corresponds to condition (3), respectively; both are extension of Theorem 3.1 in [11]. Then, on the basis of Theorems 4.1 and 4.2 and Theorem 3.2 in [11], a local theorem of a special additive function $\Omega^*(a)$, Theorem 5.1, will be proved.

We note that the theory of additive arithmetic semigroups shows essential divergence from classical number theory when the zeta function $Z(y)$ of such a semigroup has a zero at $y = -q^{-1}$ (cf. [4, 8–10]). This divergence exists even when $Z(y)$ has no zeros at $y = -q^{-1}$ as we observe in Theorem 4.2. We shall give examples, following Theorem 4.2, to explore this phenomenon. Furthermore, Theorem 5.1 reveals a new fact that this divergence is actually not related to whether $Z(y)$ has zeros at $y = -q^{-1}$. This is shown by an analysis of $\Omega^*(a)$ on algebraic function fields in the last section.

Throughout this paper, we assume, as in [11], that

$$G(n) = Aq^n + O(q^n n^{-\gamma}), \quad n \geq 1 \tag{1.1}$$

holds as $n \to \infty$ with constants $A > 0, q > 1$, and $\gamma > 2$. Hence, constants, explicit or implicit, in the future discussion may well depend on constants in condition (1.1). If it is not obvious whether a constant in a particular lemma or a theorem also depends on other constants given there then we shall indicate the dependence or no dependence in each case.

Then the zeta function $Z(y)$ of an additive arithmetic semigroup $\mathcal{G}$ is defined by

$$Z(y) = \sum_{a \in \mathcal{G}} y^{\partial(a)} = \prod_{p \in \mathcal{P}} \left(1 - y^{\partial(p)}\right)^{-1},$$

in which both the infinite series and the infinite product are absolutely convergent for $|y| < q^{-1}$.

The discussion in this paper, as that in our previous paper [11], will be based on the prime element theorem proved in [8, 10]. The theorem states that, in an equivalent form, either $Z(y)$ has no zero at $y = -q^{-1}$ and

$$P(n) = q^n n^{-1}(1 + O(n^{-\gamma+1})) \tag{1.2}$$

or $Z(y)$ has a zero of order 1 at $y = -q^{-1}$ and

$$P(n) = q^n n^{-1}(1 - (-1)^n + O(n^{-\gamma+2})). \tag{1.3}$$

In particular, $P(n) \ll q^n n^{-1}$. We remark that if we assume (1.1) and (1.2) with $\gamma > 1$ the discussion in this paper, as well as the discussion in the previous one, is still valid without any changes. Therefore the condition (1.1) with $\gamma > 2$ can be "weakened". However, if we ask whether condition (1.1) implies (1.2) then we have to assume (1.1) with $\gamma > 2$ and thus the alternative prime element theorem holds. This is our best knowledge at the current stage.

This paper will combine new ideas with the old ideas in our previous paper. Therefore, in many places of this paper, we shall save the details which have been given before in [11].

2. Lemmas

Let

$$S(t) := \sum_{p} \left(1 - \Re e^{i(\theta_p + \partial(p)\psi)}\right) |f(p)|\, t^{\partial(p)} q^{-\partial(p)} \tag{2.1}$$

for $0 \leq t < 1$ and real number ψ. We begin with lower estimates of $S(t)$.

We observe that when ψ is near $s\pi/r$, where s/r is a rational number in lowest terms, $1 - \Re e^{i(\theta_p + \partial(p)\psi)}$ is almost periodic in values of $\partial(p)$ with period $2r$. In this case, for values of $\partial(p)$ in certain arithmetic progressions modulo $2r$, $1 - \Re e^{i(\theta_p + \partial(p)\psi)}$ has a positive lower bound and a lower estimate for $S(t)$ can be obtained directly.

Lemma 2.1. *Let $r, s \in \mathbb{N}$, $(r, s) = 1$ and $r \geq 2$. Then there exist odd integers l and k such that $1 \leq l \leq 2r - 1$ and*

$$k \leq \frac{ls}{r} \leq k + \frac{1}{2}. \tag{2.2}$$

Also, there exist odd integers l' and k' such that $1 \leq l' \leq 2r - 1$ and

$$k' - \frac{1}{2} \leq \frac{l's}{r} \leq k'. \tag{2.3}$$

Proof: Since $(r, s) = 1$ and $r \geq 2$, there exist integers m and n such that $ms + nr = 1$ and $1 \leq m \leq r - 1$. We set l, k and l', k' as follows.

If s, r are both odd, set $l = r = l'$ and $k = s = k'$. Then (2.2), (2.3) are satisfied.

If s is even, then r and n are odd. If m is odd too, set $l = m$ and $k = -n$. Then

$$k \leq \frac{ls}{r} = -n + \frac{1}{r} \leq k + \frac{1}{2}. \tag{2.4}$$

Otherwise, m is even, set $l = m + r$ and $k = -n + s$. Then

$$k \leq \frac{ls}{r} = \frac{(m+r)s}{r} = -n + s + \frac{1}{r} \leq k + \frac{1}{2}. \tag{2.5}$$

Finally, if r is even then s and m are odd. If n is odd too, set $l = m$ and $k = -n$ then (2.4) is true. Otherwise, n is even, set $l = m + r$ and $k = -n + s$. Then (2.5) is true. This proves (2.2).

Similarly, to prove (2.3), if s is even and m is odd or if r is even and n is odd, set $l' = 2r - m$ and $k' = 2s + n$. Then

$$k' - \frac{1}{2} \leq \frac{l's}{r} = \frac{(2r - m)s}{r} = 2s + n - \frac{1}{r} \leq k'. \tag{2.6}$$

Then if s is even and m is even too or if r is even and n is too, set $l' = r - m$ and $k' = s + n$. Then

$$k' - \frac{1}{2} \leq \frac{l's}{r} = \frac{(r - m)s}{r} = s + n - \frac{1}{r} \leq k'. \tag{2.7}$$

Note that, in all cases, l, k and l', k' are all odd integers and $1 \leq l \leq 2r - 1$, $1 \leq l' \leq 2r - 1$. □

Lemma 2.2. *Let $r, s \in \mathbb{N}$, $s \leq r$, $(r, s) = 1$. Suppose that f satisfies conditions* (1) *and either* (2) *or* (2*). *There exists a positive constant c_1 such that*

$$S(t) \geq c_1 \frac{\lambda_1}{r} \log \frac{1}{1-t} - \lambda_1 \sum_{f(p)=0} t^{\partial(p)} q^{-\partial(p)} + (1 + \lambda_1) O(1) \tag{2.8}$$

for real numbers t satisfying $0 \le t < 1$ and ψ satisfying

$$\left|\psi - \frac{s}{r}\pi\right| \le \frac{\mu}{4r} \quad \textit{or} \quad \left|\psi + \frac{s}{r}\pi\right| \le \frac{\mu}{4r}.$$

Here c_1 and the O-constant depend on the constant μ of condition (2) *or* (2*) *only.*

Remark. We may take

$$c_1 = \frac{\mu}{16\pi}\left(1 - \cos\frac{\mu}{2}\right).$$

Proof: It suffices to prove the lemma for a function f satisfying conditions (1) and (2). If f satisfies (1) and (2*) then $\bar{f}$, the complex conjugate of f, satisfies conditions (1) and (2) and (2.8) follows directly from the one for $\bar{f}$.

We first assume $|\psi - \frac{s}{r}\pi| \le \frac{\mu}{4r}$. By condition (2) and (2.3) of Lemma 2.1, there exist odd integers l and k such that

$$(k-1)\pi + \mu \le \theta_p + \frac{ls}{r}\pi \le (k+1)\pi - \mu \tag{2.9}$$

(if $r = 1 = s$, just set $l = 1 = k$). With l, k defined in this way, we have

$$S(t) \ge \sum_{\partial(p) \equiv l \pmod{2r}} \left(1 - \Re e^{i(\theta_p + ls\pi/r + \partial(p)(\psi - s\pi/r))}\right) |f(p)|\, t^{\partial(p)} q^{-\partial(p)}. \tag{2.10}$$

To prove (2.8), if $\psi - \frac{s}{r}\pi = 0$, by (2.9),

$$1 - \Re e^{i(\theta_p + ls\pi/r + \partial(p)(\psi - s\pi/r))} = 1 - \cos\left(\theta_p + \frac{ls\pi}{r}\right) \ge 1 - \cos\mu = K_1,$$

say. Then

$$\begin{aligned} S(t) &\ge \lambda_1 K_1 \sum_{\partial(p) \equiv l \pmod{2r}} t^{\partial(p)} q^{-\partial(p)} - \lambda_1 K_1 \sum_{\substack{\partial(p) \equiv l \pmod{2r} \\ f(p) = 0}} t^{\partial(p)} q^{-\partial(p)} \\ &\ge \frac{\lambda_1 K_1}{2r} \log\frac{1}{1-t} - \lambda_1 K_1 \sum_{\substack{\partial(p) \equiv l \pmod{2r} \\ f(p) = 0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O(1) \end{aligned} \tag{2.11}$$

by the prime element theorem (1.2) and (1.3) since l is odd. Thus (2.8) holds with

$$c_1 = \frac{1}{2}(1 - \cos\mu).$$

We may now assume $\psi - \frac{s}{r}\pi \ne 0$ and consider the case $0 < \psi - \frac{s}{r}\pi \le \frac{\mu}{4r}$ and the case $-\frac{\mu}{4r} \le \psi - \frac{s}{r}\pi < 0$ separately. Accordingly, let

$$I_m^+ := \left\{p : \partial(p) \equiv l \pmod{2r}, \left|\partial(p)\left(\psi - \frac{s}{r}\pi\right) - 2m\pi\right| \le \frac{\mu}{2}\right\}$$

and

$$I_m^- := \left\{ p : \partial(p) \equiv l \pmod{2r}, \left| \partial(p)\left(\psi - \frac{s}{r}\pi\right) + 2m\pi \right| \le \frac{\mu}{2} \right\}$$

for $m \in \mathbb{Z}$. From (2.9), for $p \in I_m^+$ and $p \in I_m^-$,

$$1 - \Re e^{i(\theta_p + ls\pi/r + \partial(p)(\psi - s\pi/r))} \ge 1 - \cos\frac{\mu}{2} = K_2, \tag{2.12}$$

say. We shall use I_m^+ in the proof in case $0 < \psi - \frac{s}{r}\pi \le \frac{\mu}{4r}$. The proof in case $-\frac{\mu}{4r} \le \psi - \frac{s}{r}\pi < 0$ is the same but with I_m^- in place of I_m^+ and $\frac{s}{r}\pi - \psi$ in place of $\psi - \frac{s}{r}\pi$. Hence, we shall give only the proof for $0 < \psi - \frac{s}{r}\pi \le \frac{\mu}{4r}$. Note that I_m^+, $m = 0, 1, \ldots$ are pairwise disjoint. From (2.10) and (2.12),

$$\begin{aligned} S(t) &\ge \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{p \in I_m^+} t^{\partial(p)} q^{-\partial(p)} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} \\ &\ge \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{k \in J_m} \frac{t^{l+2rk}}{l+2rk} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O(1) \end{aligned} \tag{2.13}$$

by the prime element theorem (1.2) and (1.3). Here the sum $\sum_{k \in J_m}$ is taken over nonnegative integers k satisfying

$$\frac{2m\pi - \frac{\mu}{2}}{\psi - \frac{s}{r}\pi} \le l + 2rk \le \frac{2m\pi + \frac{\mu}{2}}{\psi - \frac{s}{r}\pi}. \tag{2.14}$$

If $0 < \psi - \frac{s}{r}\pi \le (1-t)^{\frac{1}{2}}$,

$$\sum_{k \in J_0} \frac{t^{l+2rk}}{l+2rk} \ge \sum_{1 \le k \le \frac{\mu/2}{2r(1-t)^{1/2}}} \frac{t^{2rk}}{2rk}.$$

Note that, for $1 \le k \le \frac{\mu/2}{2r(1-t)^{1/2}}$ and $\frac{1}{2} \le t < 1$,

$$t^{2rk} = \exp\left\{-2rk \log\frac{1}{t}\right\} \ge \exp\left\{-\frac{\frac{\mu}{2}}{(1-t)^{\frac{1}{2}}} \log\frac{1}{t}\right\} \ge \exp\left\{-\frac{\mu}{\sqrt{2}} \log 2\right\} = K_3,$$

say. Hence

$$\sum_{k \in J_0} \frac{t^{l+2rk}}{l+2rk} \ge K_3 \sum_{1 \le k \le \frac{\mu/2}{2r(1-t)^{1/2}}} \frac{1}{2rk} \ge \frac{K_3}{4r} \log\frac{1}{1-t} + O_\mu(1).$$

Thus,

$$S(t) \geq \frac{\lambda_1 K_2 K_3}{4r} \log \frac{1}{1-t} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O_\mu(1)$$

and (2.8) holds with

$$c_1 = \frac{1}{4}\left(1 - \cos\frac{\mu}{2}\right) \exp\left\{-\frac{\mu}{\sqrt{2}} \log 2\right\}.$$

If $(1-t)^{\frac{1}{2}} < \psi - \frac{s}{r}\pi \leq \frac{\mu}{4r}$, by (2.14),

$$\sum_{k \in J_m} \frac{t^{l+2rk}}{l+2rk} \geq t^{\frac{2m\pi+\mu/2}{\psi - s\pi/r}} \sum_{k \in J_m} \frac{1}{l+2rk}$$

for $m \geq 1$. The sum on the right-hand side is

$$\geq \int_{\frac{2m\pi-\mu/2}{2r(\psi-s\pi/r)} - \frac{l}{2r}+1}^{\frac{2m\pi+\mu/2}{2r(\psi-s\pi/r)} - \frac{l}{2r}} \frac{dx}{l+2rx} \geq \frac{1}{2r} \log \frac{2m\pi + \frac{\mu}{2}}{2m\pi} \geq \frac{1}{2r}\left(\frac{\mu}{4m\pi} - \left(\frac{\mu}{4m\pi}\right)^2\right).$$

It follows that

$$S(t) \geq \frac{\mu\lambda_1 K_2}{8r\pi} \sum_{m=1}^{\infty} \frac{1}{m} t^{\frac{2m\pi+\mu/2}{\psi-s\pi/r}} - \lambda_1 K_2 \sum_{\substack{p \in I_m^+,\, m \geq 0 \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + (1+\lambda_1) O(1).$$

The first sum on the right-hand side is

$$\begin{aligned} \geq \sum_{m=1}^{\infty} \frac{1}{m+1} \left(t^{\frac{2\pi}{\psi-s\pi/r}}\right)^{m+1} &= \log\left(1 - t^{\frac{2\pi}{\psi-s\pi/r}}\right)^{-1} - t^{\frac{2\pi}{\psi-s\pi/r}} \\ &\geq \frac{1}{2} \log \frac{1}{1-t} + O(1). \end{aligned}$$

It follows that

$$S(t) \geq \frac{\mu\lambda_1 K_2}{16r\pi} \log \frac{1}{1-t} - \lambda_1 K_2 \sum_{\substack{p \in I_m^+,\, m \geq 0 \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + (1+\lambda_1) O(1),$$

i.e., (2.8) holds with

$$c_1 = \frac{\mu}{16\pi}\left(1 - \cos\frac{\mu}{2}\right).$$

This proves (2.8) for $|\psi - \frac{s}{r}\pi| \leq \frac{\mu}{4r}$.

Then assume $|\psi + \frac{s}{r}\pi| \le \frac{\mu}{4r}$. By conditions (2) and (2.2) of Lemma 2.1, there exist odd integers l and k such that

$$-(k+1)\pi + \mu \le \theta_p - \frac{ls\pi}{r} \le -(k-1)\pi - \mu.$$

With l, k defined in this way,

$$S(t) \ge \sum_{\partial(p) \equiv l \pmod{2r}} \left(1 - \Re e^{i(\theta_p - ls\pi/r + \partial(p)(\psi + s\pi/r))}\right) |f(p)|\, t^{\partial(p)} q^{-\partial(p)}.$$

The same argument with $\theta_p - \frac{ls\pi}{r}$ in place of $\theta_p + \frac{ls\pi}{r}$ and $\psi + \frac{s}{r}\pi$ in place of $\psi - \frac{s}{r}\pi$ proves (2.8) for $|\psi + \frac{s}{r}\pi| \le \frac{\mu}{4r}$. □

A similar lower estimate holds for a function f satisfies condition (1) and (3) as the following lemma shows.

Lemma 2.3. *Let $r, s \in \mathbb{N}$, $s \le r$, $(r, s) = 1$, and $r \ge 2$. Suppose that f satisfies conditions* (1) *and* (3)*. There exists a positive constant c_2 such that*

$$S(t) \ge c_2 \frac{\lambda_1}{r} \log \frac{1}{1-t} - \frac{\lambda_1}{29} \sum_{f(p)=0} t^{\partial(p)} q^{-\partial(p)} + (1+\lambda_1) O(1) \tag{2.15}$$

for real numbers t satisfying $0 < t < 1$ and ψ satisfying

$$\left|\psi - \frac{s}{r}\pi\right| \le \frac{\pi}{96r} \quad or \quad \left|\psi + \frac{s}{r}\pi\right| \le \frac{\pi}{96r}.$$

Here c_2 and the O-constant are independent of $s, r, \lambda_1, \lambda_2$.

Proof: With each $\frac{s}{r}$, where $r, s \in \mathbb{N}$, $(r, s) = 1$, in case $r = 2$, we associate integers $l = 1, k = 0$ and, in cases $r \ge 3$, a pair of integers l and k defined as follows. Since $(r, s) = 1$, there exist integers l' and k' such that $l's - k'r = 1$ and $1 \le l' \le r - 1$. In case that s is odd, r is even and l' is odd. If $\frac{r}{2}$ is odd let $l = \frac{r}{2}l'$ and $k = \frac{r}{2}k'$; otherwise, let $l = (\frac{r}{2} + 1)l'$ and $k = k'(1 + \frac{r}{2})$. Then l is odd and

$$\frac{ls}{r} = \begin{cases} k + \dfrac{1}{2}, & \text{if } \dfrac{r}{2} \text{ is odd;} \\[2ex] k + \dfrac{1}{2} + \dfrac{1}{r}, & \text{if } \dfrac{r}{2} \text{ is even.} \end{cases} \tag{2.16}$$

In case that s is even, r is odd. If l' is odd too, let $l = l'[\frac{r}{2}]$ and $k = k'[\frac{r}{2}]$ when $[\frac{r}{2}]$ is odd; otherwise, $\frac{r+1}{2}$ is odd and let $l = l'\frac{r+1}{2}$ and $k = k'\frac{r+1}{2}$. Then l is odd and

$$\frac{ls}{r} = \begin{cases} k + \dfrac{1}{2} - \dfrac{1}{2r}, & \text{if } \left[\dfrac{r}{2}\right] \text{ is odd;} \\[2ex] k + \dfrac{1}{2} + \dfrac{1}{2r}, & \text{if } \dfrac{r+1}{2} \text{ is odd.} \end{cases} \tag{2.17}$$

Finally, if l' is even, let $l = (r - l')[\frac{r}{2}]$ and $k = (s - k')[\frac{r}{2}] - 1$ when $[\frac{r}{2}]$ is odd; otherwise, $\frac{r+1}{2}$ is odd and let $l = (r - l')\frac{r+1}{2}$ and $k = (s - k')\frac{r+1}{2} - 1$. Then l is odd and

$$\frac{ls}{r} = \begin{cases} k + \dfrac{1}{2} + \dfrac{1}{2r}, & \text{if } \left[\dfrac{r}{2}\right] \text{ is odd;} \\ k + \dfrac{1}{2} - \dfrac{1}{2r}, & \text{if } \dfrac{r+1}{2} \text{ is odd.} \end{cases} \tag{2.18}$$

To prove (2.15), we first assume $|\psi - \frac{s}{r}\pi| < \frac{\pi}{96r}$. From condition (3) and (2.16), (2.17), (2.18), there exist an integer k and an odd integer l satisfying $1 \le l \le (r-1)r$ such that

$$k\pi + \frac{\pi}{6} \le \theta_p + \frac{ls}{r}\pi \le (k+1)\pi - \frac{\pi}{12} \tag{2.19a}$$

or

$$(k+1)\pi + \frac{\pi}{6} \le \theta_p + \frac{ls}{r}\pi \le (k+2)\pi - \frac{\pi}{12}. \tag{2.19b}$$

Hence, with l, k defined in this way,

$$S(t) \ge \sum_{\partial(p) \equiv l \pmod{2r}} \left(1 - \Re e^{i(\theta_p + ls\pi/r + \partial(p)(\psi - s\pi/r))}\right) |f(p)|\, t^{\partial(p)} q^{-\partial(p)}.$$

If $\psi - \frac{s}{r}\pi = 0$,

$$1 - \Re e^{i(\theta_p + ls\pi/r + \partial(p)(\psi - s\pi/r))} \ge 1 - \cos\frac{\pi}{12} = K_1$$

and

$$S(t) \ge \frac{\lambda_1 K_1}{2r} \log\frac{1}{1-t} - \lambda_1 K_1 \sum_{\substack{\partial(p) \equiv l \pmod{2r} \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O(1).$$

Then we may assume $\psi - \frac{s}{r}\pi \ne 0$ and consider the case $0 < \psi - \frac{s}{r}\pi < \frac{\pi}{96r}$ and the case $-\frac{\pi}{96r} < \psi - \frac{s}{r}\pi < 0$ separately. Accordingly, let

$$I_m^+ := \left\{ p : \partial(p) \equiv l \pmod{2r}, \left| \partial(p)\left(\psi - \frac{s}{r}\pi\right) - 2m\pi \right| < \frac{\pi}{24} \right\}$$

and

$$I_m^- := \left\{ p : \partial(p) \equiv l \pmod{2r}, \left| \partial(p)\left(\psi - \frac{s}{r}\pi\right) + 2m\pi \right| < \frac{\pi}{24} \right\}$$

for $m \in \mathbb{Z}$. From (2.19),

$$1 - \Re e^{i(\theta_p + ls\pi/r + \partial(p)(\psi - s\pi/r))} \ge 1 - \cos\frac{\pi}{24} = K_2.$$

As in the proof of Lemma 2.2, it is enough to give only the calculation for the case $0 < \psi - \frac{s}{r}\pi < \frac{\pi}{96r}$. Hence

$$S(t) \geq \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{k \in J_m} \frac{t^{l+2rk}}{l+2rk} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O(1)$$

by the prime element theorem. Here the sum $\sum_{k \in J_m}$ is taken over nonnegative integers k satisfy

$$\frac{2m\pi - \frac{\pi}{24}}{\psi - \frac{s}{r}\pi} \leq l + 2rk \leq \frac{2m\pi + \frac{\pi}{24}}{\psi - \frac{s}{r}\pi}.$$

If $0 < \psi - \frac{s}{r}\pi < (1-t)^{\frac{1}{2}}$,

$$\sum_{k \in J_0} \frac{t^{l+2rk}}{l+2rk} \geq \sum_{1 \leq k \leq \frac{\pi/24}{2r(1-t)^{1/2}}} \frac{t^{2rk}}{2rk} + O(1)$$

and hence

$$\sum_{k \in J_0} \frac{t^{l+2rk}}{l+2rk} \geq \frac{K_3}{4r} \log \frac{1}{1-t} + O(1)$$

with

$$K_3 = \exp\left\{-\frac{\pi}{24\sqrt{2}} \log 2\right\}.$$

Thus

$$S(t) \geq \frac{\lambda_1 K_2 K_3}{4r} \log \frac{1}{1-t} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O(1).$$

If $(1-t)^{\frac{1}{2}} < \psi - \frac{s}{r}\pi \leq \frac{\pi}{96r}$,

$$\sum_{k \in J_m} \frac{t^{l+2rk}}{l+2rk} \geq t^{\frac{2m\pi + \frac{\pi}{24}}{\psi - \frac{s}{r}\pi}} \sum_{k \in J_m} \frac{1}{l+2rk} \geq \frac{1}{2r}\left(\frac{1}{96m} - \left(\frac{1}{96m}\right)^2\right).$$

for $m \geq 1$. It follows that

$$\begin{aligned} S(t) &\geq \frac{\lambda_1 K_2}{192r} \sum_{m=0}^{\infty} \frac{t^{\frac{2m\pi + \frac{\pi}{24}}{\psi - \frac{s}{r}\pi}}}{m} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + \lambda_1 O(1) \\ &\geq \frac{\lambda_1 K_2}{384r} \log \frac{1}{1-t} - \lambda_1 K_2 \sum_{m=0}^{\infty} \sum_{\substack{p \in I_m^+ \\ f(p)=0}} t^{\partial(p)} q^{-\partial(p)} + (1+\lambda_1) O(1), \end{aligned}$$

i.e., (2.15) holds with

$$c_2 = \frac{1}{384}\left(1 - \cos\frac{\pi}{24}\right).$$

This proves (2.15) for $|\psi - \frac{s}{r}\pi| \le \frac{\pi}{96r}$.

Then assume $|\psi + \frac{s}{r}\pi| \le \frac{\pi}{96r}$. By condition (3) and (2.16), (2.17), (2.18), there exist integer k and odd integer l satisfying $1 \le l \le r(r-1)$ such that

$$-(k+1)\pi + \frac{\pi}{12} \le \theta_p - \frac{ls}{r}\pi \le -(k+1)\pi + \frac{5\pi}{6} \tag{2.20a}$$

or

$$-k\pi + \frac{\pi}{12} \le \theta_p - \frac{ls}{r}\pi \le -k\pi + \frac{5\pi}{6}. \tag{2.20b}$$

With l, k defined in this way, the same argument with $\theta_p - \frac{ls}{r}\pi$ in place of $\theta_p + \frac{ls}{r}\pi$ and $\psi + \frac{s}{r}\pi$ in place of $\psi - \frac{s}{r}\pi$ proves (2.15). □

Let $\mathbf{F}_k$ denote the Farey sequence of order k as in [11]. By Lemma 2.4 in [11], the closed intervals $|\psi - \frac{s}{r}\pi| \le \frac{\mu}{4r}$, $|\psi + \frac{s}{r}\pi| \le \frac{\mu}{4r}$, $\frac{s}{r} \in \mathbf{F}_{[\frac{8\pi}{\mu}]}$ form a cover of the interval $-\pi \le \psi \le \pi$. Lemma 2.2 is applicable to all these intervals but $|\psi| \le \frac{\mu}{4}$. To find a lower estimate of $S(t)$ for $|\psi| \le \frac{\mu}{4}$, we will apply Fourier analysis. Similarly, the closed intervals $|\psi - \frac{s}{r}\pi| \le \frac{\pi}{96r}$, $|\psi + \frac{s}{r}\pi| \le \frac{\pi}{96r}$, $\frac{s}{r} \in \mathbf{F}_{192}$ form a cover of the interval $-\pi \le \psi \le \pi$. Lemma 2.3 is applicable to all these intervals but $|\psi| \le \frac{\pi}{96}$ and $|\psi - \pi| \le \frac{\pi}{96}$. To find a lower estimate of $S(t)$ for $|\psi| \le \frac{\pi}{96}$ and $|\psi - \pi| \le \frac{\pi}{96}$, we will apply Fourier analysis too.

Thus we introduce a continuous function $h_0(e^{i\theta})$ of a real variable θ which is periodic with period 2π such that

$$h_0(e^{i\theta}) = \begin{cases} \delta\left(1 - \dfrac{|\theta|}{\delta}\right), & \text{for } 0 \le |\theta| \le \delta; \\ 0, & \text{for } \delta < |\theta| \le \pi. \end{cases} \tag{2.21}$$

Then define

$$h(e^{i\theta}) = h_0\big(e^{i(\theta+\theta_0)}\big).$$

Here δ is a parameter satisfying $0 < \delta < \pi$ and θ_0 is a real number. Both δ and θ_0 will be specified in each application. Then

$$h(e^{i\theta}) = \sum_{k=-\infty}^{\infty} b_k e^{ik\theta}$$

with Fourier coefficients

$$b_0 = \frac{\delta^2}{2\pi}, \quad b_k = \frac{2}{k^2\pi} e^{ik\theta_0} \sin^2\frac{k\delta}{2}, \quad k = \pm 1, \pm 2, \ldots. \tag{2.22}$$

Lemma 2.4. *Given a number α with $0 < \alpha < 1$, let*

$$R_1 := \bigcup_{m\in\mathbb{Z}} \{\theta : (1-t)^\alpha \le |\theta - 2m\pi| \le \pi - (1-t)^\alpha\}$$

for $0 \le t < 1$. Then

$$\left| \sum_{k\psi\in R_1, k\neq 0} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \right| \le \frac{2\pi\alpha}{3} \log\frac{1}{1-t} + O(1) \tag{2.23}$$

and

$$\left| \sum_{k(\psi-\pi)\in R_1, k\neq 0} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \right| \le \frac{2\pi\alpha}{3} \log\frac{1}{1-t} + O(1), \tag{2.24}$$

where the O-constants are independent of α and the parameters δ and θ_0 of the function h.

Proof: We need only to prove (2.24) since (2.23) has been proved in Lemma 2.5 of [11]. The following proof of (2.24) is similar.

Let S_k denote the inner sum on the right-hand side of (2.24). By the prime element theorem, in case (1.2),

$$S_k = \log\frac{1}{1-te^{ik\theta}} + O(1),$$

where $\log z$ is the single-valued branch with $\log 1 = 0$ of the multiple-valued logarithmic function on the domain fromed by cutting the complex plane along the negative real axis from 0 to $-\infty$. We have

$$\log\frac{1}{1-te^{ik\psi}} = \frac{1}{2}\log\frac{1}{(1-t)^2 + 2t(1-\cos k\psi)} + O(1).$$

For $k(\psi - \pi) \in R_1$,

$$1 - \cos k\psi = 1 - \cos(k(\psi - \pi) - 2m\pi)$$

when k is even and

$$1 - \cos k\psi = 1 - \cos(\pi - (k(\psi - \pi) - 2m\pi))$$

when k is odd and hence

$$1 - \cos k\psi \ge \frac{2}{\pi^2}(1-t)^{2\alpha}.$$

Then, for $\frac{1}{2} \leq t < 1$ and $k(\psi - \pi) \in R_1$,

$$(1-t)^2 + 2t(1 - \cos k\psi) \geq \frac{2}{\pi^2}(1-t)^{2\alpha}.$$

Thus

$$|S_k| \leq \alpha \log \frac{1}{1-t} + O(1). \tag{2.25}$$

In case (1.3),

$$S_k = \log \frac{1}{1 - te^{ik\theta}} + \log \frac{1}{1 + te^{ik\theta}} + O(1).$$

If $(1-t)^\alpha \leq |k(\psi-\pi)-2m\pi| \leq \frac{\pi}{2}$ and k is even or if $\frac{\pi}{2} \leq |k(\psi-\pi)-2m\pi| \leq \pi-(1-t)^\alpha$ and k is odd, we have $1 \leq |1+te^{ik\psi}| \leq 2$ and (2.25) follows. If $(1-t)^\alpha \leq |k(\psi-\pi)-2m\pi| \leq \frac{\pi}{2}$ and k is odd, $1 \leq |1 - te^{ik\psi}| \leq 2$ and

$$S_k = \log \frac{1}{1 + te^{ik\theta}} + O(1) = \log \frac{1}{(1-t)^2 + 2t(1 - \cos k(\psi - \pi))} + O(1)$$

and (2.25) follows again. Finally, if $\frac{\pi}{2} \leq |k(\psi - \pi) - 2m\pi| \leq \pi - (1-t)^\alpha$ and k is even, $1 \leq |1 - te^{ik\psi}| \leq 2$ and

$$S_k = \log \frac{1}{(1-t)^2 + 2t(1 + \cos k(\psi - \pi))} + O(1).$$

In this case,

$$1 + \cos k(\psi - \pi) = 1 - \cos(\pi - (k(\psi - \pi) - 2m\pi)) \geq \frac{2}{\pi^2}(1-t)^{2\alpha}$$

and (2.25) follows too.

From (2.25), it now follows that

$$\begin{aligned}\left| \sum_{k(\psi-\pi)\in R_1, k\neq 0} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \right| &\leq \alpha \log \frac{1}{1-t} \sum_{k(\psi-\pi)\in R_1, k\neq 0} |b_k| + O(1) \\ &\leq \frac{2\pi\alpha}{3} \log \frac{1}{1-t} + O(1). \qquad \square\end{aligned}$$

Lemma 2.5. *Given* $0 < \alpha < 1$, *let*

$$R_2 := \bigcup_{m\in\mathbb{Z}} (\{\theta : |\theta - 2m\pi| < (1-t)^\alpha\} \cup \{\theta : |\theta - (2m+1)\pi| \leq (1-t)^\alpha\})$$

for $0 \le t < 1$. *Then, for* $(1-t)^\alpha \le |\psi|$,

$$\left| \sum_{k\psi \in R_2, k\neq 0} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \right| \le \frac{8\psi^2}{\pi} \log \frac{1}{1-t} + O(1)$$

and, for $(1-t)^\alpha \le |\psi - \pi|$,

$$\left| \sum_{k(\psi-\pi) \in R_2, k\neq 0} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \right| \le \frac{8(\psi-\pi)^2}{\pi} \log \frac{1}{1-t} + O(1),$$

where the O*-constants are independent of* α *and* δ.

Proof: The proof of the first inequality is given in Lemma 2.6 of [11]. The proof of the second one is almost the same with $\psi - \pi$ in place of ψ. $\square$

Lemma 2.6. *Let* $\alpha = \frac{3\delta^2}{25\pi^2}$, *where* δ *is the parameter of the function* $h_0(e^{i\theta})$. *For* $(1-t)^\alpha \le |\psi| \le \frac{\delta}{5}$ *and for* $(1-t)^\alpha \le |\psi - \pi| \le \frac{\delta}{5}$,

$$\sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \ge \frac{\delta^2}{10\pi} \log \frac{1}{1-t} + O(1), \tag{2.26}$$

where the O*-constant is independent of* δ *and* θ_0.

Proof: The proof of (2.26) for $(1-t)^\alpha \le |\psi| \le \frac{\delta}{5}$ is given in Lemma 2.7 of [11]. For $(1-t)^\alpha \le |\psi - \pi| \le \frac{\delta}{5}$, by the prime element theorem and Lemmas 2.4 and 2.5, the left-hand side of (2.26) is

$$\ge \left(\frac{\delta^2}{2\pi} - \frac{2\pi\alpha}{3} - \frac{8(\psi-\pi)^2}{\pi} \right) \log \frac{1}{1-t} + O(1) \ge \frac{\delta^2}{10\pi} \log \frac{1}{1-t} + O(1). \qquad \square$$

Lemma 2.7. *Let* $0 < \alpha < 1$. *For* $0 \le t < 1$, $|\psi| \le (1-t)^\alpha$,

$$\sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \ge \frac{\delta^2}{4\pi} \log\left(1 + \frac{2\psi^2}{\pi^2(1-t)^2}\right) + O_\alpha(1) + O(1). \tag{2.27}$$

For $0 \le t < 1$, $|\psi - \pi| \le (1-t)^\alpha$,

$$\sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}te^{ik\psi})^n \ge \frac{\delta^2}{8\pi} \log\left(1 + \frac{2(\psi-\pi)^2}{\pi^2(1-t)^2}\right) + O_\alpha(1) + O(1). \tag{2.28}$$

Proof: The inequality (2.27) is proved in Lemma 2.8 of [11]. We need only to prove (2.28). The current proof parallels the one given there.

For $\psi = \pi$, (2.28) is trivial, since the left-hand side is nonnegative. Hence we may assume $|\psi - \pi| > 0$. By the prime element theorem, in case (1.2), the left-hand side of (2.28) is

$$\geq \sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(2n)(q^{-1}te^{ik\psi})^{2n} = \frac{1}{4}\sum_{k=-\infty}^{\infty} b_k \log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2k\psi)} + O(1).$$

We have

$$\begin{aligned}
&\left| \sum_{|k| \geq \frac{\pi}{2|\psi-\pi|}} b_k \log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2k\psi)} \right| \\
&\quad \leq 2 \sum_{|k| \geq \frac{\pi}{2|\psi-\pi|}} b_k \log \frac{1}{1-t^2} \ll 1 + \frac{2|\psi-\pi|}{\pi} \log \frac{1}{1-t} = O_\alpha(1).
\end{aligned}$$

Also,

$$\begin{aligned}
&\sum_{0<|k|<\frac{\pi}{2|\psi-\pi|}} b_k \log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2k\psi)} \\
&\quad = \sum_{0<|k|<\frac{\pi}{2|\psi-\pi|}} b_k \left(\log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2(\psi-\pi))} + \log Q \right) \\
&\quad = \sum_{0<|k|<\frac{\pi}{2|\psi-\pi|}} b_k \log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2(\psi-\pi))} + O(1),
\end{aligned}$$

where

$$\log Q = \log \frac{(1-t^2)^2 + 2t^2(1-\cos 2(\psi-\pi))}{(1-t^2)^2 + 2t^2(1-\cos 2k(\psi-\pi))} \ll 1 + \log k.$$

Hence

$$\begin{aligned}
&\sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(2n)(q^{-1}te^{ik\psi})^{2n} \\
&\quad \geq \frac{b_0}{2} \log \frac{1}{1-t^2} + \frac{1}{4} \log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2(\psi-\pi))} \\
&\qquad \times \sum_{|k|>0} b_k + O_\alpha(1) + O(1) \\
&\quad = \frac{b_0}{2} \log \frac{1}{1-t^2} + \frac{h_0(e^{i\theta_0}) - b_0}{4} \log \frac{1}{(1-t^2)^2 + 2t^2(1-\cos 2(\psi-\pi))} \\
&\qquad + O_\alpha(1) + O(1) \\
&\quad \geq \frac{b_0}{4} \log \frac{(1-t^2)^2 + 2t^2(1-\cos 2(\psi-\pi))}{(1-t^2)^2} + O_\alpha(1) + O(1) \\
&\quad \geq \frac{\delta^2}{8\pi} \log\left(1 + \frac{(\psi-\pi)^2}{\pi^2(1-t)^2}\right) + O_\alpha(1) + O(1).
\end{aligned}$$

In case (1.3), the left-hand side of (2.28) equals

$$\sum_{k=-\infty}^{\infty} b_k\left(\log\frac{1}{1-te^{ik\psi}}+\log\frac{1}{1+te^{ik\psi}}\right)+O(1).$$

As is shown above,

$$\sum_{|k|\geq\frac{\pi}{2|\psi-\pi|}} b_k\left(\log\frac{1}{1-te^{ik\psi}}+\log\frac{1}{1+te^{ik\psi}}\right)\ll 1+\frac{2|\psi-\pi|}{\pi}\log\frac{1}{1-t}=O_\alpha(1).$$

To estimate the sum over k in the range $0<|k|<\frac{\pi}{2|\psi-\pi|}$, note that $1\pm te^{ik\psi}=1\pm te^{ik(\psi-\pi)}$ and $1\leq|1+te^{ik(\psi-\pi)}|\leq 2$ if k is even and that $1\pm te^{ik\psi}=1\mp te^{ik(\psi-\pi)}$ if k is odd. Hence

$$\begin{aligned}&\sum_{0<|k|<\frac{\pi}{2|\psi-\pi|}} b_k\left(\log\frac{1}{1-te^{ik\psi}}+\log\frac{1}{1+te^{ik\psi}}\right)\\&=\sum_{0<|k|<\frac{\pi}{2|\psi-\pi|}} b_k\log\frac{1}{1-te^{ik(\psi-\pi)}}+O(1)\\&=\frac{1}{2}\sum_{0<|k|<\frac{\pi}{2|\psi-\pi|}} b_k\log\frac{1}{(1-t)^2+2t(1-\cos(\psi-\pi))}+O(1)\end{aligned}$$

and we arrive at

$$\sum_{k=-\infty}^{\infty} b_k\sum_{n=1}^{\infty}P(n)(q^{-1}te^{ik\psi})^n\geq\frac{\delta^2}{4\pi}\log\left(1+\frac{2(\psi-\pi)^2}{\pi^2(1-t)^2}\right)+O_\alpha(1)+O(1).\qquad\square$$

3. Some inequalities

Let

$$\hat{F}(y):=\sum_{a\in\mathcal{G}}f(a)y^{\partial(a)}=\prod_{p\in\mathcal{P}}\left(1-f(p)y^{\partial(p)}\right)^{-1}$$

be the generating function of $f(a)$. Also, let

$$\hat{F}_v(y):=\sum_{a\in\mathcal{G}}|f(a)|y^{\partial(a)}=\prod_{p\in\mathcal{P}}\left(1-|f(p)|y^{\partial(p)}\right)^{-1}.$$

This section is devoted to the proofs of some inequalities for $\hat{F}(y)$ and $\hat{F}_v(y)$, corresponding to conditions (2), (2*), and (3), respectively. These inequalities prepare the ground for proofs of the quantitative mean-value theorems in Section 4.

Lemma 3.1. *If f satisfies conditions* (1) *and either* (2) *or* (2*), *then there exist positive constants α and c_3 such that*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)}\leq\exp\left\{-c_3\lambda_1\log\frac{1}{1-|z|}+\lambda_1\sum_{f(p)=0}q^{-\partial(p)}|z|^{\partial(p)}+O(1)\right\}\tag{3.1}$$

for $z = |z|e^{i\psi}$ *satisfying* $|z| < 1$ *and* $(1-|z|)^{\alpha} \le |\psi| \le \pi$. *Also*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \le \exp\left\{-c_3\lambda_1 \log\left(1+\frac{2\psi^2}{\pi^2(1-|z|)^2}\right) + \lambda_1 \sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)} + O(1)\right\} \tag{3.2}$$

for $z = |z|e^{i\psi}$ *satisfying* $|z| < 1$ *and* $|\psi| < (1-|z|)^{\alpha}$. *Here the constants* α *and* c_3 *depend at most on the constants* μ *of condition* (2) *or* (2*) *and the* O*-constants depend at most on* λ_2.

Remark. If $\mu \le \frac{\pi}{40}$, we may take

$$\alpha = \frac{12}{2025\pi^2}, \quad c_3 = \frac{\mu^2}{128\pi^2}\left(1-\cos\frac{\mu}{4}\right).$$

Proof: We shall give only the proof for f satisfying conditions (1) and (2). Then the proof for f satisfying conditions (1) and (2*) is the same with the parameter $\theta_0 = \frac{3\pi}{4}$ in place of $\theta_0 = -\frac{3\pi}{4}$ of the function $h(e^{i\theta})$. Without loss of generality, we may assume $\mu \le \frac{\pi}{40}$.

Let

$$S := S(|z|) = \sum_p \left(1 - \Re e^{i(\theta_p + \partial(p)\psi)}\right)|f(p)||z|^{\partial(p)}q^{-\partial(p)}$$

for $|z| < 1$. The intervals

$$\left|\psi - \frac{s}{r}\pi\right| \le \frac{\mu}{4r} \quad \text{and} \quad \left|\psi + \frac{s}{r}\pi\right| \le \frac{\mu}{4r}, \quad \frac{s}{r} \in \mathbf{F}_{[\frac{8\pi}{\mu}]}$$

form a cover of the interval $-\pi \le \psi \le \pi$. By Lemma 2.2, there exists a constants $c_3 = \frac{\mu^2}{128\pi^2}(1-\cos\frac{\mu}{2})$ such that

$$S \ge c_3\lambda_1 \log\frac{1}{1-|z|} - \lambda_1 \sum_{f(p)=0} |z|^{\partial(p)}q^{-\partial(p)} + (1+\lambda_1)O(1)$$

for all $\psi \in [-\pi, \pi]$ but $|\psi| \le \frac{\mu}{4}$.

Then, by condition (2),

$$\left|e^{i\theta_p} - e^{i\frac{3\pi}{4}}\right| > \frac{1}{\sqrt{2}}.$$

Hence

$$1 - \Re e^{i(\theta_p+\partial(p)\psi)} \ge \begin{cases} \frac{1}{10}, & \text{if } \left|\partial(p)\psi - \left(2m+\frac{3}{4}\right)\pi\right| < \frac{1}{3\sqrt{2}} \text{ for some } m \in \mathbb{Z}; \\ 0, & \text{otherwise.} \end{cases}$$

Let $h_0(e^{i\theta})$ be the periodic function defined in (2.21) with $\delta = \frac{1}{10}$ and $h(e^{i\theta}) = h_0(e^{i(\theta - \frac{3\pi}{4})})$ with $\theta_0 = -\frac{3\pi}{4}$. then

$$1 - \Re e^{i(\theta_p + \partial(p)\psi)} \geq h_0(e^{i(\partial(p)\psi - \frac{3\pi}{4})}) = h\big(e^{i\partial(p)\psi}\big).$$

It follows that

$$S \geq \lambda_1 \sum_p h\big(e^{i\partial(p)\psi}\big)(q^{-1}|z|)^{\partial(p)} - \frac{\lambda_1}{10} \sum_{f(p)=0} (q^{-1}|z|)^{\partial(p)}. \tag{3.3}$$

By Lemma 2.6 with $\delta = \frac{1}{10}$, the first sum on the right-hand side equals

$$\sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}|z|e^{ik\psi})^n \geq \frac{1}{1000\pi} \log \frac{1}{1-|z|} + O(1).$$

Hence

$$S \geq \frac{\lambda_1}{1000\pi} \log \frac{1}{1-|z|} - \frac{\lambda_1}{10} \sum_{f(p)=0} (q^{-1}|z|)^{\partial(p)} + O(1)$$

for $(1-|z|)^\alpha \leq |\psi| \leq \frac{\mu}{4}$ $(<\frac{\delta}{5} = \frac{1}{50})$ with $\alpha = \frac{3}{2500\pi^2}$. This proves (3.1) with

$$c_3 = \frac{\mu^2}{128\pi^2}\left(1 - \cos\frac{\mu}{4}\right)$$

when $\mu \leq \frac{\pi}{40}$.

Finally, for $|\psi| < (1-|z|)^\alpha$, by Lemma 2.7, the first sum on the right-hand side of (3.3) is

$$\geq \frac{1}{400\pi} \log\left(1 + \frac{2\psi^2}{\pi^2(1-|z|)^2}\right) + O(1)$$

and (3.2) follows with

$$c_3 = \frac{1}{400\pi}. \qquad \square$$

Lemma 3.2. *If f satisfyies conditions* (1) *and* (3), *then there exist positive constants α and c_4 such that*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \leq \exp\left\{-c_4\lambda_1 \log\frac{1}{1-|z|} + \frac{\lambda_1}{5}\sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)} + O(1)\right\} \tag{3.4}$$

for $z = |z|e^{i\psi}$ *satisfying* $|z| < 1$ *and* $(1-|z|)^\alpha \le |\psi| \le \pi - (1-|z|)^\alpha$. *Also*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \le \exp\left\{-c_4\lambda_1 \log\left(1 + \frac{2\psi^2}{\pi^2(1-|z|)^2}\right) + \frac{\lambda_1}{5}\sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)} + O(1)\right\} \tag{3.5}$$

for $z = |z|e^{i\psi}$ *satisfying* $|z| < 1$ *and* $|\psi| < (1-|z|)^\alpha$ *and*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \le \exp\left\{-c_4\lambda_1 \log\left(1 + \frac{2(\psi-\pi)^2}{\pi^2(1-|z|)^2}\right) + \frac{\lambda_1}{5}\sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)} + O(1)\right\} \tag{3.6}$$

for $z = |z|e^{i\psi}$ *satisfying* $|z| < 1$ *and* $|\psi - \pi| < (1-|z|)^\alpha$. *Here the constants* α *and* c_4 *are independent of* λ_1 *and* λ_2 *and the* O*-constants depend at most on* λ_2.

Proof: Let S be as defined in the proof of Lemma 3.1. The intervals

$$\left|\psi - \frac{s}{r}\pi\right| \le \frac{\pi}{96r} \quad \text{and} \quad \left|\psi + \frac{s}{r}\pi\right| \le \frac{\pi}{96r}$$

with $\frac{s}{r} \in \mathbf{F}_{192}$ form a cover of the interval $-\pi \le \psi \le \pi$. By Lemma 2.3, there exists a constant $c_4 = \frac{c_2}{192}$ such that

$$S \ge c_4\lambda_1 \log\frac{1}{1-|z|} - \frac{\lambda_1}{29}\sum_{f(p)=0}|z|^{\partial(p)}q^{-\partial(p)} + (1+\lambda_1)O(1)$$

for all $\psi \in [-\pi, \pi]$ except that $|\psi| \le \frac{\pi}{96}$, $(1-\frac{1}{96})\pi \le \psi \le \pi$, and $-\pi \le \psi \le -(1-\frac{1}{96})\pi$.

Then, by condition (3),

$$\left|e^{i\theta_p} - e^{i\frac{\pi}{2}}\right| \ge 1.$$

Hence

$$1 - \Re e^{i(\theta_p + \partial(p)\psi)} \ge \begin{cases} \frac{1}{5}, & \text{if } \left|\partial(p)\psi + \left(2m + \frac{1}{2}\right)\pi\right| < \frac{\pi}{10} \text{ for some } m \in \mathbb{Z}; \\ 0, & \text{otherwise.}\end{cases}$$

Let $h_0(e^{i\theta})$ be the periodic function defined in (2.21) with $\delta = \frac{1}{5}$ and $h(e^{i\theta}) = h_0(e^{i(\theta+\pi/2)})$. Hence

$$1 - \Re e^{i(\theta_p + \partial(p)\psi)} \ge h_0\big(e^{i(\theta+\pi/2)}\big) = h\big(e^{i\partial(p)\psi}\big)$$

with $\theta_0 = \frac{1}{2}\pi$. It follows that

$$S \geq \lambda_1 \sum_p h\big(e^{i\partial(p)\psi}\big)(q^{-1}|z|)^{\partial(p)} - \frac{\lambda_1}{5} \sum_{f(p)=0} (q^{-1}|z|)^{\partial(p)}$$
$$= \lambda_1 \sum_{k=-\infty}^{\infty} b_k \sum_{n=1}^{\infty} P(n)(q^{-1}|z|e^{ik\psi})^n - \frac{\lambda_1}{5} \sum_{f(p)=0} (q^{-1}|z|)^{\partial(p)}.$$

By Lemma 2.6 with $\delta = \frac{1}{5}$, the first sum on the right-hand side is

$$\geq \frac{1}{250\pi} \log \frac{1}{1-|z|} + O(1)$$

for $(1-|z|)^\alpha \leq |\psi| \leq \frac{1}{25}$ and for $(1-|z|)^\alpha \leq |\psi - \pi| \leq \frac{1}{25}$, where $\alpha = \frac{3}{625\pi^2}$. Hence

$$S \geq \frac{\lambda_1}{250\pi} \log \frac{1}{1-|z|} - \frac{\lambda_1}{5} \sum_{f(p)=0} (q^{-1}|z|)^{\partial(p)} + \lambda_1 O(1)$$

for $(1-|z|)^\alpha \leq |\psi| \leq \frac{\pi}{96}$ $(<\frac{\delta}{5} = \frac{1}{25})$ and for $(1-|z|)^\alpha \leq |\psi - \pi| \leq \frac{\pi}{96}$. This proves (3.4).

Then, for $|\psi| \leq (1-|z|)^\alpha$, by Lemma 2.7 with $\delta = \frac{1}{5}$ and $\alpha = \frac{3}{625\pi^2}$,

$$S \geq \frac{\lambda_1}{100\pi} \log\left(1 + \frac{2\psi^2}{\pi^2(1-|z|)^2}\right) - \frac{\lambda_1}{5} \sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)} + \lambda_1 O(1)$$

and (3.5) follows.

Finally, for $|\psi - \pi| \leq (1-|z|)^\alpha$, by Lemma 2.7 again,

$$S \geq \frac{\lambda_1}{200\pi} \log\left(1 + \frac{2(\psi-\pi)^2}{\pi^2(1-|z|)^2}\right) - \frac{\lambda_1}{5} \sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)} + \lambda_1 O(1)$$

and (3.6) follows. □

Lemma 3.3. *If f satisfies conditions* (1) *and either* (2) *or* (2*) *then there exist constants* $0 < \eta < 1$ *and* $c_5 > 0$ *such that*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \leq c_5 \left(\frac{|\hat{F}(q^{-1}|z|)|}{\hat{F}_v(q^{-1}|z|)}\right)^{\eta\lambda_1/\lambda_2} \exp\left\{\lambda_1 \sum_{f(p)=0} q^{-\partial(p)}|z|^{\partial(p)}\right\} \tag{3.7}$$

for $|z| < 1$. *Here the constant η depends at most on the constant μ of condition* (2) *or* (2*) *and the constant c_5 depends at most on λ_2.*

Proof: This lemma can be proved from Lemma 3.1 in the same way as Lemma 2.9 in [11] is. □

Lemma 3.4. *If f satisfies conditions* (1) *and* (3), *then*

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \leq c_6 \left(\frac{|\hat{F}(q^{-1}|z|)|}{\hat{F}_v(q^{-1}|z|)} \right)^{\eta\lambda_1/\lambda_2} \exp\left\{ \frac{\lambda_1}{5} \sum_{f(p)=0} q^{-\partial(p)} |z|^{\partial(p)} \right\} \tag{3.8}$$

for $z = |z|e^{i\psi}$ with $|z| < 1$ and $|\psi| \leq \frac{\pi}{2}$ and

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \leq c_6 \left(\frac{|\hat{F}(-q^{-1}|z|)|}{\hat{F}_v(q^{-1}|z|)} \right)^{\eta\lambda_1/\lambda_2} \exp\left\{ \frac{\lambda_1}{5} \sum_{f(p)=0} q^{-\partial(p)} |z|^{\partial(p)} \right\} \tag{3.9}$$

for $z = |z|e^{i\psi}$ with $|z| < 1$ and $|\psi - \pi| \leq \frac{\pi}{2}$, where the constant η is independent of λ_1 and λ_2 and the constant c_6 depends at most on λ_2.

Proof: We will give the proof of (3.9) only. The proof of (3.8) is almost the same.
Thus we first show that

$$\frac{|\hat{F}(q^{-1}z)|}{|\hat{F}(-q^{-1}|z|)|} \leq \exp\{\lambda_2 \, O(1)\} \tag{3.10}$$

for $z = |z|e^{i\psi}$ with $|z| < 1$ and $|\psi - \pi| \leq 1 - |z|$. Actually, for $|z| < 1$,

$$\frac{\hat{F}(q^{-1}z)}{\hat{F}(-q^{-1}|z|)} = \exp\left\{ \sum_p \left(z^{\partial(p)} - (-1)^{\partial(p)} |z|^{\partial(p)} \right) f(p) q^{-\partial(p)} + O(1) \right\}.$$

Hence,

$$\frac{|\hat{F}(q^{-1}z)|}{|\hat{F}(-q^{-1}|z|)|} \leq \exp\left\{ \sum_p \left| z^{\partial(p)} - (-1)^{\partial(p)} |z|^{\partial(p)} \right| |f(p)| q^{-\partial(p)} + O(1) \right\}.$$

We have

$$\sum_{\partial(p) \geq M} \left| z^{\partial(p)} - (-1)^{\partial(p)} |z|^{\partial(p)} \right| |f(p)| q^{-\partial(p)} \ll \lambda_2 \sum_{m \geq M} P(m) q^{-m} |z|^m \ll \frac{\lambda_2}{M(1 - |z|)} \ll \lambda_2$$

if $M = \frac{1}{1-|z|}$. Then we have

$$\begin{aligned}
&\sum_{\partial(p) \leq \frac{1}{|\psi - \pi|}} \left| z^{\partial(p)} - (-1)^{\partial(p)} |z|^{\partial(p)} \right| |f(p)| q^{-\partial(p)} \\
&= \sum_{\partial(p) \leq \frac{1}{|\psi - \pi|}} |z|^{\partial(p)} \left| e^{i\partial(p)(\psi - \pi)} - 1 \right| |f(p)| q^{-\partial(p)} \\
&\leq \lambda_2 \sum_{\partial(p) \leq \frac{1}{|\psi - \pi|}} \partial(p) |\psi - \pi| \, q^{-\partial(p)}
\end{aligned}$$

$$= \lambda_2 |\psi - \pi| \sum_{m \le \frac{1}{|\psi - \pi|}} m P(m) q^{-m}$$
$$\ll \lambda_2.$$

Hence (3.10) follows when $|\psi - \pi| \le 1 - |z|$.

Then we show that

$$\frac{|\hat{F}(q^{-1}z)|}{|\hat{F}(-q^{-1}|z|)|} \le \exp\left\{2\lambda_2 \log \frac{|\psi - \pi|}{1 - |z|} + \lambda_2 O(1)\right\} \tag{3.11}$$

for $z = |z|e^{i\psi}$ with $|z| < 1$ and $1 - |z| < |\psi - \pi| \le \frac{\pi}{2}$. Actually,

$$\sum_{\frac{1}{|\psi - \pi|} < \partial(p) < M} \left|z^{\partial(p)} - (-1)^{\partial(p)}|z|^{\partial(p)}\right| |f(p)| q^{-\partial(p)}$$
$$\le 2\lambda_2 \sum_{\frac{1}{|\psi - \pi|} < \partial(p) < M} q^{-\partial(p)} = 2\lambda_2 \log \frac{|\psi - \pi|}{1 - |z|} + \lambda_2 O(1)$$

and (3.11) follows.

Then, to prove (3.9), we write

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} = \frac{|\hat{F}(q^{-1}z)|}{|\hat{F}(-q^{-1}|z|)|} \frac{|\hat{F}(-q^{-1}|z|)|}{\hat{F}_v(q^{-1}|z|)}. \tag{3.12}$$

We first note that

$$\frac{|\hat{F}(q^{-1}z)|}{|\hat{F}(-q^{-1}|z|)|} \ll 1$$

for $|\psi - \pi| \le 1 - |z|$. In this case, (3.9) is certainly true with any $\eta \le 1$. Hence, we may assume $1 - |z| < |\psi - \pi| \le \frac{\pi}{2}$ and then (3.11) holds. If

$$2\lambda_2 \log \frac{|\psi - \pi|}{1 - |z|} \le \frac{1}{2} \log \frac{\hat{F}_v(q^{-1}|z|)}{|\hat{F}(-q^{-1}|z|)|}$$

then from (3.12) and (3.11),

$$\frac{|\hat{F}(q^{-1}z)|}{\hat{F}_v(q^{-1}|z|)} \le \left(\frac{|\hat{F}(-q^{-1}|z|)|}{\hat{F}_v(q^{-1}|z|)}\right)^{\frac{1}{2}} \exp\{O(1)\}$$

and (3.9) holds with any $\eta \le \frac{1}{2}$. Hence, we may assume that

$$2\lambda_2 \log \frac{|\psi - \pi|}{1 - |z|} \ge \frac{1}{2} \log \frac{\hat{F}_v(q^{-1}|z|)}{|\hat{F}(-q^{-1}|z|)|} \tag{3.13}$$

For $|\psi - \pi| \leq (1 - |z|)^\alpha$, we have

$$\log\left(1 + \frac{2|\psi - \pi|^2}{\pi^2(1 - |z|)^2}\right) \geq 2\log\frac{|\psi - \pi|}{1 - |z|} - \log\frac{\pi^2}{2}$$

and then (3.9) with any $\eta \leq c_4/2$ follows from (3.7) and (3.13). Finally, for $(1 - |z|)^\alpha \leq |\psi - \pi| \leq \frac{\pi}{2}$, from (3.13),

$$\log\frac{1}{1 - |z|} \geq \log\frac{|\psi - \pi|}{1 - |z|} - \log\frac{\pi}{2} \geq \frac{1}{4\lambda_2}\log\frac{\hat{F}_v(q^{-1}|z|)}{|\hat{F}(-q^{-1}|z|)|} - \log\frac{\pi}{2}$$

and then, from (3.4), (3.8) follows with $\eta \leq c_4/4$. Therefore, in all cases, (3.9) holds with $\eta = \min\{\frac{1}{2}, c_4/4\}$. □

4. Quantitative mean-value theorems

We now prove two quantitative mean-value theorems.

Theorem 4.1. *If f satisfies condition* (1) *with $\lambda_1 > \frac{1}{2}$ and either condition* (2) *or* (2*) *then there exist positive constants δ and c_7 such that*

$$\left|\sum_{\partial(a)=m} f(a)\right| \leq c_7\left(\Gamma\left(\lambda_1 - \frac{1}{2}\right)\right)^{\frac{1}{2}} q^m \exp\left\{\sum_{\partial(p)\leq m}(|f(p)| - 1)q^{-\partial(p)} - \delta\delta_1\sum_{\partial(p)\leq m}(|f(p)| - \Re f(p))q^{-\partial(p)} + \lambda_1\sum_{\partial(p)\leq m,\, f(p)=0} q^{-\partial(p)}\right\}, \quad (4.1)$$

where $\Gamma(x)$ is the Euler gamma function and

$$\delta_1 = \left(\frac{\lambda_1 - \frac{1}{2}}{2\lambda_1}\right)^2\frac{\lambda_1}{\lambda_2}.$$

The constant δ depends only on the constant μ of condition (2) *or* (2*) *and c_7 depends at most on μ and λ_2.*

Remark. If f satisfies condition (1) with $\lambda_1 > \frac{1}{2}$ except for $p \in \mathcal{P}_1 \subset \mathcal{P}$ such that $\sum_{p\in\mathcal{P}_1} q^{-\partial(p)} < \infty$ then Theorem 4.1 still holds true. The same can be said about Theorem 4.2.

Proof: This theorem can be proved, on the basis of Lemma 3.3, in the same way as Theorem 3.1 in [11] is. □

Theorem 4.2. *If f satisfies condition* (1) *with $\lambda_1 > \frac{1}{2}$ and condition* (3) *then there exist positive constants δ and c_8 such that*

$$\begin{aligned}\left|\sum_{\partial(a)=m} f(a)\right| \le c_8\left(\Gamma\left(\lambda_1-\frac{1}{2}\right)\right)^{\frac{1}{2}} q^m \exp\Bigg\{ &\sum_{\partial(p)\le m}(|f(p)|-1)q^{-\partial(p)} \\ &- \delta\delta_1 \sum_{\partial(p)\le m} \min\bigl\{|f(p)|-\Re f(p), |f(p)|-(-1)^{\partial(p)}\Re f(p)\bigr\}q^{-\partial(p)} \\ &+ \delta_2 \sum_{\partial(p)\le m, f(p)=0} q^{-\partial(p)}\Bigg\},\end{aligned} \tag{4.2}$$

where the constant δ_1 is as defined in Theorem 4.1 *and*

$$\delta_2 = \frac{3}{5}\lambda_1 + \frac{1}{5}.$$

The constant δ is independent of λ_1 and λ_2 and the constant c_8 depends at most on λ_2.

Proof: We have, by Lemma 3.2 in [11],

$$m\left|\sum_{\partial(a)=m} f(a)\right| \ll m^{\frac{1}{2}} q^m \left(\int_{-\pi}^{\pi} |\hat{F}(q^{-1}re^{i\theta})|^2\, d\theta\right)^{\frac{1}{2}}$$

for $r = 1 - \frac{1}{m}$. To estimate the integral on the right-hand side, we introduce two auxiliary functions $f_1(a)$ and $h(a)$ which are both completely multiplicative and are defined as follows.

Let

$$\tau = \frac{2\lambda_1+1}{4}.$$

Then $\frac{1}{2} < \tau < \lambda_1$. First, the function $f_1(a)$ satisfies

$$f_1(p) = \begin{cases} f(p)\dfrac{\tau}{|f(p)|}, & \text{if } f(p) \ne 0; \\ 0, & \text{if } f(p) = 0 \end{cases}$$

for $p \in \mathcal{P}$. Then the function $h(a)$ satisfies

$$h(p) := f(p) - f_1(p) = \begin{cases} \left(1 - \dfrac{\tau}{|f(p)|}\right) f(p), & \text{if } \lambda_1 \le |f(p)| \le \lambda_2; \\ 0, & \text{if } f(p) = 0 \end{cases}$$

for $p \in \mathcal{P}$.

Let

$$\hat{F}_1(y) := \sum_{a\in\mathcal{G}} f_1(a) y^{\partial(a)} = \prod_{p\in\mathcal{P}} \left(1 - f_1(p) y^{\partial(p)}\right)^{-1}$$

and

$$\hat{H}(y) := \sum_{a\in\mathcal{G}} h(a) y^{\partial(a)} = \prod_{p\in\mathcal{P}} \left(1 - h(p) y^{\partial(p)}\right)^{-1}$$

be the generating functions of f_1 and h, respectively. Note that $|f_1(p)| = \tau$ or 0 and

$$0 < \lambda_1^* = \frac{\lambda_1 - \frac{1}{2}}{2} \le |h(p)| \le \lambda_2 - \frac{\lambda_1 + \frac{1}{2}}{2}.$$

Then we can write

$$(\hat{F}(q^{-1}z))^2 = (\hat{H}(q^{-1}z))^2 (\hat{F}_1(q^{-1}z))^2 T(z),$$

where $T(z)$ is analytic for $|z| < \min\{q\lambda_2^{-1}, q^{\frac{1}{2}}\}$ and is $\ll_\epsilon 1$ for $|z| < \min\{q\lambda_2^{-1}, q^{\frac{1}{2}}\} - \epsilon$. Therefore

$$\int_{-\pi}^{\pi} |\hat{F}(q^{-1}re^{i\theta})|^2 d\theta \ll \max_{|\theta|\le\pi} |\hat{H}(q^{-1}re^{i\theta})|^2 \int_{-\pi}^{\pi} |\hat{F}_1(q^{-1}re^{i\theta})|^2\, d\theta. \tag{4.3}$$

To estimate the integral on the right-hand side of (4.3), let

$$\hat{A}(z) := \exp\left\{ \sum_p \sum_{k=1}^{\infty} \frac{\tau^k}{k} (q^{-1}z)^{k\partial(p)} \right\}.$$

Then $\hat{A}(z)$ is a majorant function of $\hat{F}_1(q^{-1}z)$ in the disk $|z| < 1$ and hence

$$\int_{-\pi}^{\pi} |\hat{F}_1(q^{-1}re^{i\theta})|^2\, d\theta \le \int_{-\pi}^{\pi} |\hat{A}(re^{i\theta})|^2\, d\theta.$$

We may write

$$\hat{A}(z) = e^{R_1(z)} \prod_p \left(1 - (q^{-1}z)^{\partial(p)}\right)^{-\tau},$$

where the function $R_1(z)$ is analytic for $|z| < \min\{q\lambda_2^{-1}, q^{\frac{1}{2}}\}$, and is $\ll_\epsilon 1$ for $|z| \le \min\{q\lambda_2^{-1}, q^{\frac{1}{2}}\} - \epsilon$. Hence

$$\hat{A}(z) = (Z(q^{-1}z))^\tau\, e^{R_1(z)}.$$

It follows that

$$\int_{-\pi}^{\pi} |\hat{A}(re^{i\theta})|^2 \, d\theta \ll \int_{-\pi}^{\pi} |Z(q^{-1}re^{i\theta})|^{2\tau} d\theta \ll 1 + \int_{-\pi}^{\pi} \left| \frac{1}{(1 - re^{i\theta})^{\tau}} \right|^2 d\theta,$$

where the constants implied by $\ll$ depend at most on λ_2. Note that $\tau > \frac{1}{2}$. The same argument in the proof of (3.18) in [11] shows that

$$\int_{-\pi}^{\pi} |\hat{A}(re^{i\theta})|^2 \, d\theta \ll 1 + \frac{\Gamma\left(\lambda_1 - \frac{1}{2}\right)}{\Gamma^2(\tau)} m^{\lambda_1 - \frac{1}{2}}$$

and hence

$$\int_{-\pi}^{\pi} |\hat{F}_1(q^{-1}re^{i\theta})|^2 \, d\theta \ll \Gamma\left(\lambda_1 - \frac{1}{2}\right) m^{\lambda_1 - \frac{1}{2}}. \tag{4.4}$$

To estimate the maximum on the right-hand side of (4.3), note that $h(a)$ satisfies the conditions of Lemma 3.4 with λ_1^* in place of λ_1. Hence

$$|\hat{H}(q^{-1}re^{i\theta})| \ll (\hat{H}_v(q^{-1}r))^{1-\delta} |\hat{H}(q^{-1}r)|^{\delta} \exp\left\{ \frac{\lambda_1 - \frac{1}{2}}{10} \sum_{f(p)=0} (q^{-1}r)^{\partial(p)} \right\}$$

when $|\theta| \le \frac{\pi}{2}$, where $\delta = \eta \lambda_1^* \lambda_2^{-1}$. The same argument in the proof of Theorem 3.1 in [11] yields

$$\begin{aligned} \max_{|\theta| \le \frac{\pi}{2}} |\hat{H}(q^{-1}re^{i\theta})| \ll m^{\frac{3}{4} - \frac{\lambda_1}{2}} \exp\Bigg\{ & \sum_{\partial(p) \le m} (|f(p)| - 1) q^{-\partial(p)} \\ & - \delta^* \delta_1 \sum_{\partial(p) \le m} (|f(p)| - \Re f(p)) q^{-\partial(p)} + \delta_2 \sum_{\substack{\partial(p) \le m \\ f(p)=0}} q^{-\partial(p)} \Bigg\}. \end{aligned} \tag{4.5}$$

Then, by Lemma 3.4 again,

$$|\hat{H}(q^{-1}re^{i\theta})| \ll (\hat{H}_v(q^{-1}r))^{1-\delta} |\hat{H}(-q^{-1}r)|^{\delta} \exp\left\{ \frac{\lambda_1 - \frac{1}{2}}{10} \sum_{f(p)=0} (q^{-1}r)^{\partial(p)} \right\}$$

when $|\theta - \pi| \le \frac{\pi}{2}$. We have

$$|\hat{H}(-q^{-1}r)|^{\delta} = \exp\left\{ \delta \sum_{\partial(p) \le m} \Re h(p)(-q^{-1}r)^{\partial(p)} + O(1) \right\}.$$

It follows that

$$
\begin{aligned}
(\hat{H}_v(q^{-1}r))^{1-\delta}|\hat{H}(-q^{-1}r)|^{\delta} &= \exp\Bigg\{(1-2\delta)\sum_{\partial(p)\le m}|h(p)|(q^{-1}r)^{\partial(p)} \\
&\qquad + \delta\sum_{\partial(p)\le m}(|h(p)|+(-1)^{\partial(p)}\Re h(p))(q^{-1}r)^{\partial(p)}+O(1)\Bigg\} \\
&\le \exp\Bigg\{\sum_{\partial(p)\le m}|h(p)|q^{-\partial(p)} \\
&\qquad - \delta\sum_{\partial(p)\le m}(|h(p)|-(-1)^{\partial(p)}\Re h(p))q^{-\partial(p)}+O(1)\Bigg\}.
\end{aligned}
$$

Note that

$$
\begin{aligned}
&|h(p)|-(-1)^{\partial(p)}\Re h(p) \\
&\quad = \begin{cases} \left(1-\dfrac{\tau}{|f(p)|}\right)\big(|f(p)|-(-1)^{\partial(p)}\Re f(p)\big), & \text{if } f(p)\ne 0; \\ 0, & \text{if} f(p)=0. \end{cases}
\end{aligned}
$$

Then we obtain

$$
\begin{aligned}
&\sum_{\partial(p)\le m}\big(|h(p)|-(-1)^{\partial(p)}\Re h(p)\big)q^{-\partial(p)} \\
&\quad \ge \frac{\lambda_1-\frac{1}{2}}{2\lambda_1}\sum_{\partial(p)\le m}\big(|f(p)|-(-1)^{\partial(p)}\Re f(p)\big)q^{-\partial(p)}
\end{aligned}
$$

and hence

$$
\begin{aligned}
&\max_{\frac{\pi}{2}\le|\theta|\le\pi}|\hat{H}(q^{-1}re^{i\theta})| \\
&\quad \ll m^{\frac{3}{4}-\frac{\lambda_1}{2}}\exp\Bigg\{\sum_{\partial(p)\le m}(|f(p)|-1)q^{-\partial(p)} \\
&\qquad -\delta^*\delta_1\sum_{\partial(p)\le m}\big(|f(p)|-(-1)^{\partial(p)}\Re f(p)\big)q^{-\partial(p)}+\delta_2\sum_{\substack{\partial(p)\le m\\ f(p)=0}}q^{-\partial(p)}\Bigg\}.
\end{aligned}
$$

Thus

$$
\begin{aligned}
&\max_{|\theta|\le\pi}|\hat{H}(q^{-1}re^{i\theta})|\\
&\quad\ll m^{\frac{3}{4}-\frac{\lambda_1}{2}}\exp\Bigg\{\sum_{\partial(p)\le m}(|f(p)|-1)q^{-\partial(p)}\,-\,\delta^*\delta_1\sum_{\partial(p)\le m}\min\big\{|f(p)|\\
&\qquad -\Re f(p),\,|f(p)|-(-1)^{\partial(p)}\Re f(p)\big\}q^{-\partial(p)}+\delta_2\sum_{\substack{\partial(p)\le m\\ f(p)=0}}q^{-\partial(p)}\Bigg\} \qquad (4.6)
\end{aligned}
$$

and (4.2) follows from (4.3), (4.4), (4.5), and (4.6). □

Theorems 4.1 and 4.2 are generalizations of Theorem 3.1 in [11].

We note that the expression

$$
\min\big\{|f(p)|-\Re f(p),\,|f(p)|-(-1)^{\partial(p)}\Re f(p)\big\} \qquad (4.7)
$$

on the right-hand side of (4.2) of Theorem 4.2 cannot be replaced by $|f(p)|-\Re f(p)$ as the latter appears on the right-hand side of (4.1) of Theorem 4.1 no matter whether the zeta function $Z(y)$ of the semigroup has zeros at $y=-q^{-1}$. This is shown by the following examples. Thus, Theorem 4.2 shows an essential divergence from classical probabilistic number theory.

Example 4.1. Consider the function $f(a)=\lambda(a):=(-1)^{\Omega(a)}$ on an additive arithmetic semigroup $\mathcal{G}$, where $\Omega(a)$ denotes the number of prime divisors of a, counted according to multiplicity. Thus $\lambda(a)$ is an analogue of Liouville's function $\lambda(n)$ in classical number theory. We have $f(p)=-1$ and hence

$$
\min\big\{1-\Re f(p),\,1-(-1)^{\partial(p)}\Re f(p)\big\}=\begin{cases}2, & \text{if } \partial(p) \text{ is even;}\\ 0, & \text{if } \partial(p) \text{ is odd.}\end{cases}
$$

By Theorem 4.2,

$$
\begin{aligned}
\frac{1}{G(m)}\sum_{\partial(a)=m}\lambda(a)&\ll\exp\Bigg\{-\eta\sum_{\partial(p)\le m}\min\big\{1-\Re f(p),\,1-(-1)^{\partial(p)}\Re f(p)\big\}q^{-\partial(p)}\Bigg\}\\
&\ll\exp\Bigg\{-2\eta\sum_{1\le n\le m,n\ even}P(n)q^{-n}\Bigg\} \qquad (4.8)
\end{aligned}
$$

for some constant $\eta>0$. If $Z(y)$, the zeta function of $\mathcal{G}$, has no zeros at $y=-q^{-1}$, then

$$
P(n)=q^n n^{-1}(1+O(n^{1-\gamma})).
$$

Hence

$$\frac{1}{G(m)}\sum_{\partial(a)=m}\lambda(a)\ll m^{-\eta};$$

which gives a quantitative form of the mean-value theorem of $\lambda(a)$ proved in [9]. If $Z(y)$ has a zero at $y=-q^{-1}$,

$$P(n)=q^n n^{-1}(1-(-1)^n+O(n^{2-\gamma})).$$

In this case, (4.8) gives only the trivial estimate

$$\frac{1}{G(m)}\sum_{\partial(a)=m}\lambda(a)\ll 1.$$

A better estimate, without regarding the constant implied by $\ll$, is impossible in general. This can be shown by revisiting the example given in [9]. Actually, for the semigroup $\mathcal{G}$ given in the example [9],

$$G(m)=Aq^m+O\big(q^{\frac{m}{2}}m^{-\frac{1}{2}}\big)$$

and

$$Z(y)=\frac{1+qy}{1-qy}\left(\frac{1+qy^2}{1-qy^2}\right)^{\frac{1}{2}}e^{F(y)},\quad |y|<q^{-1}$$

Here the function $F(y)$ is holomorphic in the disk $\{|y|<q^{-\frac{1}{3}}\}$ and the function

$$H(y):=\left(\frac{1+qy^2}{1-qy^2}\right)^{\frac{1}{2}}$$

is the single-valued branch with $H(0)=1$ of the associated multiple-valued function in the domain $\mathcal{D}$ formed by cutting the complex plane along the real axis from $-\infty$ to $-q^{-\frac{1}{2}}$ and from $q^{-\frac{1}{2}}$ to $+\infty$ and along the imaginary axis from $-i\infty$ to $-iq^{-\frac{1}{2}}$ and from $iq^{-\frac{1}{2}}$ to $i\infty$. In this case, the generating function of λ is

$$\begin{aligned}M(y):&=\sum_{a\in\mathcal{G}}\lambda(a)y^{\partial(a)}=\frac{Z(y^2)}{Z(y)}\\&=\left(\frac{1-qy}{1+qy}\right)\left(\frac{1+qy^2}{1-qy^2}\right)^{\frac{1}{2}}\left(\frac{1+qy^4}{1-qy^4}\right)^{\frac{1}{2}}e^{-F(y)+F(y^2)}.\end{aligned}$$

Hence

$$\begin{aligned}\sum_{\partial(a)=m}\lambda(a)&=\frac{1}{2\pi i}\int_{|y|=r}\frac{M(y)}{y^{m+1}}\,dy=Res_{y=-q^{-1}}\frac{M(y)}{y^{m+1}}+\frac{1}{2\pi i}\int_{|y|=q^{-\frac{1}{2}-\epsilon}}\frac{M(y)}{y^{m+1}}\,dy\\&=(-1)^{m+1}2q^m\left(\frac{q+1}{q-1}\right)^{\frac{1}{2}}\left(\frac{q^3+1}{q^3-1}\right)^{\frac{1}{2}}e^{-F(-q^{-1})+F(q^{-2})}+\ O_\epsilon\big(q^{(\frac{1}{2}+\epsilon)m}\big)\end{aligned}$$

by shifting the integration path from the circle $|y| = r$ with $0 < r < q^{-1}$ to the circle $|y| = q^{-\frac{1}{2}-\epsilon}$. Therefore

$$\lim_{m\to\infty} \frac{(-1)^m}{G(m)} \sum_{\partial(a)=m} \lambda(a)$$

exists and is nonzero.

Example 4.2. Let $\mathcal{G}$ be the additive arithmetic semigroup of monic polynomials in $\mathbb{F}_q[x]$, where $\mathbb{F}_q$ is a finite field of q elements. Then the zeta function of $\mathcal{G}$ is

$$Z(y) = \frac{1}{1-qy}.$$

Consider the completely multiplicative function f on $\mathcal{G}$ with $f(p) = (-1)^{\partial(p)}$. Then the generating function of f is

$$\hat{F}(y) = \prod_{k=1}^{\infty}(1+y^{2k-1})^{-P(2k-1)} \prod_{k=1}^{\infty}(1-y^{2k})^{-P(2k)} = \frac{1}{1+qy}e^{R(y)}, \quad |y| < q^{-1},$$

where

$$\begin{aligned} R(y) = \sum_{k=1}^{\infty} \big[&P(2k-1)\log(1+y^{2k-1})^{-1} + P(2k-1)y^{2k-1} \\ &+ P(2k)\log(1-y^{2k})^{-1} - P(2k)y^{2k}\big] \\ &+ \sum_{k=1}^{\infty}\left[\left(-P(2k-1) + \frac{q^{2k-1}}{2k-1}\right)y^{2k-1} + \left(P(2k) - \frac{q^{2k}}{2k}\right)y^{2k}\right] \end{aligned}$$

is holomorphic in the disk $\{|y| < q^{-\frac{1}{2}}\}$. Therefore, as in Example 4.1,

$$F(m) = (-1)^m q^m e^{-R(-q^{-1})} + O_\epsilon\big(q^{(\frac{1}{2}+\epsilon)m}\big).$$

This shows that the expression (4.8) on the right-hand side of (4.2) of Theorem 4.2 cannot be replaced by $|f(p)| - \Re f(p)$ even if the zeta function has no zeros at $y = -q^{-1}$. Otherwise,

$$\sum_{\partial(p)\le m} (|f(p)| - \Re f(p))q^{-\partial(p)} = 2 \sum_{\substack{\partial(p)\le m \\ \partial(p)\,\mathrm{odd}}} q^{-\partial(p)} = \log m + O(1)$$

and (4.2) gave $|F(m)| \ll q^m m^{-\eta}$; which is certainly not true.

5. A local theorem

We now apply the quantitative mean-value theorems of completely multiplicative functions, established in this paper and in [11] respectively, to the investigation of density of a special additive function $\Omega^*(a)$.

Theorem 5.1. *Let $\mathcal{P}^* \subset \mathcal{P}$ and*

$$E(m) := \sum_{\substack{p\in\mathcal{P}^*\\ \partial(p)\le m}} q^{-\partial(p)}.$$

Let $\Omega^*(a)$ denote the number of prime divisors $p \in \mathcal{P}^*$ of $a \in \mathcal{G}$, counted according to multiplicity of p. Let δ be a real number satisfying $\max\{\frac{1}{2}, 2-q\} < \delta < 1$. Then there exists a positive constant c_9 such that

$$\frac{1}{G(m)} \sum_{\substack{\partial(a)=m\\ \Omega^*(a)=k}} 1 = \frac{E^k(m)}{k!}\exp\{-E(m)\}\left\{1 + \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^5} O\left(\frac{|k-E(m)|}{E(m)} + \frac{1}{\sqrt{E(m)}}\right)\right.$$
$$\left. + O\left(\exp\left\{-c_9\left(\delta-\frac{1}{2}\right)^2 \sum_{\substack{p\in\mathcal{P}^*\\ \partial(p)\le m, \partial(p)\, even}} q^{-\partial(p)}\right\}\right)\right\} \tag{5.1}$$

for $E(m) > (q-2+\delta)^3$ and $\delta \le k/E(m) \le 2-\delta$. Here c_9 and the O-constants are independant of δ.

Remark. For function fields, $q \ge 2$. The condition on δ becomes $\frac{1}{2} < \delta < 1$. This might be replaced by $0 < \delta < 1$ if the corresponding condition $\frac{1}{2} < \lambda_1$ in Theorems 4.1 and 4.2 and in Theorems 3.1 and 3.2 of [11] can be replaced by $0 < \lambda_1$.

Proof: Let $f(a) := z^{\Omega^*(a)}$. Then f is a completely multiplicative function of $a \in \mathcal{G}$ and satisfies

$$f(p) = \begin{cases} z, & \text{if } p \in \mathcal{P}^*;\\ 1, & \text{if } p \notin \mathcal{P}^*. \end{cases}$$

Consider

$$F(z) = F(z;m) := \sum_{\partial(a)=m} f(a) = \sum_{0\le k\le m} N(m,k) z^k,$$

where

$$N(m,k) = \sum_{\partial(a)=m, \Omega^*(a)=k} 1.$$

Then

$$N(m,k) = \frac{1}{2\pi i}\int_{|z|=r} F(z) z^{-k-1}\, dz = \frac{1}{2\pi r^k}\int_{-\pi}^{\pi} F(re^{i\theta}) e^{-ik\theta}\, d\theta. \tag{5.2}$$

The proof of (5.1) is an application of the quantitative mean-value theorems to $F(z;m)$ in (5.2).

For $E(m) \geq (q-2+\delta)^3$, define

$$\theta_0 = \begin{cases} \sqrt{2}(|r-1|E(m))^{-\frac{1}{2}}, & \text{if } |r-1| \geq 2(E(m))^{-\frac{1}{3}}; \\ (E(m))^{-\frac{1}{3}}, & \text{if } |r-1| < 2(E(m))^{-\frac{1}{3}}. \end{cases}$$

Then $0 < \theta_0 \leq (E(m))^{-\frac{1}{3}}$. Also, we set $r = k/E(m)$.

We split the integration interval in (5.2) into two parts: $|\theta| \leq \theta_0$ and $\theta_0 < |\theta| \leq \pi$ and evaluate the contribution toward $N(m,k)$ of the integral on each part separately.

I. *On the range* $|\theta| \leq \theta_0$.

By Theorem 3.2 of [11],

$$\begin{aligned} F(re^{i\theta};m) = {} & A^* G(m) + q^m O\left(|re^{i\theta}-1| \exp\left\{\sum_{\partial(p)\leq m,\, p\in\mathcal{P}^*} (r\cos\theta - 1)q^{-\partial(p)}\right\}\right) \\ & + q^m \left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}} O\left(\exp\left\{-c_{11}\frac{\delta-\frac{1}{2}}{2}|re^{i\theta}-1|^{-\frac{1}{3}}\right\}\right. \\ & \times \exp\left\{\sum_{\partial(p)\leq m,\, p\in\mathcal{P}^*} (r-1)\,q^{-\partial(p)}\right. \\ & \left.\left. - c_{12}\left(\delta-\frac{1}{2}\right)^2 \sum_{\partial(p)\leq m,\, p\in\mathcal{P}^*} r(1-\cos\theta)\,q^{-\partial(p)}\right\}\right) \end{aligned} \tag{5.3}$$

where

$$A^* = \exp\left\{\sum_{\partial(p)\leq m,\, p\in\mathcal{P}^*} (re^{i\theta}-1)q^{-\partial(p)}\right\} = \exp\{(re^{i\theta}-1)E(m)\}.$$

The main term on the right-hand side of (5.3) contributes

$$\begin{aligned} & \frac{1}{2\pi r^k} G(m) \int_{|\theta|\leq\theta_0} A^* e^{-ik\theta}\, d\theta \\ & = G(m)\left(\frac{1}{2\pi i}\int_{|z|=r} e^{(z-1)E(m)} z^{-k-1}\, dz - \frac{1}{2\pi r^k}\int_{\theta_0<|\theta|\leq\pi} A^* e^{-ik\theta}\, d\theta\right) \\ & = G(m)\left(\frac{E^k(m)}{k!}e^{-E(m)} - \frac{1}{2\pi r^k}\int_{\theta_0<|\theta|\leq\pi} A^* e^{-ik\theta}\, d\theta\right). \end{aligned} \tag{5.4}$$

Note that

$$|A^*| = \exp\{k - E(m) - k(1-\cos\theta)\} \leq \exp\left\{k - E(m) - \frac{2k\theta^2}{\pi^2}\right\}.$$

The modulus of the last integral in (5.4) is

$$\le 2\exp\{k - E(m)\} \int_{\theta_0}^{\pi} \exp\left\{-\frac{2k\theta^2}{\pi^2}\right\} d\theta \le \sqrt{\frac{\pi^3}{2k}} \exp\{k - E(m)\} \exp\left\{-\frac{2k\theta_0^2}{\pi^2}\right\}.$$

If $|r-1| \ge 2(E(m))^{-\frac{1}{3}}$ then $\theta_0 = \sqrt{2}|k - E(m)|^{-\frac{1}{2}}$ and

$$\exp\left\{-\frac{2k\theta_0^2}{\pi^2}\right\} \le \frac{\pi^2}{4\delta} \frac{|k - E(m)|}{E(m)}.$$

If $|r-1| < 2(E(m))^{-\frac{1}{3}}$ then $\theta_0 = (E(m))^{-\frac{1}{3}}$ and

$$\exp\left\{-\frac{2k\theta_0^2}{\pi^2}\right\} \le \frac{\pi^4}{2\delta^2}(E(m))^{-\frac{2}{3}}.$$

Hence

$$\begin{aligned}
&\left|\frac{1}{2\pi r^k} \int_{\theta_0 < |\theta| \le \pi} A^* e^{-ik\theta} d\theta\right| \\
&\quad \le \frac{1}{2\pi} \frac{E^k(m)}{k^k} \sqrt{\frac{\pi^3}{2k}} \exp\{k - E(m)\} \left(\frac{\pi^2}{4\delta} \frac{|k - E(m)|}{E(m)} + \frac{\pi^4}{2\delta^2}(E(m))^{-\frac{2}{3}}\right) \\
&\quad = \frac{E^k(m)}{k!} e^{-E(m)} O_\delta\left(\frac{|k - E(m)|}{E(m)} + (E(m))^{-\frac{2}{3}}\right).
\end{aligned}$$

Thus the main term on the right-hand side of (5.3) contributes

$$G(m)\frac{E^k(m)}{k!} e^{-E(m)} \left(1 + O\left(\frac{|k - E(m)|}{E(m)} + (E(m))^{-\frac{2}{3}}\right)\right) \tag{5.5}$$

Then, apart from a factor $q^m r^{-k} \exp\{k - E(m)\}$, the first remainder on the right-hand side of (5.3) contributes

$$\begin{aligned}
O\left(\int_{|\theta| \le \theta_0} |re^{i\theta} - 1| \exp\left\{-\frac{2k\theta^2}{\pi^2}\right\} d\theta\right) &\ll \int_{|\theta| \le \theta_0} (|r - 1| + |\theta|) \exp\left\{-\frac{2k\theta^2}{\pi^2}\right\} d\theta \\
&\le \sqrt{\frac{\pi^3}{2k}} \frac{|k - E(m)|}{E(m)} + \frac{\pi^2}{2\sqrt{k}} \frac{1}{(\delta E(m))^{\frac{1}{2}}}.
\end{aligned}$$

The second remainder contributes

$$\ll \left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}} \int_{|\theta| \le \theta_0} \exp\left\{-c_{11} \frac{\delta - \frac{1}{2}}{2|re^{i\theta} - 1|^{\frac{1}{3}}} - c_{12}\left(\delta - \frac{1}{2}\right)^2 \frac{2k\theta^2}{\pi^2}\right\} d\theta.$$

If $|r-1| \geq 2(E(m))^{-\frac{1}{3}}$, then $|r-1| > 2\theta_0$ and

$$|re^{i\theta}-1| \leq |r-1|+|\theta| < \frac{3}{2}|r-1|.$$

The contribution is

$$\ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} \exp\left\{-c_{13}\frac{\delta-\frac{1}{2}}{2}\frac{1}{|r-1|^{\frac{1}{3}}}\right\}\frac{1}{\left(\delta-\frac{1}{2}\right)\sqrt{k}} \ll \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^4}\frac{1}{\sqrt{k}}\frac{|k-E(m)|}{E(m)}.$$

If $|r-1| < 2(E(m))^{-\frac{1}{3}}$, then $|r-1| < 2\theta_0$ and

$$|re^{i\theta}-1| \leq |r-1|+|\theta| < 3\theta_0.$$

The contribution is

$$\begin{aligned}
&\ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} \exp\left\{-c_{14}\frac{\delta-\frac{1}{2}}{2}\frac{1}{(3\theta_0)^{\frac{1}{3}}}\right\}\frac{1}{\left(\delta-\frac{1}{2}\right)\sqrt{k}}\\
&\ll \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^5}\frac{1}{\sqrt{k}}\theta_0^2 \leq \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^5}\frac{1}{\sqrt{k}}\frac{1}{\sqrt{E(m)}}.
\end{aligned}$$

Hence the two remainders on the right-hand side of (5.3) contribute

$$q^m\frac{E^k(m)}{k!}e^{-E(m)}O_\delta\left(\frac{|k-E(m)|}{E(m)}+\frac{1}{\sqrt{E(m)}}\right). \tag{5.6}$$

II. *On the range* $\theta_0 < |\theta| \leq \pi$.

For $\theta_0 < |\theta| \leq \frac{5\pi}{6}$, Theorem 4.1 with $\mu = \frac{\pi}{6}$ gives

$$F(re^{i\theta};m) \ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} q^m \exp\{k-E(m)\}\exp\left\{-c_{15}\left(\delta-\frac{1}{2}\right)^2 k(1-\cos\theta)\right\}.$$

Hence the contribution from this range, apart from a factor $q^m r^{-k}\exp\{k-E(m)\}$ again, is

$$\begin{aligned}
&\ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} \int_{\theta_0}^{\frac{5\pi}{6}} \exp\left\{-2c_{15}\left(\delta-\frac{1}{2}\right)^2 k\frac{\theta^2}{\pi^2}\right\}d\theta\\
&\ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} \exp\left\{-2c_{15}\left(\delta-\frac{1}{2}\right)^2 k\frac{\theta_0^2}{\pi^2}\right\}\sqrt{\frac{\pi}{2k}}\frac{1}{\delta-\frac{1}{2}}\\
&\ll \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^3}\frac{1}{\sqrt{k}}\left(\frac{|k-E(m)|}{E(m)}+(E(m))^{-\frac{2}{3}}\right).
\end{aligned}$$

Finally, for $\frac{5\pi}{6} < |\theta| \leq \pi$, Theorem 4.2 gives

$$\begin{aligned}
&F(re^{i\theta}; m)\\
&\quad \ll \left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}} q^m \exp\{k - E(m)\}\\
&\qquad \times \exp\left\{-c_{15}\left(\delta - \frac{1}{2}\right)^2 \sum_{\substack{\partial(p)\leq m\\ p\in\mathcal{P}^*}} \min\bigl\{r - r\cos\theta, r - (-1)^{\partial(p)} r\cos\theta\bigr\} q^{-\partial(p)}\right\}.
\end{aligned}$$

We have

$$\min\bigl\{r - r\cos\theta, r - (-1)^{\partial(p)} r\cos\theta\bigr\} = \begin{cases} r(1+\cos\theta) - 2r\cos\theta, & \text{if } \partial(p) \text{ is even;}\\ r(1+\cos\theta), & \text{if } \partial(p) \text{ is odd.}\end{cases}$$

Hence

$$\begin{aligned}
&\sum_{\partial(p)\leq m,\, p\in\mathcal{P}^*} \min\bigl\{r - r\cos\theta, r - (-1)^{\partial(p)} r\cos\theta\bigr\} q^{-\partial(p)}\\
&\quad = (1+\cos\theta)k - 2r\cos\theta \sum_{\substack{p\in\mathcal{P}^*\\ \partial(p)\leq m,\partial(p)\, even}} q^{-\partial(p)}\\
&\quad \geq (1+\cos\theta)k + 2r\cos\frac{\pi}{6} \sum_{\substack{p\in\mathcal{P}^*\\ \partial(p)\leq m,\partial(p)\, even}} q^{-\partial(p)}.
\end{aligned}$$

The contribution from this range is, apart from the factor again,

$$\begin{aligned}
&\ll \left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}} \exp\left\{-\sqrt{3}c_{15}\left(\delta - \frac{1}{2}\right)^2 r \sum_{\substack{p\in\mathcal{P}^*\\ \partial(p)\leq m,\partial(p)\, even}} q^{-\partial(p)}\right\}\\
&\quad \times \int_{\frac{5}{6}\pi}^{\pi} \exp\left\{-c_{15}\left(\delta - \frac{1}{2}\right)^2 k(1+\cos\theta)\right\} d\theta.
\end{aligned}$$

The last integral equals

$$\begin{aligned}
\int_0^{\frac{\pi}{6}} \exp\left\{-c_{15}\left(\delta - \frac{1}{2}\right)^2 k(1-\cos\theta)\right\} d\theta &\leq \int_0^{\frac{\pi}{6}} \exp\left\{-c_{15}\left(\delta - \frac{1}{2}\right)^2 k\frac{\theta^2}{\pi^2}\right\} d\theta\\
&\ll \frac{1}{\left(\delta - \frac{1}{2}\right)\sqrt{k}}.
\end{aligned}$$

Therefore the contribution is

$$q^m \frac{E^k(m)}{k!} \frac{e^{-E(m)}}{\delta - \frac{1}{2}} O\left(\exp\left\{ -c_{16}\left(\delta - \frac{1}{2}\right)^2 \sum_{\substack{p\in\mathcal{P}^* \\ \partial(p)\le m, \partial(p)\, even}} q^{-\partial(p)} \right\} \right). \tag{5.7}$$

Now (5.1) follows from (5.5), (5.6), and (5.7). □

In particular, if $\mathcal{P}^* = \mathcal{P}$ and the zeta function $Z(y)$ of the additive arithmetic semigroup $\mathcal{G}$ has no zero at $y = -q^{-1}$, then

$$E(m) = \sum_{\partial(p)\le m} q^{-\partial(p)} = \log m + d_1 + O\left(\frac{1}{m}\right)$$

and

$$\sum_{\partial(p)\le m, \partial(p)\, even} q^{-\partial(p)} = \sum_{2n\le m} P(2n) q^{-2n} = \frac{1}{2}\log m + d_2 + O\left(\frac{1}{m^{\gamma-2}}\right),$$

where d_1 and d_2 are constants. In this case,

$$\frac{1}{G(m)} N(k; m) = \frac{e^{-d_1}\log^k m}{k!\, m}\left\{1 + O\left(\frac{|k - \log m|}{\log m} + \frac{1}{\sqrt{\log m}}\right)\right\}$$

for $\delta \log m < k < (2 - \delta)\log m$.

6. An example

The last O-term on the right-hand side of (5.1) might be unexpected from the point of view of classical probabilistic number theory. The following example shows, besides its independent interest, that the existence of the last O-term is actually quite natural and is not related to whether the zeta function $Z(y)$ of an additive arithmetic semigroup has a zero at $y = -q^{-1}$. Thus the theory of additive arithmetic semigroups shows essential divergence from classical number theory as we observe in Theorem 5.1.

Example 6.1. Let $\mathcal{K}$ be an algebraic function field of transcendence degree one, i.e., in one variable, over a finite constant field $\mathbb{F}_q$. Consider the additive arithmetic semigroup $\mathcal{G}$ of integral divisors in $\mathcal{K}$. Let $\mathcal{P}^*$ denote the subset of $\mathcal{P}$ consisting of primes p with $\partial(p)$ odd and $\Omega^*(a)$ denote the number of prime divisors $p \in \mathcal{P}^*$ of $a \in \mathcal{G}$, counted according to multiplicity of p. Since $\mathcal{P}$ and $\mathcal{P}^*$ are quite regular in some sense, it is possible to replace the last O-term on the right-hand side of (5.1) by an asymptotic estimate.

Proposition 6.1. *Let $E(m)$ be defined as in Theorem 5.1 and δ be a real number satisfying $\frac{1}{2} < \delta < 1$. Then we have*

$$\frac{1}{G(m)} \sum_{\substack{\partial(a)=m \\ \Omega^*(a)=k}} 1 = \frac{E^k(m)}{k!} e^{-E(m)} \left\{ 2 + O\left(\frac{|k - E(m)|}{E(m)} + \frac{1}{\sqrt{E(m)}} \right) \right\} \tag{6.1}$$

for $E(m) > (q - 2 + \delta)^3$ and $\delta \le k/E(m) \le 2 - \delta$ when $m + k$ is even. Otherwise, the sum on the left-hand side is zero when $m + k$ is odd.

The last part of Proposition 6.1 is obvious because no $a \in \mathcal{G}$ of even (resp., odd) degree has an odd (resp., even) number of prime factors of odd degree.

To prove (6.1), as in the proof of Theorem 5.1, we consider

$$F(z) = F(z; m) := \sum_{\partial(a)=m} f(a)$$

with f defined there.

Lemma 6.1. *Assume $|z + 1| \le \eta < 1$ and $\delta \le |z| \le 2 - \delta$. Then*

$$\begin{aligned} &|F(z; m) - (-1)^m A G(m)| \\ &\quad \le c_{17} q^m \left(\eta \exp\left\{ - \sum_{\substack{\partial(p) \le m \\ \partial(p)\, odd}} (\Re z + 1) q^{-\partial(p)} \right\} + \left(\Gamma\left(\delta - \frac{1}{2} \right) \right)^{\frac{1}{2}} \exp\left\{ -c_{18} \frac{\delta - \frac{1}{2}}{2} \eta^{-\frac{1}{3}} \right\} \right. \\ &\qquad \left. \times \exp\left\{ \sum_{\substack{\partial(p) \le m \\ \partial(p)\, odd}} (|z| - 1) q^{-\partial(p)} - \frac{\delta_2 \delta_1}{2} \sum_{\substack{\partial(p) \le m \\ \partial(p)\, odd}} (|z| + \Re z) q^{-\partial(p)} \right\} \right), \end{aligned} \tag{6.2}$$

where

$$A = A(z; m) := \exp\left\{ - \sum_{\substack{\partial(p) \le m \\ \partial(p)\, odd}} (z + 1) q^{-\partial(p)} \right\}$$

and

$$\delta_1 = \frac{\left(\delta - \frac{1}{2} \right)^2}{4\delta(2 - \delta)} > 0.$$

The conatants δ_2 and c_{18} are independent of δ and η while the constant c_{17} depends at most on δ and is uniform for $0 < \eta \le \eta_0$ with any fixed $\eta_0 < 1$.

Proof: We note that, by the Riemann-Roch theorem [6, 7],

$$G(m) = \frac{h}{q-1}(q^{m-g+1} - 1), \quad \text{for} \quad m \geq 2g - 1$$

where h is the class number and g the genus of $\mathcal{K}$. Also, the zeta function is

$$Z(y) = \frac{L(y)}{(1-qy)(1-y)},$$

where $L(y)$ is a polynomial of degree $2g$ and with algebraic integer coefficients. Let

$$\hat{T}(y) = \hat{T}(y; z, m) := \prod_{\substack{\partial(p) \leq m \\ \partial(p)\, odd}} \left(1 - zy^{\partial(p)}\right)^{-1} \prod_{\substack{\partial(p) \leq m \\ \partial(p)\, even}} \left(1 - y^{\partial(p)}\right)^{-1} \prod_{\partial(p) > m} \left(1 - (-y)^{\partial(p)}\right)^{-1}.$$

Then

$$mF(z; m) = \frac{q^{m-1}}{2\pi i} \int_{|s|=r} \frac{\hat{T}'(q^{-1}s)}{s^m}\, ds,$$

where $0 < r < 1$. Hence

$$m(F(z; m) - (-1)^m AG(m)) = \frac{q^{m-1}}{2\pi i} \int_{|s|=r} \frac{\hat{T}'(q^{-1}s) + AZ'(-q^{-1}s)}{s^m}\, ds.$$

The integral on the right-hand side equals

$$(-1)^m \int_{|s|=r} \frac{\hat{T}'(-q^{-1}s) + AZ'(q^{-1}s)}{s^m} ds.$$

Note that

$$\hat{T}(y) = Z(-y) \prod_{\substack{\partial(p) \leq m \\ \partial(p)\, odd}} \left(1 - zy^{\partial(p)}\right)^{-1}\left(1 + y^{\partial(p)}\right).$$

We can write

$$-\hat{T}(-q^{-1}s) + AZ(q^{-1}s) = -AZ(q^{-1}s)\,(\exp\{S(s) + R(s)\} - 1),$$

where

$$S(s) = -\sum_{\substack{\partial(p) \leq m \\ \partial(p)\, odd}} (z+1) q^{-\partial(p)}\left(s^{\partial(p)} - 1\right)$$

and

$$R(s) = \sum_{\substack{\partial(p) \leq m \\ \partial(p)\, odd}} \left[-\log\left(1 + zq^{-\partial(p)}s^{\partial(p)}\right) + \log\left(1 - q^{-\partial(p)}s^{\partial(p)}\right) + (z+1)q^{-\partial(p)}s^{\partial(p)}\right].$$

Thus

$$\hat{T}'(-q^{-1}s) + AZ'(q^{-1}s) = -q\left\{Ae^{R(r)}\frac{d}{ds}\left[Z(q^{-1}s)e^{S(s)}\left(e^{R(s)-R(r)} - 1\right)\right]\right.$$
$$\left. + Ae^{R(r)}\frac{d}{ds}\left[Z(q^{-1}s)(e^{S(s)} - 1)\right]\right\} - A\left(e^{R(r)} - 1\right)Z'(q^{-1}s).$$

The same argument given in the proof of Theorem 3.2 in [11], with $-z$ in place of $f(p)$ when $\partial(p)$ is odd and 1 in place of $f(p)$ when $\partial(p)$ is even, yields the required estimate on the right-hand side of (6.2). □

We now proof Proposition 6.1. As in the proof of Theorem 5.1,

$$N(m,k) = \frac{1}{2\pi r^k}\int_{-\pi}^{\pi} F(re^{i\theta})e^{-ik\theta}d\theta,$$

where $r = \frac{r}{E(m)}$, and the contribution of the integral on the part $|\theta| \le \frac{5}{6}\pi$ is

$$G(m)\frac{E^k(m)}{k!}e^{-E(m)}\left(1 + \frac{\left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta - \frac{1}{2}\right)^5}O\left(\frac{|k - E(m)|}{E(m)} + \frac{1}{\sqrt{E(m)}}\right)\right). \tag{6.3}$$

Then, to evaluate the contribution of the integral on the part $|\theta - \pi| \le \frac{\pi}{6}$, we also split the intervals into two parts: $|\theta - \pi| \le \theta_0$ and $\theta_0 \le |\theta - \pi| \le \frac{\pi}{6}$ and apply (6.2), where θ_0 is defined as in the proof of Theorem 5.1.

I. *On the range* $|\theta - \pi| \le \theta_0$. We have

$$\begin{aligned} F(re^{i\theta};m) &= (-1)^m A(re^{i\theta})G(m) \\ &\quad + q^m O\left(|re^{i\theta} + 1|\exp\left\{-\sum_{\substack{\partial(p)\le m\\ \partial(p)\, odd}}(r\cos\theta + 1)q^{-\partial(p)}\right\}\right) \\ &\quad + q^m\left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}} O\left(\exp\left\{-c_{18}\frac{\delta - \frac{1}{2}}{2}|re^{i\theta} + 1|^{-\frac{1}{3}}\right\}\right. \\ &\quad \times \exp\left\{\sum_{\substack{\partial(p)\le m\\ \partial(p)\, odd}}(r - 1)q^{-\partial(p)} - c_{19}\left(\delta - \frac{1}{2}\right)^2\right. \\ &\quad \left.\left.\times \sum_{\substack{\partial(p)\le m\\ \partial(p)\, odd}} r(\cos\theta + 1)q^{-\partial(p)}\right\}\right), \end{aligned} \tag{6.4}$$

where

$$A(re^{i\theta}) = A(re^{i\theta}; m) = \exp\{-(re^{i\theta} + 1)E(m)\}.$$

The main term on the right-hand side of (6.4) contributes

$$\begin{aligned}
&\frac{(-1)^m G(m)}{2\pi r^k} \int_{|\theta-\pi|\le\theta_0} A(re^{i\theta}) e^{-ik\theta} d\theta \\
&= (-1)^m G(m) \left(\frac{1}{2\pi i} \int_{|z|=r} e^{-(z+1)E(m)} z^{-k-1} dz - \frac{1}{2\pi r^k} \int_{\theta_0 \le |\theta-\pi| \le \pi} A(re^{i\theta}) e^{-ik\theta} d\theta \right) \\
&= (-1)^{m+k} G(m) \left(\frac{E^k(m)}{k!} e^{-E(m)} - \frac{1}{2\pi r^k} \int_{\theta_0 \le |\theta| \le \pi} A(-re^{i\theta}) e^{-ik\theta} d\theta \right)
\end{aligned}$$

by changing the integration variable. Note that

$$|A(-re^{i\theta})| = \exp\{k \cos\theta - E(m)\}.$$

Hence, as in the proof of Theorem 5.1,

$$\frac{1}{2\pi r^k} \int_{\theta_0 \le |\theta| \le \pi} A(-re^{i\theta}) e^{-ik\theta} d\theta = \frac{E^k(m)}{k!} e^{-E(m)} O\left(\frac{|k - E(m)|}{E(m)} + (E(m))^{-\frac{2}{3}} \right).$$

Similarly, apart from a factor q^m, the integral on the interval $|\theta - \pi| \le \theta_0$ of the first remainder on the right-hand side of (6.4) is

$$\begin{aligned}
&\ll \int_{|\theta-\pi|\le\theta_0} |re^{i\theta} + 1| \exp\left\{ - \sum_{\substack{\partial(p)\le m \\ \partial(p)\, odd}} (r\cos\theta + 1) q^{-\partial(p)} \right\} d\theta \\
&= \int_{|\theta|\le\theta_0} |-re^{i\theta} + 1| \exp\left\{ - \sum_{\substack{\partial(p)\le m \\ \partial(p)\, odd}} (1 - r\cos\theta) q^{-\partial(p)} \right\} d\theta \\
&= \frac{E^k(m)}{k!} e^{-E(m)} O\left(\frac{|k - E(m)|}{E(m)} + \frac{1}{\sqrt{E(m)}} \right),
\end{aligned}$$

and the integral on the same interval of the second remainder is

$$\begin{aligned}
&\ll \left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}} \exp\{k - E(m)\} \\
&\quad \times \int_{|\theta|\le\theta_0} \exp\left\{ -c_{18} \frac{\delta - \frac{1}{2}}{2|1 - re^{i\theta}|^{\frac{1}{3}}} - c_{19} \left(\delta - \frac{1}{2}\right)^2 k(1 - \cos\theta) \right\} d\theta \\
&\ll \frac{\left(\Gamma\left(\delta - \frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta - \frac{1}{2}\right)^5} \frac{E^k(m)}{k!} e^{-E(m)} \left(\frac{|k - E(m)|}{E(m)} + \frac{1}{\sqrt{E(m)}} \right).
\end{aligned}$$

Therefore, the contribution from this range is

$$G(m)\frac{E^k(m)}{k!}e^{-E(m)}\left((-1)^{m+k}+\frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^5}\left(\frac{|k-E(m)|}{E(m)}+\frac{1}{\sqrt{E(m)}}\right)\right). \tag{6.5}$$

II. *On the range* $\theta_0 \le |\theta-\pi| \le \frac{\pi}{6}$. Theorem 4.2 gives

$$F(re^{i\theta};m) \ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} q^m \exp\{k-E(m)\}$$
$$\times\exp\left\{-c_{20}\left(\delta-\frac{1}{2}\right)^2 \sum_{\substack{\partial(p)\le m\\ \partial(p)\, odd}} r(\cos\theta+1)q^{-\partial(p)}\right\}.$$

Hence

$$\int_{\theta_0\le|\theta-\pi|\le\frac{\pi}{6}} F(re^{i\theta};m)e^{-ik\theta}d\theta$$
$$\ll \left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}} q^m\exp\{k-E(m)\}\int_{\theta_0}^{\frac{\pi}{6}}\exp\left\{-c_{20}\left(\delta-\frac{1}{2}\right)^2 k(1-\cos\theta)\right\}d\theta$$
$$\ll \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^3} q^m\exp\{k-E(m)\}\left(\frac{|k-E(m)|}{E(m)}+(E(m))^{-\frac{2}{3}}\right)$$

as shown in the proof of Theorem 5.1. Therefore, the contribution from this range is

$$\ll \frac{\left(\Gamma\left(\delta-\frac{1}{2}\right)\right)^{\frac{1}{2}}}{\left(\delta-\frac{1}{2}\right)^3}\frac{q^m E^k(m)}{k!}e^{-E(m)}\left(\frac{|k-E(m)|}{E(m)}+(E(m))^{-\frac{2}{3}}\right). \tag{6.6}$$

Thus (6.1) follows from (6.3), (6.5), and (6.6). This completes the proof of Proposition 6.1.

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An Expansion Formula for q-Series and Applications

ZHI-GUO LIU xxlzg@public.xxptt.ha.cn; liuzg18@hotmail.com
Xinxiang Education College, Xinxiang, Henan 453000, People's Republic of China

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Abstract. In this paper the author proves a q-expansion formula which utilizes the Leibniz formula for the q-differential operator. This expansion leads to new proofs of the Rogers–Fine identity, the nonterminating ${}_6\phi_5$ summation formula, and Watson's q-analog of Whipple's theorem. Andrews' identities for sums of three squares and sums of three triangular numbers are also derived. Other identities of Andrews and new identities for Hecke type series are also discussed.

Key words: q-series, q-differential operator, Rogers–Fine identity, ${}_6\phi_5$ summation formula, Watson's q-analog of Whipple's theorem, Andrews' identities, Hecke type series

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1. Introduction

In [3], G.E. Andrews used the Bailey chain technique to study the fifth and seventh order mock theta functions. His study yielded many deep and beautiful identities for Hecke type series. In this paper we shall derive a q-expansion formula which utilizes the Leibniz formula for the q-differential operator. Using this new q-expansion formula, we will provide new proofs of the Rogers–Fine formula, the nonterminating ${}_6\phi_5$ summation formula, and Watson's q-analog of Whipple's theorem. We will also derive Andrews' important identities for sums of three squares and sums of three triangular numbers [3, Eqs. (5.16) and (5.17)]. We also derive other identities of Andrews and obtain new identities for Hecke type series.

Let $f(b)$ be a formal series in b. In Section 2 of this paper we will prove the above mentioned expansion formula:

$$f(b) = \sum_{n=0}^{\infty} \frac{(1-aq^{2n})(aq/b;q)_n b^n}{(q,b;q)_n} \left[D^n_{q,x}\{f(x)(x;q)_{n-1}\}\right]_{x=aq}, \tag{1.1}$$

where $(x;q)_n$ is defined by (2.1) below; and $D_{q,x}$ is defined by (3.1) below.

In Sections 3, 4 and 5, we will prove the Rogers–Fine identity, the nonterminating ${}_6\phi_5$ summation formula, and Watson's q-analog of Whipple's theorem, by using the above expansion.

In Section 6 we will prove the following formula:

$$\frac{(bt, aq, asq; q)_\infty}{(b, sb, atq; q)_\infty} = \sum_{n=0}^{\infty} \frac{(1-aq^{2n})(a, aq/b; q)_n b^n}{(1-a)(q, b, atq; q)_n} \times \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix} q^{(n-j)(n-1)} s^j (asq; q)_{n-j} (t/s; q)_j. \tag{1.2}$$

Using this formula we will derive Andrews' identities for sums of three squares and sums of three triangular numbers [3, Eqs. (5.16) and (5.17)]. We also derive the following interesting identity:

$$(q; q)_\infty^2 = \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{2n+1}) q^{2n^2+n-j(3j+1)/2}, \tag{1.3}$$

which is equivalent to an identity of Hecke [3, Eq. (5.1)].

In Section 7 we prove the following transformation formula:

$$\sum_{n=0}^{\infty} \frac{a^n q^{n^2}}{(cq; q)_n} = \frac{1}{(a; q)_\infty} \sum_{n=0}^{\infty} \frac{(-1)^n (1-aq^{2n})(a; q)_n a^{2n} q^{(5n^2-n)/2}}{(cq; q)_n} \times \sum_{j=0}^{n} \frac{(c; q)_j a^{-j} q^{j(1-n)}}{(q; q)_j}, \tag{1.4}$$

which we use to derive the following two beautiful identities of Andrews [3, Eqs. (6.1) and (6.5)] for the fifth order mock theta functions:

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(-q; q)_n} = \frac{1}{(q; q)_\infty} \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{4n+2}) q^{n(5n+1)/2-j^2}, \tag{1.5}$$

$$\sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(-q; q)_n} = \frac{1}{(q; q)_\infty} \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{2n+1}) q^{n(5n+3)/2-j^2}. \tag{1.6}$$

We also prove the following identity

$$(q; q)_\infty (q^2; q^2)_\infty = \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{2n+1}) q^{n(2n+1)-j^2}, \tag{1.7}$$

which is equivalent to an identity of Kac and Peterson [2, Eq. (1.3)], [10].

2. Some basic facts about q-series

We shall follow the notation and terminology in [9]. The q-shifted factorials are defined by

$$(a;q)_0 = 1, \quad (a;q)_n = \prod_{k=0}^{n-1}(1-aq^k), \quad (a;q)_\infty = \prod_{k=0}^{\infty}(1-aq^k). \tag{2.1}$$

For any complex α, we define

$$(a;q)_\alpha = (a;q)_\infty/(aq^\alpha;q)_\infty. \tag{2.2}$$

Hence

$$(a;q)_{-1} = (1-a/q)^{-1}. \tag{2.3}$$

We shall also adopt the following notation for multiple q-shifted factorials:

$$(a_1, a_2, \ldots, a_m; q)_n = \prod_{k=1}^{m}(a_k;q)_n. \quad n = 0, 1, \ldots \text{or} \quad \infty. \tag{2.4}$$

The q-binomial coefficient is defined by

$$\begin{bmatrix} n \\ k \end{bmatrix} = \frac{(q;q)_n}{(q;q)_k(q;q)_{n-k}}. \tag{2.5}$$

It is easy to show that

$$\begin{bmatrix} n \\ k \end{bmatrix} = (-1)^k q^{nk-\binom{k}{2}} \frac{(q^{-n};q)_k}{(q;q)_k}. \tag{2.6}$$

The basic hypergeometric series ${}_{r+1}\phi_r$ is defined by

$${}_{r+1}\phi_r\left(\begin{matrix} a_1, a_2, \ldots, a_{r+1} \\ b_1, b_2, \ldots, b_r \end{matrix}; q, x\right) = \sum_{n=0}^{\infty} \frac{(a_1, a_2, \ldots, a_{r+1}; q)_n x^n}{(q, b_1, b_2, \ldots, b_r; q)_n}. \tag{2.7}$$

In this paper we shall use the q-Pfaff-Saalschütz formula [5, Eq. (3.25), p. 25], [9, Eq. (1.7.2), p. 13]

$${}_3\phi_2\left(\begin{matrix} q^{-n}, a, b \\ c, abc^{-1}q^{1-n} \end{matrix}; q, q\right) = \frac{(c/a, c/b; q)_n}{(c, c/ab; q)_n}, \tag{2.8}$$

and the q-Gauss summation formula [4, Theorem 2.6], [9, Eq. (1.51), p. 10]

$${}_2\phi_1\left(\begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab}\right) = \frac{(c/a, c/b; q)_\infty}{(c, c/ab; q)_\infty}. \tag{2.9}$$

We will also need the terminating ${}_6\phi_5$ summation formula [9, Eq. (2.4.2), p. 34]:

$$ {}_6\phi_5\left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, aq/b, aq/c, q^{-n} \\ \sqrt{a}, -\sqrt{a}, b, c, aq^{1+n} \end{matrix}; q, \frac{bcq^{n-1}}{a}\right) = \frac{(aq, bc/aq; q)_n}{(b, c; q)_n}. \tag{2.10} $$

3. Expansion formula

The q-differential operator about x is defined by

$$ D_{q,x}\{f(x)\} = \frac{f(x) - f(xq)}{x}, \tag{3.1} $$

By convention, $D^0_{q,x}$ is understood as the identity.

The Leibniz rule for $D_{q,x}$ is the following identity [12, p. 233]

$$ D^n_{q,x}\{g(x)h(x)\} = \sum_{k=0}^{n} q^{k(k-n)} \begin{bmatrix} n \\ k \end{bmatrix} D^k_{q,x}\{g(x)\} D^{n-k}_{q,x}\{h(q^k x)\}. \tag{3.2} $$

From the definition of $D_{q,x}$, we immediately have the following

Theorem 1. *If $f(x)$ is a polynomial of degree less than n in x, then*

$$ D^n_{q,x}\{f(x)\} = 0. \tag{3.3} $$

The Eulerian family of polynomials $\{P_n(b, a)\}$ is defined as [1, p. 350], [4, pp. 470–471]

$$ P_0(b, a) = 1, \quad P_n(b, a) = (b - aq)(b - aq^2)\dots(b - aq^n). \tag{3.4} $$

Note that

$$ P_n(b, a) = b^n (aq/b; q)_n \tag{3.5} $$

and

$$ P_n(aq, a) = 0, n \geq 1. \tag{3.6} $$

It is easy to show that

$$ D^k_{q,b}\{P_n(b, a)\} = \begin{cases} \dfrac{(q; q)_n}{(q; q)_{n-k}} P_{n-k}(b, a) & 0 \leq k \leq n - 1, \\ (q; q)_n, & k = n, \\ 0 & k \geq n + 1. \end{cases} \tag{3.7} $$

Hence

$$ \left[D^k_{q,b}\{P_n(b, a)\}\right]_{b=aq} = \begin{cases} 0 & k \neq n, \\ (q; q)_n & k = n. \end{cases} \tag{3.8} $$

The following lemma follows easily from (3.6) and the Leibniz rule.

Lemma 1. *If $h(b)$ is a formal series in b, then we have*

$$\left[D_{q,b}^n\{P_k(b,a)h(b)\}\right]_{b=aq} = \begin{cases} 0 & k > n, \\ (q;q)_n h(aq^{n+1}) & k = n. \end{cases} \tag{3.9}$$

We also need the following

Lemma 2. *If $f(b)$ is a formal series in b, then we have the expansion formula*

$$f(b) = \sum_{n=0}^{\infty} A_n \frac{P_n(b,a)}{(b;q)_n}, \tag{3.10}$$

where A_n is independent of the variable b.

Proof: Carlitz' q-analog of a special case of the Lagrange inversion formula [7, p. 206, Eq. (1.11)], [12, p. 253, Eq. (8.4)] is

$$f(b) = \sum_{n=0}^{\infty} \frac{b^n}{(q,b;q)_n} \left[D_{q,x}^n\{f(x)(x;q)_{n-1}\}\right]_{x=0}. \tag{3.11}$$

Setting $c = \infty$ in (2.10) we obtain

$$\frac{(aq)^n}{(aq;q)_n} \sum_{k=0}^{n} \frac{(-1)^k(1-aq^{2k})(a,q^{-n};q)_k a^{-k} q^{nk-k(k+1)/2}}{(1-a)(q,aq^{n+1};q)_k} \frac{P_k(b,a)}{(b;q)_k} = \frac{b^n}{(b;q)_n}. \tag{3.12}$$

Substituting the above equation into (3.11) we find that $f(b)$ can be expanded in terms of $P_n(b,a)/(b;q)_n$. This completes the proof. ☐

The main result of this paper is the following theorem.

Theorem 2. *Suppose $f(b)$ is a formal series in b, then we have*

$$f(b) = \sum_{n=0}^{\infty} \frac{(1-aq^{2n})(aq/b;q)_n b^n}{(q,b;q)_n} \left[D_{q,x}^n\{f(x)(x;q)_{n-1}\}\right]_{x=aq}. \tag{3.13}$$

Remark. Equation (3.11) is the special case $a = 0$.

Proof: From Lemma 2 we know that

$$f(b) = \sum_{n=0}^{\infty} A_n \frac{P_n(b,a)}{(b;q)_n}, \tag{3.14}$$

where A_n is independent of b.

Assuming $n \geq 1$, and by multiplying both sides by $(b;q)_{n-1}$ we obtain

$$f(b)(b;q)_{n-1} = \sum_{k=0}^{n-1} A_k P_k(b,a)(1-bq^k)\dots(1-bq^{n-2}) + A_n \frac{P_n(b,a)}{(1-bq^{n-1})} + \sum_{k=n+1}^{\infty} \frac{A_k P_k(b,a)}{(1-bq^{n-1})\dots(1-bq^{k-1})}. \tag{3.15}$$

Hence

$$\begin{aligned}&\big[D_{q,b}^n\{f(b)(b;q)_{n-1}\}\big]_{b=aq}\\ &= \sum_{k=0}^{n-1} A_k \big[D_{q,b}^n\{P_k(b,a)(1-bq^k)\dots(1-bq^{n-1})\}\big]_{b=aq}\\ &\quad + A_n\left[D_{q,b}^n\left\{\frac{P_n(b,a)}{(1-bq^{n-1})}\right\}\right]_{b=aq}\\ &\quad + \sum_{k=n+1}^{\infty} A_k\left[D_{q,b}^n\left\{\frac{P_k(b,a)}{(1-bq^{n-1})\dots(1-bq^{k-1})}\right\}\right]_{b=aq}.\end{aligned} \tag{3.16}$$

Since $P_k(b,a)(1-bq^k)\dots(1-bq^{n-2})$ is a polynomial in b of degree $n-1$, by Theorem 1, we have

$$\big[D_{q,b}^n\{P_k(b,a)(1-bq^k)\dots(1-bq^{n-2})\}\big]_{b=aq} = 0. \tag{3.17}$$

From Lemma 1 we know that

$$\left[D_{q,b}^n\left\{\frac{P_k(b,a)}{(1-bq^{n-1})\dots(1-bq^{k-1})}\right\}\right]_{b=aq} = 0 \tag{3.18}$$

and

$$\left[D_{q,b}^n\left\{\frac{P_n(b,a)}{(1-bq^{n-1})}\right\}\right]_{b=aq} = \frac{(q;q)_n}{1-aq^{2n}}. \tag{3.19}$$

So we have

$$\big[D_{q,b}^n\{f(b)(b;q)_{n-1}\}\big]_{b=aq} = \frac{(q;q)_n}{1-aq^{2n}} A_n. \tag{3.20}$$

Hence

$$\begin{aligned}A_n &= \frac{1-aq^{2n}}{(q;q)_n}\big[D_{q,b}^n\{f(b)(b;q)_{n-1}\}\big]_{b=aq}\\ &= \frac{1-aq^{2n}}{(q;q)_n}\big[D_{q,x}^n\{f(x)(x;q)_{n-1}\}\big]_{x=aq}.\end{aligned} \tag{3.21}$$

By using (2.3), it is easy to show that the above equation is also true for $n = 0$. This completes the proof. $\square$

Identity (3.13) is an interesting expansion formula for q-series. In the subsequent sections of this paper, we will prove several well-known identities using this expansion. We will also use it to derive many identities for Hecke type series. We believe that this expansion formula will have many applications in the theory of q-series and lead to new and important results.

4. The Rogers–Fine identity

It is easy to verify the following property of $D_{q,x}$:

$$D_{q,x}^{n}\left\{\frac{(tx;q)_\infty}{(sx;q)_\infty}\right\} = s^n(t/s;q)_n\frac{(txq^n;q)_\infty}{(sx;q)_\infty}. \tag{4.1}$$

Hence we have

$$\begin{aligned} D_{q,x}^{n}\left\{\frac{(x;q)_{n-1}}{(x;q)_{n+m}}\right\} &= D_{q,x}^{n}\left\{\frac{(xq^{n+m};q)_\infty}{(xq^{n-1};q)_\infty}\right\} \\ &= q^{n(n-1)}\frac{(x;q)_{n-1}(q;q)_{n+m}}{(q;q)_m(x;q)_{2n+m}}. \end{aligned} \tag{4.2}$$

On setting $t = 0$, we obtain

$$D_{q,x}^{n}\left\{\frac{1}{(sx;q)_\infty}\right\} = \frac{s^n}{(sx;q)_\infty}. \tag{4.3}$$

Applying Theorem 2 to the function

$$f(b) = \sum_{n=0}^{\infty}\frac{(a/z;q)_n z^n}{(b;q)_n}, \tag{4.4}$$

and by using (4.2) and the q-Gauss summation we have

$$\begin{aligned} \left[D_{q,x}^{n}\{(x;q)_{n-1}f(x)\}\right]_{x=aq} &= \sum_{k=n}^{\infty}(a/z;q)_k z^k\left[D_{q,x}^{n}\left\{\frac{(x;q)_{n-1}}{(x;q)_k}\right\}\right]_{x=aq} \\ &= \sum_{m=0}^{\infty}(a/z;q)_{n+m}z^{m+n}\left[D_{q,x}^{n}\left\{\frac{(xq^{n+m};q)_\infty}{(xq^{n-1};q)_\infty}\right\}\right]_{x=aq} \\ &= \frac{(q,a/z,a;q)_n q^{n(n-1)}z^n}{(a;q)_{2n+1}}\,{}_2\phi_1\left(\begin{matrix}aq^n/z,\ q^{n+1} \\ aq^{2n+1}\end{matrix};q,z\right) \\ &= \frac{(q,a/z,a;q)_n q^{n(n-1)}z^n}{(a;q)_{2n+1}}\frac{(zq^{n+1},aq^n;q)_\infty}{(aq^{2n+1},z;q)_\infty} \\ &= \frac{(a/z,q;q)_n q^{n(n-1)}z^n}{(z;q)_{n+1}}. \end{aligned} \tag{4.5}$$

Substituting this into (3.13) and replacing a by az we obtain the Rogers–Fine identity [8, p. 15, Eq. (14.1)]:

Theorem 3.

$$\sum_{n=0}^{\infty}\frac{(a;q)_n z^n}{(b;q)_n}=\sum_{n=0}^{\infty}\frac{(1-azq^{2n})(azq/b,a;q)_n q^{n(n-1)}(bz)^n}{(b;q)_n(z;q)_{n+1}}. \tag{4.6}$$

This identity has been used by N.J. Fine [8] to derive many important and diverse results in number theory, partitions, mock theta functions and modular equations.

5. The nonterminating ${}_6\phi_5$ summation formula

We apply Theorem 2 to the function

$$f(b)=\frac{(b/c,b/d;q)_\infty}{(b,b/cd;q)_\infty}, \tag{5.1}$$

and by using (4.1), the Leibniz formula, and the q-Gauss summation formula, we have

$$\begin{aligned} D_{q,x}^n\{f(x)(x;q)_{n-1}\} &= \sum_{j=0}^{n}\begin{bmatrix} n\\ j\end{bmatrix} q^{j(j-n)}D_{q,x}^j\left\{\frac{(x/c;q)_\infty}{(xq^{n-1};q)_\infty}\right\}D_{q,x}^{n-j}\left\{\frac{(xq^j/d;q)_\infty}{(xq^j/cd;)_\infty}\right\}\\ &= \sum_{j=0}^{n}\begin{bmatrix} n\\ j\end{bmatrix} q^{j(j-n)}q^{j(n-1)}(q^j/cd)^{n-j}(q^{1-n}/c;q)_j(c;q)_{n-j}\\ &\quad\times\frac{(xq^j/c,xq^n/d;q)_\infty}{(xq^{n-1},xq^j/cd;q)_\infty}\\ &= \frac{(x/c,xq^n/d;q)_\infty}{(xq^{n-1},x/cd;q)_\infty}(c;q)_n(cd)^{-n}{}_2\phi_1\left(\begin{matrix} q^{-n},x/cd\\ x/c\end{matrix};q,q^nd\right)\\ &= \frac{(c,d,x/q;q)_n(cd)^{-n}}{(x/c,x/d;q)_n}\frac{(x/c,x/d;q)_\infty}{(x/q,x/cd;q)_\infty}. \end{aligned} \tag{5.2}$$

So we have

$$\left[D_{q,x}^n\{f(x)(x;q)_{n-1}\}\right]_{x=aq}=\frac{(c,d,a;q)_n(cd)^{-n}}{(aq/c,aq/d;q)_n}\frac{(aq/c,aq/d;q)_\infty}{(a,aq/cd;q)_\infty}. \tag{5.3}$$

Substituting this into (3.13) and then writing aq/b as b we obtain the well-known nonterminating ${}_6\phi_5$ summation formula [4, Theorem 3.2], [9, Eq. (2.7.1), p. 36]:

Theorem 4.

$${}_6\phi_5\left(\begin{matrix} a,q\sqrt{a},-q\sqrt{a},b,c,d\\ \sqrt{a},-\sqrt{a},aq/b,aq/c,aq/d\end{matrix};q,\frac{aq}{bcd}\right)=\frac{(aq,aq/bc,aq/cd,aq/bd;q)_\infty}{(aq/b,aq/c,aq/d,aq/bcd;q)_\infty}. \tag{5.4}$$

The above Theorem is a very important result in the theory of q-series. R. Askey and M. Ismail [6] used this result and their clever analytical continuation techniques to obtain an elegant and elementary proof of Bailey's fundamental ${}_6\psi_6$ summation formula [4, Eq. (3.1)], [9, Eq. (5.3.1), p. 128] for bilateral basic hypergeometric series. Andrews [4] deduces many important and diverse results in number theory from the ${}_6\psi_6$ summation formula.

6. Watson's q-analog of Whipple's theorem

Next, we apply Theorem 2 to the following function:

$$f(b) = {}_4\phi_3\left(\begin{matrix} bc/aq, d, e, q^{-N} \\ b, c, de/aq^N \end{matrix}; q, q\right). \tag{6.1}$$

When $k < n$, $(b;q)_{n-1}(bc/aq;q)_k/(b;q)_k$ is a polynomial in b of degree $n-1$, and so by Theorem 1, we have

$$\left[D_{q,x}^n\left\{\frac{(x;q)_{n-1}(ac/xq;q)_k}{(x;q)_k}\right\}\right]_{x=aq} = 0. \tag{6.2}$$

When $k \geq n$, replacing c by q^{-k} in (5.3) and then replacing d by aq/c in the resulting equation we find that

$$\begin{aligned}\left[D_{q,x}^n\left\{\frac{(x;q)_{n-1}(xc/aq;q)_k}{(x;q)_k}\right\}\right]_{x=aq} &= \left[D_{q,x}^n\left\{\frac{(xq^k, xc/aq;q)_\infty}{(xq^{n-1}, xcq^{k-1}/a)_\infty}\right\}\right]_{x=aq} \\ &= q^{\binom{n}{2}}\left(-\frac{c}{aq}\right)^n \frac{(a, aq/c;q)_n(q, c;q)_k}{(c;q)_n(a;q)_{2n+1}(q, aq^{2n+1};q)_{k-n}}.\end{aligned} \tag{6.3}$$

So, by using the above two equations and the q-Pfaff–Saalschütz formula, we have

$$\begin{aligned}&\left[D_{q,x}^n\{f(x)(x;q)_{n-1}\}\right]_{x=aq} \\ &= q^{\binom{n}{2}}\left(-\frac{c}{aq}\right)^n \frac{(a, aq/c;q)_n}{(c;q)_n(a;q)_{2n+1}} \sum_{k=n}^{N} \frac{(q^{-N}, d, e;q)_k q^k}{(de/aq^N;q)_k(q, aq^{2n+1};q)_{k-n}} \\ &= q^{\binom{n}{2}}\left(-\frac{c}{a}\right)^n \frac{(a, aq/c, d, e, q^{-N};q)_n}{(c, de/aq^N;q)_n(a;q)_{2n+1}} {}_3\phi_2\left(\begin{matrix} dq^n, eq^n, q^{-N+n} \\ aq^{2n+1}, de/aq^{N-n} \end{matrix}; q, q\right) \\ &= \frac{(aq/d, aq/e;q)_N(a, aq/c, d, e, q^{-N};q)_n}{(a;q)_{N+1}(aq/de;q)_N(c, aq^{N+1}, aq/d, aq/e;q)_n}\left(\frac{cq^N}{de}\right)^n.\end{aligned} \tag{6.4}$$

Substituting this into (3.13) and then writing aq/b as b, and aq/c as c in the resulting equation, we obtain Watson's q-analog of Whipple's theorem [5, Theorem 3.6, p. 31], [9, Eq. (2.5.1), p. 35]:

Theorem 5.

$$
{}_8\phi_7\left(\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, c, d, e, q^{-N} \\ \sqrt{a}, -\sqrt{a}, aq/b, aq/c, aq/d, aq/e, aq^{1+N} \end{matrix}; q, \frac{a^2q^{N+2}}{bcde}\right)
$$
$$
= \frac{(aq, aq/de; q)_N}{(aq/d, aq/e; q)_N}\, {}_4\phi_3\left(\begin{matrix} aq/bc, d, e, q^{-N} \\ aq/b, aq/c, de/aq^N \end{matrix}; q, q\right). \tag{6.5}
$$

7. Andrews' identity for sums of three squares

Now, we apply Theorem 2 to the function

$$
f(b) = \frac{(bt; q)_\infty}{(b, sb; q)_\infty}. \tag{7.1}
$$

Taking $g(x) = 1/(q^{n-1}x; q)_\infty$ and $h(x) = (tx; q)_\infty/(sx; q)_\infty$ in the Leibniz formula, and using (4.1) and (4.3), we find that

$$
\begin{aligned}
\left[D_{q,x}^n\{f(x)(x;q)_{n-1}\}\right]_{x=aq} &= \frac{(atq^{n+1}; q)_\infty}{(aq^n, asq; q)_\infty} \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix} q^{j(n-1)} s^{n-j} (asq; q)_j (t/s; q)_{n-j} \\
&= \frac{(atq^{n+1}; q)_\infty}{(aq^n, asq; q)_\infty} \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix} q^{(n-j)(n-1)} s^j (asq; q)_{n-j} (t/s; q)_j.
\end{aligned} \tag{7.2}
$$

On substituting this into (3.13) we obtain

Theorem 6.

$$
\begin{aligned}
\frac{(bt, aq, asq; q)_\infty}{(b, sb, atq; q)_\infty} &= \sum_{n=0}^\infty \frac{(1-aq^{2n})(a, aq/b; q)_n b^n}{(1-a)(q, b, atq; q)_n} \\
&\quad \times \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix} q^{(n-j)(n-1)} s^j (asq; q)_{n-j} (t/s; q)_j.
\end{aligned} \tag{7.3}
$$

Setting $a = s = 1$, $t = -1$ and $b = -q$ in the equation above gives

$$
\frac{(q;q)_\infty^3}{(-q;q)_\infty^3} = 1 + 2\sum_{n=1}^\infty (-1)^n \frac{(q;q)_n q^{n^2}}{(-q;q)_n} \sum_{j=0}^n \frac{(-1;q)_j q^{j(1-n)}}{(q;q)_j}. \tag{7.4}
$$

The identity in [3, Eq. (5.7)] is equivalent to

$$
\sum_{j=0}^n \frac{(-1;q)_j q^{-jn}}{(q;q)_j} = (-1)^n \frac{(-q;q)_n}{(q;q)_n} \sum_{j=-n}^n (-1)^j q^{-j^2}. \tag{7.5}
$$

It follows that

$$\sum_{j=0}^{n}\frac{(-1;q)_j q^{j(1-n)}}{(q;q)_j}=\frac{(-1;q)_n q^{n(1-n)}}{(q;q)_n}+(-1)^{n-1}\frac{(-q;q)_{n-1}}{(q;q)_{n-1}}\sum_{j=-n+1}^{n-1}(-1)^j q^{-j^2}. \quad (7.6)$$

On substituting (7.6) into (7.4), we obtain Andrews' identity for sums of three squares [3, Eq. (5.16)]:

Theorem 7.

$$\begin{aligned}\sum_{n=0}^{\infty}(-1)^n r_3(n)q^n &= \left(\sum_{n=-\infty}^{\infty}(-1)^n q^{n^2}\right)^3=\frac{(q;q)_\infty^3}{(-q;q)_\infty^3}\\ &= 1+4\sum_{n=1}^{\infty}(-1)^n\frac{q^n}{1+q^n}-2\sum_{n=1}^{\infty}\frac{1-q^n}{1+q^n}\sum_{|j|<n}(-1)^j q^{n^2-j^2}. \quad (7.7)\end{aligned}$$

It should be pointed out that D. Krammer [11] derived an equivalent form of (7.7), which he used to give a simple proof of Legendre's three squares theorem. Legendre's theorem states that a natural number is the sum of three squares if and only if it is not of the form $4^k(8l+7)$. Krammer's proof only used Jacobi's triple identity and a special case of Ramanujan's ${}_1\psi_1$-summation formula.

Taking $a=q$, $s=q^{-1}$, $t=-q^{-1}$ and letting $b\to 0$ in (7.3), and then applying (7.5) to the resulting equation, we obtain the following result [3, Eq. (5.15)]:

$$(q;q)_\infty^2(q;q^2)_\infty=\sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-1)^j(1-q^{2n+1})q^{n(3n+1)/2-j^2}. \quad (7.8)$$

Taking $t=0$, $b\to 0$, $a=q$ and $s=q^{-1}$ in (7.3) we obtain

$$(q;q)_\infty^2=\sum_{n=0}^{\infty}(-1)^n(1-q^{2n+1})(q;q)_n q^{(3n^2+n)/2}\sum_{j=0}^{n}\frac{q^{-nj}}{(q;q)_j}. \quad (7.9)$$

The identity in [3, Eq. (5.11)] is equivalent to

$$\sum_{j=0}^{n}\frac{q^{-jn}}{(q;q)_j}=(-1)^n\frac{q^{\binom{n+1}{2}}}{(q;q)_n}\sum_{j=-n}^{n}(-1)^j q^{-j(3j+1)/2}. \quad (7.10)$$

Substituting (7.10) in (7.9), we obtain the following interesting identity

$$(q;q)_\infty^2=\sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-1)^j(1-q^{2n+1})q^{2n^2+n-j(3j+1)/2}. \quad (7.11)$$

The Hecke identity [3, Eq. (5.1)] is

$$(q;q)_\infty^2 = \sum_{m=-\infty}^{\infty} \sum_{n\ge 2|m|} (-1)^{n-m} q^{\binom{n+1}{2}-m(3m-1)/2}. \tag{7.12}$$

After replacing n by $2n$ or $2n+1$ and writing $m=-j$, we see that the right side of (7.12) becomes the right side of (7.11), and (7.12) is equivalent to (7.11).

By replacing q by q^2 in (7.3), we have

$$\begin{aligned}\frac{(bt, aq^2, asq^2; q^2)_\infty}{(b, sb, atq^2; q^2)_\infty} &= \sum_{n=0}^{\infty} \frac{(1-aq^{4n})(a, aq^2/b; q^2)_n b^n}{(1-a)(q^2, b, atq^2; q^2)_n} \\ &\quad \times \sum_{j=0}^{n} \frac{(q^2;q^2)_n q^{2(n-j)(n-1)} s^j}{(q^2;q^2)_j (q^2;q^2)_{n-j}} (asq^2;q^2)_{n-j} (t/s;q^2)_j. \end{aligned} \tag{7.13}$$

By taking $b=q^3$, $a=q^2$, $s=q^{-2}$ and $t=q^{-1}$ we obtain

$$\frac{(q^2, q^2, q^4; q^2)_\infty}{(q, q^3, q^3; q^2)_\infty} = \sum_{n=0}^{\infty} \frac{(1-q^{4n+2})(q, q^2; q^2)_n q^{2n^2+2n}}{(1-q^2)(q^3, q^3; q^2)_n} \sum_{j=0}^{n} \frac{(q;q^2)_j q^{-n(2j+1)}}{(q^2;q^2)_j}. \tag{7.14}$$

The identity in [5, Eq. (5.9)] is equivalent to

$$\sum_{j=0}^{n} \frac{(q;q^2)_j q^{-n(2j+1)}}{(q^2;q^2)_j} = \frac{(q;q^2)_n}{(q^2;q^2)_n} \sum_{j=0}^{2n} q^{-\binom{j+1}{2}}. \tag{7.15}$$

On substituting (7.15) into (7.14), we obtain Andrews' identity for sums of three triangular numbers [3, Eq. (5.17)]:

Theorem 8.

$$\begin{aligned}\left(\sum_{n=0}^{\infty} q^{\binom{n+1}{2}}\right)^3 &= \frac{(q^2;q^2)_\infty^3}{(q;q^2)_\infty^3} \\ &= \sum_{n=0}^{\infty} \sum_{j=0}^{2n} \frac{(1+q^{2n+1}) q^{2n^2+2n-\binom{j+1}{2}}}{(1-q^{2n+1})} \\ &= \sum_{n=0}^{\infty} \sum_{j=0}^{2n} \frac{(1+q^{2n+1}) q^{n+j(4n+1-j)/2}}{(1-q^{2n+1})}, \end{aligned} \tag{7.16}$$

which trivially implies Gauss' famous theorem that every integer is a sum of three triangular numbers.

By taking $a = q^2$, $b \to 0$, $s = q^{-2}$ and $t = q^{-1}$ in (7.13), and then using (7.15), we have

$$\frac{(q^2;q^2)_\infty^2}{(q;q^2)_\infty} = \sum_{n=0}^{\infty}\sum_{j=0}^{2n}(-1)^n(1+q^{2n+1})q^{3n^2+2n-\binom{j+1}{2}}. \tag{7.17}$$

8. Identities for Hecke type series

On setting $b = q^{-n}$ in (2.9) and then replacing a by c/a, we obtain

$${}_2\phi_1\left(\begin{matrix} q^{-n}, c/a \\ c \end{matrix}; q, aq^n\right) = \frac{(a;q)_n}{(c;q)_n}. \tag{8.1}$$

Setting $a = q$, replacing c by cq, and using (2.6) we find that

$$\sum_{j=0}^{n}(-1)^j \begin{bmatrix} n \\ j \end{bmatrix} q^{\binom{j+1}{2}} \frac{(c;q)_j}{(cq;q)_j} = \frac{(q;q)_n}{(cq;q)_n}. \tag{8.2}$$

Letting $b \to 0$ and $c \to 0$ in (2.8) we have

$$a^n = \sum_{k=0}^{n}\begin{bmatrix} n \\ k \end{bmatrix}(-1)^k q^{\binom{k+1}{2}-nk}(a;q)_k. \tag{8.3}$$

We now prove the following Lemma:

Lemma 3.

$$\begin{aligned}&\sum_{k=0}^{n}\begin{bmatrix} n \\ k \end{bmatrix}(-1)^k (aq^n;q)_k q^{(n-1)(n-k)+\binom{k}{2}} \frac{(q;q)_k}{(cq;q)_k} \\ &\quad = \frac{(q;q)_n a^n q^{2n^2-n}}{(cq;q)_n}\sum_{j=0}^{n}\frac{(c;q)_j a^{-j} q^{j(1-n)}}{(q;q)_j}.\end{aligned} \tag{8.4}$$

Proof: From (8.2) and (8.3) we find that

$$\begin{aligned}&\sum_{k=0}^{n}\begin{bmatrix} n \\ k \end{bmatrix}(-1)^k (aq^n;q)_k q^{(n-1)(n-k)+\binom{k}{2}} \frac{(q;q)_k}{(cq;q)_k} \\ &\quad = \sum_{k=0}^{n}\begin{bmatrix} n \\ k \end{bmatrix}(-1)^k (aq^n;q)_k q^{(n-1)(n-k)+\binom{k}{2}} \\ &\qquad \times \sum_{j=0}^{k}\begin{bmatrix} k \\ j \end{bmatrix}(-1)^j q^{\binom{j+1}{2}} \frac{(c;q)_j}{(cq;q)_j}\end{aligned}$$

$$
\begin{aligned}
&= \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix} \frac{(aq^n, c; q)_j}{(cq; q)_j} q^{j^2+(n-1)(n-j)} \\
&\quad \times \sum_{m=0}^{n-j} \begin{bmatrix} n-j \\ m \end{bmatrix} (-1)^m (aq^{n+j}; q)_m q^{\binom{m+1}{2}-m(n-j)} \\
&= \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix} \frac{(aq^n, c; q)_j}{(cq; q)_j} q^{j^2+(n-1)(n-j)} (aq^{n+j})^{n-j} \\
&= a^n q^{2n^2-n} \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix} \frac{(aq^n, c; q)_j}{(cq; q)_j} a^{-j} q^{j(1-n)}. \qquad (8.5)
\end{aligned}
$$

Using the identity [3, p. 118]

$$
\sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix} \frac{(aq^n, c; q)_j}{(cq; q)_j} a^{-j} q^{j(1-n)} = \frac{(q; q)_n}{(cq; q)_n} \sum_{j=0}^{n} \frac{(c; q)_j a^{-j} q^{j(1-n)}}{(q; q)_j}, \qquad (8.6)
$$

in the above equation, we obtain (8.4). □

Now, we apply Theorem 2 to the function

$$
f(b) = \frac{1}{(b; q)_\infty} \sum_{k=0}^{\infty} A_k P_k(b, a), \qquad (8.7)
$$

where the $\{A_k\}$ are constant sequences. Using (2.5), (4.3), and the Leibniz rule , we find that

$$
\left[D_{q,x}^n \{f(x)(x; q)_{n-1}\}\right]_{x=aq} = \frac{(a; q)_n}{(a; q)_\infty} \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} q^{(n-1)(n-k)} (q, aq^n; q)_k A_k. \qquad (8.8)
$$

On substituting the above equation into (3.13), we obtain the following important transformation formula.

Theorem 9.

$$
\begin{aligned}
\sum_{n=0}^{\infty} A_n b^n (aq/b; q)_n &= \frac{(b; q)_\infty}{(a; q)_\infty} \sum_{n=0}^{\infty} \frac{(1-aq^{2n})(a, aq/b; q)_n b^n}{(q, b; q)_n} \\
&\quad \times \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} q^{(n-1)(n-k)} (q, aq^n; q)_k A_k. \qquad (8.9)
\end{aligned}
$$

Taking $A_k = (-1)^k q^{\binom{k}{2}}/(cq; q)_k$ and using (8.4), the above theorem reduces to the following transformation formula, which contains many interesting special cases.

Theorem 10.

$$\sum_{n=0}^{\infty}\frac{(aq/b;q)_n q^{\binom{n}{2}}(-b)^n}{(cq;q)_n}=\frac{(b;q)_\infty}{(a;q)_\infty}\sum_{n=0}^{\infty}\frac{(1-aq^{2n})(a,aq/b;q)_n(ab)^n q^{2n^2-n}}{(b,cq;q)_n}\times\sum_{j=0}^{n}\frac{(c;q)_j a^{-j}q^{j(1-n)}}{(q;q)_j}. \tag{8.10}$$

When $b\to 0$ the above equation reduces to the following transformation formula

Theorem 11.

$$\sum_{n=0}^{\infty}\frac{a^n q^{n^2}}{(cq;q)_n}=\frac{1}{(a;q)_\infty}\sum_{n=0}^{\infty}\frac{(-1)^n(1-aq^{2n})(a;q)_n a^{2n}q^{(5n^2-n)/2}}{(cq;q)_n}\times\sum_{j=0}^{n}\frac{(c;q)_j a^{-j}q^{j(1-n)}}{(q;q)_j}. \tag{8.11}$$

Setting $c=1$ we immediately get the following identity from which we can derive the Rogers-Ramanujan identities [9, Eq. (2.7.6), p. 37]

$$\sum_{n=0}^{\infty}\frac{a^n q^{n^2}}{(q;q)_n}=\frac{1}{(a;q)_\infty}\sum_{n=0}^{\infty}\frac{(-1)^n(1-aq^{2n})(a;q)_n a^{2n}q^{n(5n-1)/2}}{(q;q)_n}. \tag{8.12}$$

Letting $a\to 1$ and $c=-1$ in (8.11) we find that

$$\sum_{n=0}^{\infty}\frac{q^{n^2}}{(-q;q)_n}=\frac{1}{(q;q)_\infty}\sum_{n=0}^{\infty}(-1)^n(1-q^{2n})\frac{(q;q)_{n-1}}{(-q;q)_n}q^{n(5n-1)/2}\times\sum_{j=0}^{n}\frac{(-1;q)_j q^{j(1-n)}}{(q;q)_j}. \tag{8.13}$$

From (7.6), by a straightforward calculation, we find that

$$\sum_{j=0}^{n}\frac{(-1;q)_j q^{j(1-n)}}{(q;q)_j}=(-1)^n\frac{(-q;q)_{n-1}}{(q;q)_n}\left\{q^n\sum_{j=-n}^{n}(-1)^j q^{-j^2}-\sum_{j=-n+1}^{n-1}(-1)^j q^{-j^2}\right\}. \tag{8.14}$$

Substituting (8.14) into (8.13) we find that

$$\begin{aligned}\sum_{n=0}^{\infty}\frac{q^{n^2}}{(-q;q)_n}&=\frac{1}{(q;q)_\infty}\sum_{n=0}^{\infty}q^{n(5n-1)/2}\left\{q^n\sum_{j=-n}^{n}(-)^j q^{-j^2}-\sum_{j=-n+1}^{n-1}(-)^j q^{-j^2}\right\}\\&=\frac{1}{(q;q)_\infty}\left\{\sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-)^j q^{n(5n+1)/2-j^2}-\sum_{n=0}^{\infty}\sum_{j=-n+1}^{n-1}(-)^j q^{n(5n-1)/2-j^2}\right\}\\&=\frac{1}{(q;q)_\infty}\sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-1)^j(1-q^{4n+2})q^{n(5n+1)/2-j^2}.\end{aligned} \tag{8.15}$$

Hence we have the following identity of Andrews [3, Eq. (6.1)]:

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(-q;q)_n} = \frac{1}{(q;q)_\infty} \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{4n+2}) q^{n(5n+1)/2-j^2}. \tag{8.16}$$

Letting $a = q$ and $c = -1$ in (8.11) and then using (7.5) we find that

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(-q;q)_n} &= \frac{1}{(q;q)_\infty} \sum_{n=0}^{\infty} (-1)^n (1-q^{2n+1}) \frac{(q;q)_n}{(-q;q)_n} q^{n(5n+3)/2} \sum_{j=0}^{n} \frac{(-1;q)_j q^{-jn}}{(q;q)_j} \\
&= \frac{1}{(q;q)_\infty} \sum_{n=0}^{\infty} (1-q^{2n+1}) q^{n(5n+3)/2} \sum_{j=-n}^{n} (-1)^j q^{-j^2} \\
&= \frac{1}{(q;q)_\infty} \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{2n+1}) q^{n(5n+3)/2-j^2}.
\end{aligned} \tag{8.17}$$

Hence we have the following identity of Andrews [3, Eq. (6.5)]:

$$\sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(-q;q)_n} = \frac{1}{(q;q)_\infty} \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{2n+1}) q^{n(5n+3)/2-j^2}. \tag{8.18}$$

Letting $a = q, b = q$, and $c = -1$ in (8.10), and using (7.5), we obtain the following identity:

$$\sum_{n=0}^{\infty} \frac{(q;q)_n}{(-q;q)_n} (-1)^n q^{\binom{n+1}{2}} = \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^{n+j} (1-q^{2n+1}) q^{n(2n+1)-j^2}. \tag{8.19}$$

Letting $a = q, b = -q$, and $c = -1$ in (8.10) gives

$$\sum_{n=0}^{\infty} q^{\binom{n+1}{2}} = \frac{(-q;q)_\infty}{(q;q)_\infty} \sum_{n=0}^{\infty} (-1)^n (1-q^{2n+1}) q^{2n^2+n} \sum_{j=0}^{n} \frac{(-1;q)_j q^{-jn}}{(q;q)_j}. \tag{8.20}$$

We recall that [8, p. 6, Eq. (7.321)]

$$\frac{(q^2;q^2)_\infty}{(q;q^2)_\infty} = \sum_{n=0}^{\infty} q^{\binom{n+1}{2}}. \tag{8.21}$$

Substituting (7.5) and (8.21) into (8.20) we obtain the following identity:

$$(q;q)_\infty (q^2;q^2)_\infty = \sum_{n=0}^{\infty} \sum_{j=-n}^{n} (-1)^j (1-q^{2n+1}) q^{n(2n+1)-j^2}. \tag{8.22}$$

We show that (8.22) is equivalent to the Kac-Peterson identity [2, Eq. (1.3)], [10]. The right side of (8.22) can be written as

$$\sum_{n\geq 0}\sum_{|j|\leq n}(-)^j q^{n(2n+1)-j^2}-\sum_{n\geq 0}\sum_{|j|\leq n}(-)^j q^{(n+1)(2n+1)-j^2}$$

In the first sum, write $N=2n$; in the second write $N=2n+1$. Combining the resulting sums gives

$$\sum_{N\geq 0}\sum_{|j|\leq N/2}(-)^{N+j}q^{N(N+1)/2-j^2}.$$

We separate this into two parts depending on the parity of j. In the even part, write $j=2m$; in the odd part, note that the sum for $|j|<0$ equals the sum for $|j|>0$:

$$\sum_{N\geq 0}\sum_{|m|\leq N/4}(-)^N q^{N(N+1)/2-4m^2}+2\sum_{N\geq 0}\sum_{\substack{0\leq j\leq N/2\\ j\ odd}}(-1)^{N+1}q^{N(N+1)/2-j^2}$$

In the last sum, write $N=3n-8m+1$ and $j=2n-6m+1$, so that $m=N+(1-3j)/2$ and $n=3N-4j+1$:

$$\begin{aligned}&\sum_{N\geq 0}\sum_{|m|\leq N/4}(-)^N q^{N(N+1)/2-4m^2}+2\sum_{n\geq 0}\sum_{n/4\leq m\leq n/3}(-1)^n q^{n(n+1)/2-4m^2}\\&=\sum_{n\geq 0}\sum_{|m|\leq n/4}(-)^N q^{n(n+1)/2-4m^2}+2\sum_{n\geq 0}\sum_{n/4\leq m\leq n/3}(-1)^n q^{n(n+1)/2-4m^2}\\&=\sum_{n=-\infty}^{\infty}\sum_{|n|\geq 3m}(-1)^n q^{n(n+1)/2-4m^2}.\end{aligned}$$

Hence we obtain the identity of Kac and Peterson [2, Eq. (1.3)], [10, final equation]:

$$(q;q)_\infty(q^2;q^2)_\infty=\sum_{n=-\infty}^{\infty}\sum_{|n|\geq 3m}(-1)^n q^{n(n+1)/2-4m^2}. \tag{8.23}$$

Taking $a=q$, $b=-q$, and $c=0$ in (8.10) we obtain

$$\sum_{n=0}^{\infty}(-q;q)_n q^{\binom{n+1}{2}}=\frac{(-q;q)_\infty}{(q;q)_\infty}\sum_{n=0}^{\infty}(-1)^n(q;q)_n(1-q^{2n+1})q^{2n^2+n}\sum_{j=0}^{n}\frac{q^{-nj}}{(q;q)_j}. \tag{8.24}$$

Substituting (7.10) into (8.24) we obtain the following identity of Andrews [3, Eq. (6.7)]:

$$\sum_{n=0}^{\infty}(-q;q)_n q^{\binom{n+1}{2}}=\frac{(-q;q)_\infty}{(q;q)_\infty}\sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-1)^j(1-q^{2n+1})q^{n(5n+3)/2-j(3j+1)/2}. \tag{8.25}$$

Taking $a = q, b = q$, and $c = 0$ in (8.10), and using (7.10) we obtain the following identity

$$\sum_{n=0}^{\infty}(-1)^n(q;q)_n q^{\binom{n+1}{2}} = \sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-1)^{j+n}(1-q^{2n+1})q^{n(5n+3)/2-j(3j+1)/2}. \tag{8.26}$$

Taking $a = b = q^2$ and $c = 0$ in (8.10), and multiplying by q, we find that

$$\sum_{n=0}^{\infty}(-1)^n(q;q)_n q^{\binom{n+2}{2}} = \sum_{n=0}^{\infty} q^{2n^2+3n+1}(1-q^{2n+2})(q;q)_n \sum_{j=0}^{n}\frac{q^{-j(n+1)}}{(q;q)_j}. \tag{8.27}$$

Next, replace n by $n+1$ in (7.10) and note that the $j = n+1$ terms are the same on both side. Subtracting this value gives

$$\sum_{j=0}^{n}\frac{q^{-j(n+1)}}{(q;q)_j} = (-1)^{n+1}\frac{q^{\binom{n+2}{2}}}{(q;q)_{n+1}}\sum_{j=-n-1}^{n}(-1)^j q^{-j(3j+1)/2}. \tag{8.28}$$

Combining the above equations gives

$$\sum_{n=0}^{n}(-1)^n(q;q)_n q^{\binom{n+2}{2}} = \sum_{n=0}^{\infty}\sum_{j=-n-1}^{n}(-1)^{n+j+1}(1+q^{n+1})q^{(n+1)(5n+4)/2-j(3j+1)/2}. \tag{8.29}$$

Taking $a = q$ and $c = 0$ in (8.10), we find that

$$\sum_{n=0}^{\infty} q^{n^2+n} = \frac{1}{(q;q)_\infty}\sum_{n=0}^{\infty}(-1)^n(1-q^{2n+1})(q;q)_n q^{n(5n+3)/2}\sum_{j=0}^{n}\frac{q^{-jn}}{(q;q)_j}. \tag{8.30}$$

Replacing q by q^2 in (8.21) we have

$$\sum_{n=0}^{\infty} q^{n^2+n} = \frac{(q^4;q^4)_\infty}{(q^2;q^4)_\infty}. \tag{8.31}$$

By using this and (7.10), and simplifying (8.31) we obtain

$$\sum_{n=0}^{\infty}\sum_{j=-n}^{n}(-1)^j(1-q^{2n+1})q^{3n^2+2n-j(3j+1)/2} = (q;q^2)_\infty(q^4;q^4)_\infty^2. \tag{8.32}$$

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On Some Integrals Involving the Hurwitz Zeta Function: Part 2

OLIVIER ESPINOSA olivier.espinosa@fis.utfsm.cl
Departamento de Física, Universidad Técnica Federico Santa María, Valparaíso, Chile

VICTOR H. MOLL vhm@math.tulane.edu
Department of Mathematics, Tulane University, New Orleans, Louisiana 70118

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Abstract. We establish a series of indefinite integral formulae involving the Hurwitz zeta function and other elementary and special functions related to it, such as the Bernoulli polynomials, $\ln\sin(\pi q)$, $\ln\Gamma(q)$ and the polygamma functions. Many of the results are most conveniently formulated in terms of a family of functions $A_k(q) := k\zeta'(1-k, q)$, $k \in \mathbb{N}$, and a family of polygamma functions of negative order, whose properties we study in some detail.

Key words: Hurwitz zeta function, polylogarithms, loggamma, integrals

2000 Mathematics Subject Classification: Primary—33E20

1. Introduction

The Hurwitz zeta function, defined by

$$\zeta(z, q) = \sum_{n=0}^{\infty} \frac{1}{(n+q)^z} \tag{1.1}$$

for $z \in \mathbb{C}$, $\operatorname{Re} z > 1$ and $q \neq 0, -1, -2, \ldots$, admits the integral representation

$$\zeta(z, q) = \frac{1}{\Gamma(z)} \int_0^\infty \frac{e^{-qt}}{1-e^{-t}} t^{z-1} dt, \tag{1.2}$$

where $\Gamma(z)$ is Euler's gamma function, which is valid for $\operatorname{Re} z > 1$ and $\operatorname{Re} q > 0$, and can be used to prove that $\zeta(z, q)$ has an analytic extension to the whole complex plane except for a simple pole at $z = 1$.

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For $\operatorname{Re} z < 0$, $\zeta(z, q)$ admits the following Fourier representation, originally derived by Hurwitz, in the range $0 \leq q \leq 1$:

$$\zeta(z, q) = \frac{2\Gamma(1-z)}{(2\pi)^{1-z}} \left[\sin\left(\frac{\pi z}{2}\right) \sum_{n=1}^{\infty} \frac{\cos(2\pi q n)}{n^{1-z}} + \cos\left(\frac{\pi z}{2}\right) \sum_{n=1}^{\infty} \frac{\sin(2\pi q n)}{n^{1-z}} \right]. \tag{1.3}$$

This representation was used in [5] to obtain several definite integral formulae involving $\zeta(z, q)$. A derivation of (1.3) can be found in [11, p. 268]. An alternative proof, based upon the representation

$$\zeta(z, q) = \frac{q^{1-z}}{z-1} + \frac{q^{-z}}{2} - z \int_0^{\infty} \frac{\{t\} - \frac{1}{2}}{(t+q)^{z+1}}\, dt, \tag{1.4}$$

where $\{t\}$ is the fractional part of t, has been given by Berndt [4]. The expression (1.4) is employed in [4] to give short proofs of several classical formulae, including Lerch's beautiful expression

$$\ln \Gamma(q) = \zeta'(0, q) - \zeta'(0). \tag{1.5}$$

In this paper we continue the work, initiated in [5], on the explicit evaluation of integrals involving $\zeta(z, q)$. Special cases of $\zeta(z, q)$ include the Bernoulli polynomials,

$$B_m(q) = -m\, \zeta(1-m, q), \quad m \in \mathbb{N}, \tag{1.6}$$

defined by their generating function

$$\frac{te^{qt}}{e^t - 1} = \sum_{m=0}^{\infty} B_m(q) \frac{t^m}{m!}, \tag{1.7}$$

and given explicitly in terms of the Bernoulli numbers B_k by

$$B_m(q) = \sum_{k=0}^{m} \binom{m}{k} B_k q^{m-k}; \tag{1.8}$$

the digamma function,

$$\psi(q) := \frac{d}{dq} \ln \Gamma(q) = \lim_{z \to 1} \left[\frac{1}{z-1} - \zeta(z, q) \right]; \tag{1.9}$$

and the polygamma functions,

$$\psi^{(m)}(q) = (-1)^{m+1}\, m!\, \zeta(m+1, q), \quad m \in \mathbb{N}, \tag{1.10}$$

defined by

$$\psi^{(m)}(q) := \frac{d^m}{dq^m} \psi(q), \quad m \in \mathbb{N}. \tag{1.11}$$

An important property of the Hurwitz zeta function, which will be essential for the indefinite integral evaluations presented in Section 2, is the following:

$$\frac{\partial}{\partial q}\,\zeta(z,q) = -z\,\zeta(z+1,q). \tag{1.12}$$

The rest of this paper is organized as follows. In Section 2 we consider the evaluation of indefinite integrals of functions of the form $f(q)\zeta(z,a+bq)$, using a simple integration by parts approach. In Section 3 we introduce and study some of the properties of two families of functions related to the first derivative with respect to the argument z of the Hurwitz zeta function $\zeta(z,q)$, evaluated at z equal to nonpositive integers. These functions appear in connection to the indefinite integrals involving polygamma and negapolygamma functions, as well as $\ln\Gamma(q)$ and $\ln\sin(\pi q)$, considered in Section 4. These, in turn, are derived from the formulae obtained in Section 2 either by direct differentiation or by taking the appropriate limits. Finally, in Section 5 we use some of the indefinite integral formulae to rederive some of the definite integral evaluations obtained in Ref. [5] and to present some new analogous formulae.

2. The evaluation of indefinite integrals

In this section we discuss a method to evaluate primitives of functions of the form $f(q)\zeta(z,a+bq)$. This is illustrated in the cases where f is a polynomial and an exponential function. The resulting evaluations can be taken as a starting point to derive similar formulae involving other special functions in place of $\zeta(z,a+bq)$. For instance, differentiation with respect to the parameter z leads, in view of Lerch's result (1.5), to the evaluation of primitives involving the weight $\ln\Gamma(a+bq)$, and thus also $\ln\sin\pi q$, by virtue of the reflection formula for the gamma function. Also, the limit $z\to m\in\mathbb{N}$ leads to the evaluation of primitives involving the polygamma function $\psi^{(m-1)}(q)$. We shall present these results, from a slightly more general point of view, in Section 4.

Theorem 2.1. *Let $r\in\mathbb{N}$, f be r-times differentiable and $a,\ b\in\mathbb{R}$. Then*

$$\begin{aligned}\int f(q)\zeta(z,a+bq)\,dq &= \sum_{k=1}^{r}(-1)^{k+1}\frac{f^{(k-1)}(q)\,\zeta(z-k,a+bq)}{b^k(1-z)_k}\\ &\quad+\frac{(-1)^r}{b^r(1-z)_r}\int f^{(r)}(q)\,\zeta(z-r,a+bq)\,dq.\end{aligned} \tag{2.1}$$

Proof: Observe that

$$\frac{\partial}{\partial q}\zeta(z-1,a+bq) = b(1-z)\zeta(z,a+bq), \tag{2.2}$$

so that integration by parts yields

$$\int f(q)\zeta(z,a+bq)\,dq = \frac{f(q)\zeta(z-1,a+bq)}{b(1-z)} - \frac{1}{b(1-z)}\int f'(q)\,\zeta(z-1,a+bq)\,dq.$$

The expression (2.1) follows by repeating this procedure. □

We now produce the evaluation of certain indefinite integrals by choosing appropriate functions f in Theorem 2.1.

Example 2.2. Let $n \in \mathbb{N}_0$ and $a,\ b \in \mathbb{R}$. Then the moments of $\zeta(z,q)$ are given by

$$\int q^n \zeta(z,a+bq)\,dq = n!\sum_{j=0}^{n} \frac{(-1)^j q^{n-j}}{b^{j+1}(1-z)_{j+1}(n-j)!}\zeta(z-j-1,a+bq). \tag{2.3}$$

Proof: The case $n=0$ is simply the known result

$$\int \zeta(z,a+bq)\,dq = \frac{\zeta(z-1,a+bq)}{b(1-z)}. \tag{2.4}$$

For $n \geq 1$, the function $f(q) = q^n$ satisfies $f^{(k-1)}(q) = n!\,q^{n-k+1}/(n-k+1)!$, for $k \leq n$. Then (2.1), with $r=n$, yields

$$\int q^n \zeta(z,a+bq)\,dq = \sum_{k=1}^{n} \frac{(-1)^{k+1} n!\,q^{n-k+1}}{(n-k+1)!\,b^k(1-z)_k}\zeta(z-k,a+bq) + \frac{(-1)^n\,n!}{b^n(1-z)_n}\int \zeta(z-n,a+bq)\,dq,$$

so that (2.3) follows from (2.4). □

In a similar fashion we obtain:

Example 2.3. Let $m \in \mathbb{N}_0$ and $a,\ b,\ c,\ d \in \mathbb{R}$. Then

$$\int B_m(c+dq)\zeta(z,a+bq)\,dq = m!\sum_{j=0}^{m} \frac{(-1)^j d^j B_{m-j}(c+dq)}{b^{j+1}(1-z)_{j+1}(m-j)!}\zeta(z-j-1,a+bq). \tag{2.5}$$

Proof: Same as the proof for Example 3.2 with

$$\frac{d^{k-1}}{dq^{k-1}}B_m(c+dq) = \frac{m!\,d^{k-1}}{(m-k+1)!}B_{m-k+1}(c+dq).$$ □

Definition. The family of functions $\mathfrak{F} := \{f_j(q) : j \in \mathbb{N}\}$ is said to be *closed under primitives* if for each j the primitive of $f_j(q)$ can be written as a *finite* linear combination of the elements of $\mathfrak{F}$. Naturally, the family $\mathfrak{F}$ is allowed to depend on a finite number of parameters, as in the next example.

Example 2.4. Example 2.2 shows that

$$\mathfrak{F}_{a,b} := \{P_j(q)\zeta(z-m, a+bq) : j,\ m \in \mathbb{N} \text{ and } P_j \text{ is a polynomial in } q \text{ of degree } j\}$$

is closed under primitives. This follows from

$$\int q^n \zeta(z-m, a+bq)\,dq = n! \sum_{j=0}^{n} \frac{(-1)^j q^{n-j} \zeta(z-m-1-j, a+bq)}{b^{j+1}(m+1-z)_{j+1}(n-j)!}, \tag{2.6}$$

which is a variation of (2.3).

Example 2.5. The moments of the Bernoulli polynomials are given by

$$\int q^n B_m(a+bq)\,dq = \frac{n!\,m!}{(n+m+1)!} \sum_{j=0}^{n} \frac{(-1)^j q^{n-j}}{b^{j+1}} \binom{m+n+1}{n-j} B_{m+j+1}(a+bq). \tag{2.7}$$

Proof: Use the identity (1.6) in (2.3). □

Example 2.6. Let $n \in \mathbb{N}$ be odd. Then

$$\int \zeta(z-n, q)\zeta(z, q)\,dq = \frac{1}{2} \sum_{k=1}^{n} \frac{(z-n)_{k-1}}{(1-z)_k} \zeta(z-k, q)\,\zeta(z-n+k-1, q). \tag{2.8}$$

Proof: The Hurwitz zeta function satisfies

$$\frac{\partial^{k-1}}{\partial q^{k-1}} \zeta(z-n, q) = (-1)^{k-1}(z-n)_{k-1}\zeta(z-n+k-1, q), \tag{2.9}$$

so the result follows from (2.1) with $r = n$, since in that case the integral on the righthand side equals the one on the left-hand side, except for the prefactor $(z-n)_n/(1-z)_n = (-1)^n$. □

Note. In view of the identity

$$\frac{(z-n)_{k-1}}{(1-z)_k} = (-1)^{n+1} \frac{(z-n)_{n-k}}{(1-z)_{n-k+1}},$$

it is easily seen that the terms in the sum on the right-hand side of (2.9) are equal in pairs, except for the central term $k = r$, where $r \in \mathbb{N}$ is defined by $n = 2r - 1$. Therefore we have the alternative formula

$$\int \zeta(z-2r+1,q)\zeta(z,q)\,dq = \frac{(z-2r+1)_{r-1}}{2\,(1-z)_r}\zeta^2(z-r,q) + \sum_{k=1}^{r-1}\frac{(z-2r+1)_{k-1}}{(1-z)_k}\zeta(z-k,q)\zeta(z-(2r-k),q). \tag{2.10}$$

Note. We have been unable to evaluate the integral in Example 2.6 for the case n even. Thus the question of whether the family

$$\mathfrak{F}_z := \{\zeta(z-n,q)\zeta(z-m,q) :\ n, m \in \mathbb{N}\} \tag{2.11}$$

is closed under primitives remains to be decided.

Example 2.7. Let $a,\ b \in \mathbb{R}$. Then

$$\int e^q \zeta(z, a+bq)\,dq = e^q \sum_{j=0}^{\infty} \frac{(-1)^j}{b^{j+1}(1-z)_{j+1}}\,\zeta(z-1-j, a+bq). \tag{2.12}$$

Proof: Divide (2.3) by $n!$ and then sum over n to produce

$$\int e^q \zeta(z, a+bq)\,dq = \sum_{n=0}^{\infty}\sum_{j=0}^{n} \frac{(-1)^j q^{n-j}}{b^{j+1}(1-z)_{j+1}(n-j)!}\,\zeta(z-1-j, a+bq).$$

The result follows by interchanging the order of summation. □

Note. We have been unable to produce a *finite expression* for the integral in (2.12).

Example 2.8. Let $m \in \mathbb{N}$. Then

$$\int e^q B_m(a+bq)\,dq = m!\,e^q(-1)^m \sum_{j=0}^{m} \frac{(-1)^j}{j!} b^{m-j} B_j(a+bq). \tag{2.13}$$

Proof: Use the identity (1.6) and $(m)_{j+1} = (m+j)!/(m-1)!$ in (2.12) to produce

$$\int e^q B_m(a+bq)\,dq = m!\,e^q(-1)^{m+1} \sum_{j=m+1}^{\infty} \frac{(-1)^j}{j!} b^{m-j} B_j(a+bq). \tag{2.14}$$

The generating function (1.7) is now employed to see that the sum from $j=0$ to infinity is independent of q, so it is absorbed into the implicit constant of integration. □

3. The function $A_k(q)$ and negapolygammas

In this section we consider the function

$$A_k(q) := k \frac{\partial}{\partial z} \zeta(z, q)\Big|_{z=1-k} \tag{3.1}$$

for $k \in \mathbb{N}$. The function $A_1(q)$ has a simple explicit form,

$$A_1(q) = \zeta'(0, q) = \ln \Gamma(q) + \zeta'(0), \tag{3.2}$$

in view of Lerch's result (1.5). These functions appear in all of the formulae for the indefinite integrals involving the loggamma and the logsine functions studied in Section 4. The derivative of the Hurwitz zeta function has appeared before in connection to integrals of $\ln \Gamma(q)$ [6], and in a number of related contexts, such as the studies of polygamma functions of negative order [1], the Barnes function [2] and the multiple gamma function [10], and other unrelated ones such as the evaluation of sums of the type $\sum_{m \geq 2} z^m \zeta(m, \alpha)$ [7]. Here, we will rediscover, from a more general point of view, the intimate relationship existing between the functions $A_k(q)$ and polygamma functions of negative order, first pointed out in Ref. [6].

The nonelementary behavior of $A_k(q)$ can be considered to be contained in the range $0 \leq q \leq 1$, since for $q > 1$ the value of $A_k(q)$ can be obtained by repeated use of the following result:

Lemma 3.1. *The function $A_k(q)$ satisfies*

$$A_k(q + 1) = A_k(q) + kq^{k-1} \ln q. \tag{3.3}$$

Proof: Differentiate both sides of the identity

$$\zeta(z, q) = \frac{1}{q^z} + \zeta(z, q + 1) \tag{3.4}$$

with respect to z at $z = 1 - k$. □

As an immediate consequence of property (3.3) and definition (3.1) we have

Lemma 3.2. *For $k \geq 2$,*

$$A_k(0) = A_k(1) = k\, \zeta'(1 - k). \tag{3.5}$$

Proof: Take the limit $q \to 0$ in (3.3) and use the fact that $\zeta(z, 1) = \zeta(z)$. □

Lemma 3.3. *For $k \in \mathbb{N}$,*

$$\int_0^1 A_k(q)\, dq = 0. \tag{3.6}$$

Proof: This is a direct consequence of

$$\int_0^1 \zeta(z,q)\,dq = 0, \tag{3.7}$$

valid for $\operatorname{Re} z < 1$. □

Lemma 3.4. *For $k \in \mathbb{N}$, the functions $A_k(q)$ satisfy*

$$A'_{k+1}(q) = (k+1)\left[A_k(q) + \frac{1}{k}B_k(q)\right]. \tag{3.8}$$

Proof: We have

$$\begin{aligned}
A'_{k+1}(q) &= (k+1)\frac{\partial}{\partial q}\frac{\partial}{\partial z}\zeta(z,q)\bigg|_{z=-k} = (k+1)\frac{\partial}{\partial z}\frac{\partial}{\partial q}\zeta(z,q)\bigg|_{z=-k} \\
&= (k+1)\frac{\partial}{\partial z}\left[-z\zeta(z+1,q)\right]\bigg|_{z=-k} \\
&= (k+1)\left[A_k(q) + \frac{1}{k}B_k(q)\right].
\end{aligned}$$

□

The following results can sometimes be used to simplify a formula:

$$\zeta'(-2n) = (-1)^n \frac{(2n)!\,\zeta(2n+1)}{2\,(2\pi)^{2n}}, \quad n \in \mathbb{N}, \tag{3.9}$$

$$\zeta'(0) = -\ln\sqrt{2\pi}. \tag{3.10}$$

Lemma 3.5. *For $k \in \mathbb{N}$,*

$$A_k\left(\frac{1}{2}\right) = (-1)^{k-1}B_k 2^{1-k}\ln 2 - (1-2^{1-k})k\,\zeta'(1-k). \tag{3.11}$$

In particular,

$$A_1\left(\frac{1}{2}\right) = -\frac{1}{2}\ln 2, \tag{3.12}$$

$$A_{2n+1}\left(\frac{1}{2}\right) = (-1)^{n+1}\frac{(1-2^{-2n})(2n+1)!\,\zeta(2n+1)}{2(2\pi)^{2n}} \quad n \geq 1. \tag{3.13}$$

Proof: Differentiate $\zeta(z,\frac{1}{2}) = (2^z-1)\,\zeta(z)$ at $z = 1-k$ and use the identity

$$\zeta(1-k) = \frac{(-1)^{k+1}B_k}{k}, \quad k \in \mathbb{N}. \tag{3.14}$$

□

Hurwitz's Fourier representation (1.3) for $\zeta(z, q)$ in the range $0 \le q \le 1$ and negative z is absolutely convergent and thus can be used to directly obtain a Fourier representation for the function $A_k(q)$ in the range $0 \le q \le 1$.

Define

$$C(z, q) = \sum_{n=1}^{\infty} \frac{\cos(2\pi nq)}{n^z} \quad \text{and} \quad S(z, q) = \sum_{n=1}^{\infty} \frac{\sin(2\pi nq)}{n^z}, \tag{3.15}$$

so that

$$\frac{\partial}{\partial z} C(z, q) = -\sum_{n=1}^{\infty} \frac{\ln n}{n^z} \cos(2\pi nq) \quad \text{and} \quad \frac{\partial}{\partial z} S(z, q) = -\sum_{n=1}^{\infty} \frac{\ln n}{n^z} \sin(2\pi nq).$$

Now replace z by $1 - z$ in the Fourier representation (1.3) and differentiate to produce

$$\begin{aligned} \zeta'(1 - z, q) = \frac{2\Gamma(z)}{(2\pi)^z} \Bigg\{ &\left[\Psi(z) \cos\frac{\pi z}{2} + \frac{\pi}{2} \sin\frac{\pi z}{2} \right] C(z, q) \\ &+ \left[\Psi(z) \sin\frac{\pi z}{2} - \frac{\pi}{2} \cos\frac{\pi z}{2} \right] S(z, q) \\ &- \cos\frac{\pi z}{2} \frac{\partial}{\partial z} C(z, q) - \sin\frac{\pi z}{2} \frac{\partial}{\partial z} S(z, q) \Bigg\}, \end{aligned} \tag{3.16}$$

where $\Psi(z) := \ln 2\pi - \psi(z)$.

For z a positive integer, the functions $S(z, q)$ and $C(z, q)$ are related to the Bernoulli polynomials (in view of (1.3) and (1.6)) and the Clausen functions. The latter are defined by

$$Cl_{2n}(x) = \sum_{k=1}^{\infty} \frac{\sin kx}{k^{2n}}, \quad n \ge 1 \tag{3.17}$$

and

$$Cl_{2n+1}(x) = \sum_{k=1}^{\infty} \frac{\cos kx}{k^{2n+1}}, \quad n \ge 0. \tag{3.18}$$

One has

$$S(2m + 1, q) = \sum_{n=1}^{\infty} \frac{\sin(2\pi nq)}{n^{2m+1}} = \frac{(-1)^{m+1} (2\pi)^{2m+1}}{2(2m + 1)!} B_{2m+1}(q), \tag{3.19}$$

$$C(2m + 1, q) = \sum_{n=1}^{\infty} \frac{\cos(2\pi nq)}{n^{2m+1}} = Cl_{2m+1}(2\pi q), \tag{3.20}$$

$$S(2m + 2, q) = \sum_{n=1}^{\infty} \frac{\sin(2\pi nq)}{n^{2m+2}} = Cl_{2m+2}(2\pi q), \tag{3.21}$$

$$C(2m + 2, q) = \sum_{n=1}^{\infty} \frac{\cos(2\pi nq)}{n^{2m+2}} = \frac{(-1)^{m} (2\pi)^{2m+2}}{2(2m + 2)!} B_{2m+2}(q). \tag{3.22}$$

The value $z = 2m + 1$ yields, upon using $\psi(k + 1) = -\gamma + H_k$, the expression

$$A_{2m+1}(q) = (H_{2m} - \gamma - \ln 2\pi)\, B_{2m+1}(q) \\ + (-1)^m \frac{2(2m+1)!}{(2\pi)^{2m+1}} \left[\sum_{n=1}^{\infty} \frac{\ln n}{n^{2m+1}} \sin(2\pi nq) + \frac{\pi}{2} \sum_{n=1}^{\infty} \frac{\cos(2\pi nq)}{n^{2m+1}} \right]. \quad (3.23)$$

Similarly, for $z = 2m + 2$ we find

$$A_{2m+2}(q) = (H_{2m+1} - \gamma - \ln 2\pi)\, B_{2m+2}(q) \\ + (-1)^{m+1} \frac{2(2m+2)!}{(2\pi)^{2m+2}} \left[\sum_{n=1}^{\infty} \frac{\ln n}{n^{2m+2}} \cos(2\pi nq) - \frac{\pi}{2} \sum_{n=1}^{\infty} \frac{\sin(2\pi nq)}{n^{2m+2}} \right]. \quad (3.24)$$

Lemma 3.6. *For $0 \le q \le 1$ the function $A_2(q)$ is given by*

$$A_2(q) = (1 - \gamma - \ln 2\pi)\left(q^2 - q + \frac{1}{6}\right) \\ - \frac{1}{\pi^2} \sum_{n=1}^{\infty} \frac{\ln n}{n^2} \cos(2\pi nq) + \frac{1}{2\pi} \sum_{n=1}^{\infty} \frac{\sin(2\pi nq)}{n^2}. \quad (3.25)$$

In particular,

$$\begin{aligned} A_2(1) &= 2\zeta'(-1), \\ A_2\left(\frac{1}{2}\right) &= -\zeta'(-1) - \frac{1}{12}\ln 2, \\ A_2\left(\frac{1}{4}\right) &= -\frac{1}{4}\zeta'(-1) + \frac{G}{2\pi}, \end{aligned}$$

where G is Catalan's constant.

Proof: The expression (3.25) follows directly from (3.24) at $m = 0$. □

Special values of the derivative of the Hurwitz zeta function at the rational arguments $q = \frac{1}{2}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{6}$ and $\frac{5}{6}$, for k odd, have been given in [8]. Special values of $A_k(q)$ for $q > 1$ can be obtained by using Lemma 3.3 a sufficient number of times.

Example 3.7.

$$\begin{aligned} A_2(2) &= 2\zeta'(-1), \\ A_2(3) &= 2\zeta'(-1) + 4\ln 2, \\ A_2\left(\frac{3}{2}\right) &= -\zeta'(-1) - \frac{13}{12}\ln 2, \\ A_2\left(\frac{5}{4}\right) &= -\frac{1}{4}\zeta'(-1) + \frac{G}{2\pi} - \ln 2. \end{aligned}$$

Several of the indefinite integrals derived in the next sections can be conveniently expressed in terms of a family of functions closely related to the *negapolygamma* family introduced by Gosper [6] as

$$\begin{aligned}\psi_{-1}(q) &:= \ln\Gamma(q),\\ \psi_{-k}(q) &:= \int_0^q \psi_{-k+1}(t)\,dt, \quad k \geq 2.\end{aligned} \tag{3.26}$$

These functions were later reconsidered by Adamchik [1] in the form

$$\psi_{-k}(q) = \frac{1}{(k-2)!}\int_0^q (q-t)^{k-2}\ln\Gamma(t)\,dt, \quad k \geq 2. \tag{3.27}$$

We introduce the *balanced negapolygamma* functions by

$$\psi^{(-k)}(q) := \frac{1}{k!}\left[A_k(q) - H_{k-1}B_k(q)\right], \tag{3.28}$$

where $k \in \mathbb{N}$ and H_r is the harmonic number ($H_0 := 0$). For instance,

$$\psi^{(-1)}(q) = A_1(q) = \ln\Gamma(q) + \zeta'(0), \tag{3.29}$$

$$\psi^{(-2)}(q) = \frac{1}{2}A_2(q) - \frac{1}{2}B_2(q). \tag{3.30}$$

Note. We can express the derivative of the Hurwitz zeta function at non-positive integers as

$$\zeta'(-r,q) = r!\psi^{(-1-r)}(q) + \frac{H_r}{1+r}B_{1+r}(q), \quad r \in \mathbb{N}_0. \tag{3.31}$$

Lemma 3.8. *The balanced negapolygammas can be expressed as*

$$\psi^{(-k)}(q) = e^{-\gamma z}\frac{\partial}{\partial z}\left[e^{\gamma z}\frac{\zeta(z,q)}{\Gamma(1-z)}\right]\bigg|_{z=1-k}. \tag{3.32}$$

Proof: Perform the derivative and use $\psi(k) = H_{k-1} - \gamma$, in addition to the definition (3.1) and the identity (1.6). □

We shall now study some of the properties of the balanced negapolygamma functions $\psi^{(-k)}(q)$. The adjective *balanced* is motivated by the following result:

Lemma 3.9. *For* $k \in \mathbb{N}$,

$$\int_0^1 \psi^{(-k)}(q)\,dq = 0. \tag{3.33}$$

Proof: Use (3.6) and the analogous result for the Bernoulli polynomials,

$$\int_0^1 B_k(q)\,dq = 0,$$

for $k \in \mathbb{N}$. □

Lemma 3.10. *For $k \in \mathbb{N}$,*

$$\frac{d}{dq}\psi^{(-k)}(q) = \psi^{(-k+1)}(q). \tag{3.34}$$

Proof: For $k \geq 2$, (3.34) is a direct consequence of Lemma 3.4 and the well-known property of the Bernoulli polynomials,

$$\frac{d}{dq}B_k(q) = kB_{k-1}(q). \tag{3.35}$$

The case $k = 1$ follows directly from (3.29) since

$$\frac{d}{dq}\ln\Gamma(q) = \psi(q) = \psi^{(0)}(q). \qquad \square$$

The functions $\psi^{(-k)}(q)$ defined by (3.28) are thus closely related to the ones introduced by Gosper. In fact, the precise relationship between both families of polygammas of negative order is

$$\psi^{(-k)}(q) = \psi_{-k}(q) + \sum_{r=0}^{k-1}\frac{q^{k-1-r}}{r!(k-1-r)!}[\zeta'(-r) + H_r\zeta(-r)], \tag{3.36}$$

where we have used the evaluation

$$\psi^{(-1-r)}(0) = \frac{1}{r!}[\zeta'(-r) + H_r\zeta(-r)], \quad r \in \mathbb{N}. \tag{3.37}$$

Note. According to [1],

$$\zeta'(-r) + H_r\zeta(-r) = -\ln A_r, \tag{3.38}$$

where the A_r are the generalized Glaisher constants defined by Bendersky [3].

Lemma 3.11. *The balanced negapolygamma functions $\psi^{(-k)}(q)$ admit the following Fourier expansion in the range $0 \leq q \leq 1$:*

$$\begin{aligned}\psi^{(-k)}(q) = \frac{2}{(2\pi)^k}\Bigg[&\sum_{n=1}^{\infty}\frac{\ln(2\pi n)+\gamma}{n^k}\cos\left(2\pi nq - \frac{k\pi}{2}\right)\\ &- \frac{\pi}{2}\sum_{n=1}^{\infty}\frac{1}{n^k}\sin\left(2\pi nq - \frac{k\pi}{2}\right)\Bigg].\end{aligned} \tag{3.39}$$

Proof: In the definition (3.28) of $\psi^{(-k)}(q)$ substitute, depending on the parity of k, the Fourier expansions (3.23) or (3.24) for the functions $A_k(q)$ and (3.19) or (3.22) for the Bernoulli polynomials. □

Lemma 3.12. *For $k \in \mathbb{N}$,*

$$\psi^{(-k)}(q+1) = \psi^{(-k)}(q) + \frac{q^{k-1}}{(k-1)!}[\ln q - H_{k-1}]. \tag{3.40}$$

Proof: In the definition (3.28) use (3.3) and the property

$$B_m(q+1) = B_m(q) + mq^{m-1},$$

satisfied by the Bernoulli polynomials. □

Corollary 3.13. *For $r \in \mathbb{N}$,*

$$\psi^{(-1-r)}(1) = \psi^{(-1-r)}(0) = \frac{1}{r!}[\zeta'(-r) + H_r\zeta(-r)]. \tag{3.41}$$

Proof: Set $k = 1 + r$ and evaluate (3.40) at $q = 0$. Then use (3.37). □

4. Indefinite integrals of polygamma functions

Integral formulae involving the polygamma functions can be obtained from the corresponding ones for $\zeta(z, q)$, like (2.3) or (2.5), by taking the limit $z \to m = 2, 3, \ldots$. This limit is not trivial in general, because of the vanishing of the Pochhammer symbol $(1-z)_{j+1}$ for $j > m - 2$, and the appearance of the function $\zeta(1, q)$ on the right-hand side of the formulae cited above. Similarly, differentiation at $z = 1 - k$ leads to evaluations involving the functions $A_k(q)$ and the negapolygamma functions. Because of the connection (3.29) between $\psi^{(-1)}(q)$ and $\ln\Gamma(q)$, and of the latter with the function $\ln\sin(\pi q)$, we also obtain indefinite integral formulae involving these last two functions.

The next theorem gives the moments of the polygamma functions in terms of themselves and the balanced negapolygammas defined in Section 3. The proof of the theorem rests on the following result:

Lemma 4.1. *Let $p \in \mathbb{N}_0$. Then*

$$\sum_{r=0}^{p}(-1)^r\binom{p+1}{r+1}\frac{q^{p-r}B_{r+1}(a+bq)}{b^{r+1}} = q^{p+1} + (-1)^p\frac{B_{p+1}(a)}{b^{p+1}}. \tag{4.1}$$

Proof: The basic identity

$$B_{p+1}(x+y) = \sum_{r=0}^{p+1}\binom{p+1}{r}B_r(x)y^{p+1-r} \tag{4.2}$$

appears in [9] 19:5:3. The identity of the lemma is obtained by replacing x by $a + bq$ and y by $-bq$ in (4.2). □

Theorem 4.2. *Let $n \in \mathbb{N}_0$, $m \in \mathbb{N}$. Then*

$$\int q^n \psi^{(m)}(a+bq)\,dq = n! \sum_{j=0}^{n} \frac{(-1)^j}{b^{j+1}(n-j)!} q^{n-j} \psi^{(m-j-1)}(a+bq). \tag{4.3}$$

Proof: In view of relationship (1.10), we set $z = m + \epsilon$ in (2.3) and take the limit $\epsilon \to 0$. For a general value of n there will be three kinds of terms to consider in the sum on the right-hand side of (2.3):

(a) For $0 \le j \le m-2$ no singularities will arise and we simply have

$$\lim_{\epsilon \to 0} \frac{\zeta(m-j+\epsilon, q)}{(-m-\epsilon)_{j+1}} = \frac{(-1)^{m+1}}{m!} \psi^{(m-j-1)}(q).$$

(b) For $j = m-1$ we use the results

$$(-m-\epsilon)_m = (-1)^m m![1 + H_m\epsilon + O(\epsilon^2)]$$

and

$$\zeta(1+\epsilon, q) = \frac{1}{\epsilon} - \psi(q) + O(\epsilon^2),$$

where $\psi(q) = \psi^{(0)}(q)$ is the usual digamma function, to obtain

$$\left.\frac{\zeta(m-j+\epsilon, q)}{(-m-\epsilon)_{j+1}}\right|_{j=m-1} = \frac{(-1)^{m+1}}{m!}\left[-\frac{1}{\epsilon} + \psi(q) + H_m + O(\epsilon)\right].$$

(c) For $j \ge m$, say $j = m + r$ with $r \ge 0$, we use

$$(-m-\epsilon)_{m+1+r} = (-1)^{m+1} m! r! \epsilon[1 + (H_m - H_r)\epsilon + O(\epsilon^2)]$$

and

$$\zeta(-r+\epsilon, q) = \zeta(-r, q) + \epsilon\zeta'(-r, q) + O(\epsilon^2)$$

to obtain

$$\left.\frac{\zeta(m-j+\epsilon, q)}{(-m-\epsilon)_{j+1}}\right|_{j=m+r} = \frac{(-1)^{m+1}}{m!}\left[-\frac{1}{\epsilon}\frac{B_{r+1}(q)}{(r+1)!} + H_m \frac{B_{r+1}(q)}{(r+1)!} \right.$$
$$\left. + \psi^{(-1-r)}(q) + O(\epsilon)\right],$$

where we have used the definition (3.28) of the balanced negapolygamma function and (1.6). When all the terms are added up, we find, in view of Lemma 4.1, that the coefficients of the $1/\epsilon$ singularity and the term proportional to the harmonic number H_m reduce to a q-independent constant, which can be dropped. This proves the theorem. □

A similar result holds for the moments of the digamma function:

Theorem 4.3. *Let $n \in \mathbb{N}_0$. Then*

$$\int q^n \psi(a+bq)\,dq = n! \sum_{j=0}^{n} \frac{(-1)^j}{b^{j+1}(n-j)!} q^{n-j} \psi^{(-j-1)}(a+bq). \tag{4.4}$$

Proof: Use (1.9) in (2.3) and proceed along the same lines as the proof above. □

Note. Result (4.4) can be identified as an extension of Theorem 4.2 to the case $m = 0$.

Note. Adamchik [1, 2] has provided the alternative representations

$$\begin{aligned}\int_0^z x^n \psi(x)\,dx = (-1)^n \left(\frac{B_{n+1} H_n}{n+1} - \zeta'(-n) \right) \\ + \sum_{k=0}^{n} (-1)^k \binom{n}{k} z^{n-k} \left(\zeta'(-k, z) - \frac{B_{k+1}(z) H_k}{k+1} \right),\end{aligned}$$

and

$$\begin{aligned}\int_0^z x^n \psi(x)\,dx = \sum_{k=0}^{n-1} (-1)^k z^{n-k} \binom{n}{k} \left(\zeta'(-k) - \frac{B_{k+1}(z) H_k}{k+1} \right) \\ - \sum_{k=1}^{n} (-1)^k k! \left\{ n \atop k \right\} \log G_{k+1}(z+1) + (-1)^n H_n \frac{B_{n+1} - B_{n+1}(z)}{n+1},\end{aligned}$$

where $\left\{ n \atop k \right\}$ are the Stirling numbers of the second kind and $G_k(z)$ is the multiple Barnes function. The first representation can be directly obtained from (4.4) at $a = 0$ and $b = 1$, by explicit evaluation of the integral between $q = 0$ and $q = z$, making use of (3.37) for $\psi^{(-j-1)}(0)$ and of the definition (3.28) of the negapolygamma functions.

By differentiating formula (2.3) at $z = 1 - m$ we can arrive at the following result for the moments of the balanced negapolygamma functions, which extends Theorem 4.2 to negative values of m:

Theorem 4.4. *Let $n \in \mathbb{N}_0$, $m \in \mathbb{N}$. Then*

$$\int q^n \psi^{(-m)}(a+bq)\,dq = n! \sum_{j=0}^{n} \frac{(-1)^j}{b^{j+1}(n-j)!} q^{n-j} \psi^{(-m-j-1)}(a+bq). \tag{4.5}$$

Proof: Differentiate (2.3) at $z = 1 - m$ and use the result

$$\frac{d}{dz}(1-z)_{j+1}\Big|_{z=1-m} = -(m)_{j+1}[H_{m+j} - H_{m-1}]$$

to obtain the following result for the moments of the function $A_m(q)$:

$$\int q^n A_m(a+bq)\,dq = m!\,n!\sum_{j=0}^{n} \frac{(-1)^j}{b^{j+1}(n-j)!(m+j+1)!} q^{n-j}[A_{m+j+1}(a+bq) - (H_{m+j} - H_{m-1})B_{m+j+1}(a+bq)]. \tag{4.6}$$

This result is equivalent to the statement of the theorem in view of definition (3.28) of the balanced negapolygammas and Formula (2.7) for the moments of the Bernoulli polynomials. □

As a particular case of the last theorem we obtain a formula for the moments of the loggamma function $\ln\Gamma(q)$.

Example 4.5. Let $n \in \mathbb{N}_0$ and $a,\ b \in \mathbb{R}$. Then

$$\int q^n \ln\Gamma(a+bq)\,dq = \ln\sqrt{2\pi}\,\frac{q^{n+1}}{n+1} + n!\sum_{j=0}^{n} \frac{(-1)^j q^{n-j}}{b^{j+1}(n-j)!}\psi^{(-2-j)}(a+bq). \tag{4.7}$$

This generalizes Gosper's result [6], which establishes that all integrals of the form $\int q^n \ln q!\,dq$ are expressible in terms of $\zeta'(-j,q)$, with $1 \le j \le n+1$.

As particular cases of (4.7) we have

$$\int q^n \ln\Gamma(q)\,dq = \ln\sqrt{2\pi}\,\frac{q^{n+1}}{n+1} + n!\sum_{j=0}^{n} \frac{(-1)^j q^{n-j}}{(n-j)!}\psi^{(-2-j)}(q) \tag{4.8}$$

and

$$\int q^n \ln\Gamma(1-q)\,dq = \ln\sqrt{2\pi}\,\frac{q^{n+1}}{n+1} - n!\sum_{j=0}^{n} \frac{q^{n-j}}{(n-j)!}\psi^{(-2-j)}(1-q). \tag{4.9}$$

Proof: Apply theorem (4.4) to $\psi^{(-1)}(q) = \ln\Gamma(q) - \ln\sqrt{2\pi}$. The expressions (4.8) and (4.9) correspond to $a = 0,\ b = 1$ and $a = 1,\ b = -1$ respectively. □

The two special cases of Example 4.5 are now combined with the reflection formula for the gamma function

$$\Gamma(q)\Gamma(1-q) = \frac{\pi}{\sin\pi q} \tag{4.10}$$

to obtain an expression for the moments of $\ln\sin\pi q$.

Example 4.6. Let $n \in \mathbb{N}_0$. Then

$$\int q^n \ln \sin \pi q \, dq = -\frac{q^{n+1} \ln 2}{n+1} - n! \sum_{j=0}^{n} \frac{q^{n-j}}{(j+2)!(n-j)!} \times [(-1)^j A_{j+2}(q) - A_{j+2}(1-q)]. \tag{4.11}$$

Proof: Use the reflection formula for $\Gamma(q)$, results (4.8, 4.9) and the definition (3.28) to produce (4.11). The term that corresponds to the Bernoulli polynomials disappears in view of

$$(-1)^j B_{j+2}(q) = B_{j+2}(1-q). \tag{4.12}$$

□

Example 4.7.

$$\int e^q \ln \sin \pi q \, dq = -e^q \left[\ln 2 + \sum_{j=0}^{\infty} \frac{(-1)^j A_{j+2}(q) - A_{j+2}(1-q)}{(j+2)!} \right]. \tag{4.13}$$

Proof: Divide (4.11) by $n!$ and sum over n. □

Example 4.8. Integrating (4.13) by parts yields

$$\int e^q \operatorname{cotg} \pi q \, dq = \frac{e^q}{\pi} \left[\ln \sin \pi q + \ln 2 + \sum_{j=0}^{\infty} \frac{(-1)^j A_{j+2}(q) - A_{j+2}(1-q)}{(j+2)!} \right]. \tag{4.14}$$

5. Some definite integrals

Some of the definite integral formulae given in [5], in the range (0, 1), can be obtained directly from the indefinite integral formulae given in Sections 2 and 4.

Example 5.1. Evaluate Eq. (2.3) between 0 and 1 to obtain

$$\int_0^1 q^n \zeta(z, a+bq) \, dq = n! \sum_{j=0}^{n-1} \frac{(-1)^j \zeta(z-j-1, a+b)}{b^{j+1}(1-z)_{j+1}(n-j)!} + \frac{n!(-1)^n}{b^{n+1}(1-z)_{n+1}} (\zeta(z-n-1, a+b) - \zeta(z-n-1, a)). \tag{5.1}$$

As a particular case we obtain formula (12.2) of [5]:

Corollary 5.2. *Let $n \in \mathbb{N}_0$ and $z \in \mathbb{R}$, with $z - n - 1 < 0$. Then*

$$\int_0^1 q^n \zeta(z, q)\, dq = n! \sum_{j=0}^{n-1} \frac{(-1)^j \zeta(z - j - 1)}{(1 - z)_{j+1}(n - j)!}. \tag{5.2}$$

Proof: Set $b = 1$ in (5.1) and use the identity (3.4) and the hypothesis $z - n - 1 < 0$ to get rid of the last term in (5.1) in the limit $a \to 0$. □

The evaluation of formula (4.11) between $q = 0$ and $q = 1$ leads to formula (5.6) of [5]:

Example 5.3. Let $n \in \mathbb{N}_0$. Then

$$\int_0^1 q^n \ln(\sin \pi q)\, dq = -\frac{\ln 2}{n + 1} + n! \sum_{k=1}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^k \zeta(2k + 1)}{(2\pi)^{2k}\,(n + 1 - 2k)!}. \tag{5.3}$$

Proof: Direct evaluation of the right-hand side of (4.11) gives, in view of property (3.5),

$$\int_0^1 q^n \ln \sin \pi q\; dq = -\frac{\ln 2}{n + 1} - n! \sum_{j=0}^{n-1} \frac{1}{(j + 1)!(n - j)!} [(-1)^j - 1]\zeta'(-j - 1). \tag{5.4}$$

Clearly only the terms with j odd, say $j = 2k - 1$, survive in the sum. (5.3) then follows directly from (3.9). □

Example 5.4. Let $n \in \mathbb{N}_0$. Then

$$\begin{aligned}\int_0^{1/2} q^n \ln(\sin \pi q)\, dq = {} & -\frac{1}{2^{n+1}} \left[\frac{\ln 2}{n + 1} + n! \sum_{k=1}^{\lfloor \frac{n+1}{2} \rfloor} \frac{(-1)^k (2^{2k} - 1)\zeta(2k + 1)}{(2\pi)^{2k}\,(n + 1 - 2k)!} \right] \\ & - \frac{1 - (-1)^n}{n + 1} \zeta'(-n - 1). \end{aligned} \tag{5.5}$$

Proof: Use (3.11) in (4.11). □

For instance,

$$\begin{aligned}
\int_0^{1/2} q \ln(\sin \pi q)\, dq &= -\frac{1}{8} \ln 2 + \frac{7\zeta(3)}{16\pi^2}, \\
\int_0^{1/2} q^2 \ln(\sin \pi q)\, dq &= -\frac{1}{24} \ln 2 + \frac{3\zeta(3)}{16\pi^2}, \\
\int_0^{1/2} q^3 \ln(\sin \pi q)\, dq &= -\frac{1}{64} \ln 2 + \frac{9\zeta(3)}{64\pi^2} - \frac{93\zeta(5)}{128\pi^4}.
\end{aligned}$$

Example 5.5. Formulae (4.7) or (4.8) allow us to derive Gosper's formulae for integrals of $\ln \Gamma(q)$ [6] in a very economical way. For instance, setting $n = 0$, $a = b = 1$ in (4.7) yields

$$\int_0^q \ln\Gamma(q+1)\,dq = q\ln\sqrt{2\pi} + \frac{1}{2}A_2(q+1) - \frac{1}{2}B_2(q+1) - \zeta'(-1) + \frac{1}{2}B_2. \quad (5.6)$$

Evaluation of the right-hand side at $q = \frac{1}{2}$ and $\frac{1}{4}$ yields, respectively,

$$\int_0^{1/2} \ln\Gamma(q+1)\,dq = -\frac{3}{8} - \frac{13}{24}\ln 2 + \frac{1}{2}\ln\sqrt{2\pi} - \frac{3}{2}\zeta'(-1),$$
$$\int_0^{1/4} \ln\Gamma(q+1)\,dq = -\frac{5}{32} - \frac{1}{2}\ln 2 + \frac{1}{4}\ln\sqrt{2\pi} - \frac{9}{8}\zeta'(-1) + \frac{G}{4\pi},$$

which can easily be seen to be equivalent to Gosper's formulae, after using Riemann's functional equation for the Riemann zeta function to express $\zeta'(-1)$ in the form

$$\zeta'(-1) = \frac{\zeta'(2)}{2\pi^2} - \frac{1}{12}(2\ln\sqrt{2\pi} + \gamma - 1). \quad (5.7)$$

Example 5.6. For $k, k' \in \mathbb{N}$,

$$\int_0^1 \psi^{(-k)}(q)\psi^{(-k')}(q)\,dq = \frac{2\cos(k-k')\frac{\pi}{2}}{(2\pi)^{k+k'}}\left[\zeta''(k+k') - 2(\gamma+\ln 2\pi)\zeta'(k+k') + \left\{(\gamma+\ln 2\pi)^2 + \frac{\pi^2}{4}\right\}\zeta(k+k')\right].$$

The special case $k = k' = 1$ reduces to

$$\int_0^1 (\ln\Gamma(q))^2\,dq = \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{1}{3}\gamma\ln\sqrt{2\pi} + \frac{4}{3}\ln^2\sqrt{2\pi} - (\gamma + 2\ln\sqrt{2\pi})\frac{\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2},$$

given in [5].

Proof: Use the representation (3.32) of the balanced negapolygammas to obtain the desired result, by direct differentiation of the formula

$$\int_0^1 \zeta(z', q)\zeta(z, q)\,dq = \frac{2\Gamma(1-z)\Gamma(1-z')}{(2\pi)^{2-z-z'}}\zeta(2-z-z')\cos\left(\frac{\pi(z-z')}{2}\right), \quad (5.8)$$

valid for real $z,\ z' \le 0$, given in [5]. □

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Sums of Five, Seven and Nine Squares

SHAUN COOPER s.cooper@massey.ac.nz
Institute of Information and Mathematical Sciences, Massey University—Albany, Private Bag 102904, North Shore Mail Centre, Auckland, New Zealand

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Abstract. Let $r_k(n)$ denote the number of representations of an integer n as a sum of k squares. We prove that

$$r_5(n) = r_5(n')\left[\frac{2^{3\lfloor\lambda/2\rfloor+3}-1}{2^3-1} - \epsilon_5(n')\,\frac{2^{3\lfloor\lambda/2\rfloor}-1}{2^3-1}\right] \times \prod_p \left[\frac{p^{3\lfloor\lambda_p/2\rfloor+3}-1}{p^3-1} - p\left(\frac{n'}{p}\right)\frac{p^{3\lfloor\lambda_p/2\rfloor}-1}{p^3-1}\right],$$

where

$$\epsilon_5(n') = \begin{cases} 0 & \text{if } n' \equiv 1 \pmod 8 \\ 4 & \text{if } n' \equiv 2 \text{ or } 3 \pmod 4 \\ 16/7 & \text{if } n' \equiv 5 \pmod 8). \end{cases}$$

Here $n = 2^\lambda \prod_p p^{\lambda_p}$ is the prime factorisation of n, n' is the square-free part of n, the products are taken over the odd primes p, and $(\frac{n}{p})$ is the Legendre symbol.

Some similar formulas for $r_7(n)$ and $r_9(n)$ are also proved.

Key words: Hecke operator, modular forms of half integer weight, sums of squares

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1. Introduction

Let $r_k(n)$ denote the number of representations of n as a sum of k squares. The generating function for $r_k(n)$ is

$$\left(\sum_{j=-\infty}^{\infty} q^{j^2}\right)^k = \sum_{n=0}^{\infty} r_k(n)q^n.$$

Let us write the prime factorisation of n as

$$n = 2^\lambda \prod_p p^{\lambda_p}.$$

Here and throughout this article p will always denote an odd prime, $\prod_p$ will denote a product over all odd primes p, and λ and λ_p are nonnegative integers. Furthermore, n' will denote the square-free part of n.

Hurwitz [7, p. 271], [14], [15, p. 751] proved that if n is a square, then

$$r_3(n) = 6\prod_p \left[\frac{p^{\lambda_p/2+1}-1}{p-1} - (-1)^{(p-1)/2}\frac{p^{\lambda_p/2}-1}{p-1}\right] \tag{1.1}$$

This was generalised to arbitrary postive integers n by Hirschhorn and Sellers [12], who proved

$$r_3(n) = r_3(n')\prod_p \left[\frac{p^{\lfloor\lambda_p/2\rfloor+1}-1}{p-1} - \left(\frac{-n'}{p}\right)\frac{p^{\lfloor\lambda_p/2\rfloor}-1}{p-1}\right]. \tag{1.2}$$

Thus Hurwitz' result (1.1) is the special case $n' = 1$ of (1.2).

In a letter to Hermite, Stieltjes [7, p. 310], [25], [27, pp. 329–331] wrote that he had computed $r_5(p^2)$ for a large number of cases, and conjectured that if p is an odd prime, then

$$r_5(p^2) = 10(p^3 - p + 1).$$

In the same letter, he wrote that he had also computed $r_5(p^4)$ for $p = 3, 5$ and 7, conjectured

$$r_5(p^4) = 10[p(p^2-1)(p^3+1)+1],$$

and remarked "les calculs deviennent trop laborieux". Hurwitz [7, p. 311], [13], [15, pp. 5–7] proved Stieltjes' conjectures as well as the generalisation

$$r_5(n) = 10\frac{2^{3\lambda/2+3}-1}{2^3-1}\prod_p\left[\frac{p^{3\lambda_p/2+3}-1}{p^3-1} - p\frac{p^{3\lambda_p/2}-1}{p^3-1}\right] \tag{1.3}$$

when n is a square. Interestingly, (1.3) appeared in print 23 years earlier than (1.1).

In this article we will prove the more general result

$$\begin{aligned} r_5(n) = r_5(n')&\left[\frac{2^{3\lfloor\lambda/2\rfloor+3}-1}{2^3-1} - \epsilon_5(n')\frac{2^{3\lfloor\lambda/2\rfloor}-1}{2^3-1}\right] \\ &\times\prod_p\left[\frac{p^{3\lfloor\lambda_p/2\rfloor+3}-1}{p^3-1} - p\left(\frac{n'}{p}\right)\frac{p^{3\lfloor\lambda_p/2\rfloor}-1}{p^3-1}\right], \end{aligned}$$

where

$$\epsilon_5(n') = \begin{cases} 0 & \text{if } n' \equiv 1 \pmod 8 \\ 4 & \text{if } n' \equiv 2 \text{ or } 3 \pmod 4 \\ 16/7 & \text{if } n' \equiv 5 \pmod 8. \end{cases}$$

We will also prove that

$$\begin{aligned} r_7(n) = r_7(n')&\left[\frac{2^{5\lfloor\lambda/2\rfloor+5}-1}{2^5-1} - \epsilon_7(n')\frac{2^{5\lfloor\lambda/2\rfloor}-1}{2^5-1}\right] \\ &\times\prod_p\left[\frac{p^{5\lfloor\lambda_p/2\rfloor+5}-1}{p^5-1} - p^2\left(\frac{-n'}{p}\right)\frac{p^{5\lfloor\lambda_p/2\rfloor}-1}{p^5-1}\right], \end{aligned}$$

where

$$\epsilon_7(n') = \begin{cases} 0 & \text{if } n' \equiv 3 \pmod 8 \\ -8 & \text{if } n' \equiv 1 \text{ or } 2 \pmod 4 \\ 64/37 & \text{if } n' \equiv 7 \pmod 8. \end{cases}$$

Furthermore, we will prove that

$$r_9(n) = r_9(n')\frac{2^{7\lfloor\lambda/2\rfloor+7}-1}{2^7-1}\prod_p\left[\frac{p^{7\lfloor\lambda_p/2\rfloor+7}-1}{p^7-1} - p^3\left(\frac{n'}{p}\right)\frac{p^{7\lfloor\lambda_p/2\rfloor}-1}{p^7-1}\right],$$

in the case that $n' \equiv 5 \pmod 8$. Some results for when $n' \not\equiv 5 \pmod 8$ will also be given, but these involve additional and more complicated number theoretic functions.

The corresponding result for sums of eleven squares is

$$r_{11}(n) = r_{11}(n')\frac{2^{9\lfloor\lambda/2\rfloor+9}-1}{2^9-1}\prod_p\left[\frac{p^{9\lfloor\lambda_p/2\rfloor+9}-1}{p^9-1} - p^4\left(\frac{-n'}{p}\right)\frac{p^{9\lfloor\lambda_p/2\rfloor}-1}{p^9-1}\right],$$

in the case that $n' \equiv 7 \pmod 8$. A proof of this result is given in [6]. Just as for sums of nine squares, results for $r_{11}(n)$ for when $n' \not\equiv 7 \pmod 8$ can also be given, but these all involve additional and more complicated number theoretic functions.

Similarly, results for $r_{13}(n), r_{15}(n), \ldots,$ for general n can be given, but none are as simple as the ones stated above for $r_3(n), r_5(n), r_7(n), r_9(n), r_{11}(n)$.

The values of $r_3(n')$, $r_5(n')$ and $r_7(n')$ can be expressed as a finite sum involving Legendre and Jacobi symbols, [11, p. 315]. For nine and eleven squares I conjecture that, for n' squarefree,

$$r_9(n') = \frac{4320}{17}\sum_{j=1}^{(n'-1)/2}\left(\frac{j}{n'}\right)j^2(4j-3n'), \quad \text{if } n' \equiv 5 \pmod 8,$$

$$r_{11}(n') = \frac{31680}{31}\sum_{j=1}^{(n'-1)/2}\left(\frac{j}{n'}\right)j^3(j-n'), \quad \text{if } n' \equiv 7 \pmod 8.$$

The methods used in this article are different from those of Hardy [9, 10], Minkowski [20] and Smith [24]. Our results for sums of nine squares are new, and are different from those given by Lomadze [18, 19].

This article is organised as follows. The main definitions are given in Section 2, and Section 3 contains some lemmas. The values of $r_5(p^2)$, $r_7(p^2)$ and $r_9(p^2)$ are computed in Section 4. The main results for $r_5(n)$ and $r_7(n)$ are given in Section 5. The details for $r_5(n)$ and $r_7(n)$ are similar, because the spaces $M_{5/2}(\tilde\Gamma_0(4))$ and $M_{7/2}(\tilde\Gamma_0(4))$ are both two dimensional. The results for $r_9(n)$ are developed in Section 6. The details are slightly more complicated because $\operatorname{Dim} M_{9/2}(\tilde\Gamma_0(4)) = 3$, and $\sum_{n=0}^{\infty} r_9(n)q^n$ is not an eigenfunction of the Hecke operator T_{p^2}.

2. Definitions

Let

$$\phi(q) = \sum_{j=-\infty}^{\infty} q^{j^2},$$
$$\psi(q) = \sum_{j=0}^{\infty} q^{j(j+1)/2},$$
$$\phi_{s,t}(q) = \phi(q)^s (2q^{1/4}\psi(q^2))^t,$$
$$z = \phi(q)^2,$$
$$x = \frac{16q\psi(q^2)^4}{\phi(q)^4}.$$

If t is a multiple of 4 then we may write

$$\phi_{s,t}(q) = \sum_{n=0}^{\infty} \phi_{s,t}(n) q^n,$$

so that $\phi_{s,t}(n)$ is the coefficient of q^n in the series expansion of $\phi_{s,t}(q)$. Geometrically, $\phi_{s,t}(n)$ counts the number of lattice points, that is, points with integer coordinates, on the sphere

$$\sum_{j=1}^{s} x_j^2 + \sum_{j=s+1}^{s+t} (x_j - 1/2)^2 = n.$$

Let $r_{s,t}(n)$ denote the number of representations of n as a sum of $s+t$ squares, of which s are even and t are odd. Then

$$\sum_{n=0}^{\infty} r_{s,t}(n) q^n = \binom{s+t}{t} \phi(q^4)^s (2q\psi(q^8))^t = \binom{s+t}{t} \phi_{s,t}(q^4)$$

and

$$r_{s,0}(4n) = r_s(n).$$

Glaisher's number theoretic function $\theta(n)$ is defined by

$$\frac{1}{16} z^8 x(1-x) = \sum_{n=1}^{\infty} \theta(n) q^n. \tag{2.1}$$

Some combinatorial interpretations of $\theta(n)$ were given by Glaisher [8].

A sequence $\{a(n)\}$, $n = 0, 1, 2, \ldots$, is said to be multiplicative if $a(mn) = a(m)a(n)$ for all positive integers m and n satisfying g.c.d.$(m, n) = 1$. Observe that a multiplicative sequence satisfies $a(1) = 0$ or 1, while $a(0)$ may have any value.

3. Some lemmas

We will require some facts about modular forms of half integer weight:

Lemma 3.1 [17, p. 184, Prop. 4]. *$M_{(2k+1)/2}(\tilde{\Gamma}_0(4))$ is the vector space consisting of all linear combinations of $\phi_{2k+1-4j,4j}(q)$, $j = 0, 1, \ldots, \lfloor k/2 \rfloor$.*

Lemma 3.2 [17, p. 206]. *The Hecke operators T_{p^2}, where p is any prime, map $M_{(2k+1)/2}(\tilde{\Gamma}_0(4))$ into itself. That is,*

$$f \in M_{(2k+1)/2}(\tilde{\Gamma}_0(4)) \Rightarrow T_{p^2} f \in M_{(2k+1)/2}(\tilde{\Gamma}_0(4)).$$

Lemma 3.3 [17, p. 207, Prop. 13 and page 210]. *Suppose*

$$f(q) = \sum_{n=0}^{\infty} a(n) q^n \in M_{(2k+1)/2}(\tilde{\Gamma}_0(4)).$$

If p is an odd prime then

$$T_{p^2} f = \sum_{n=0}^{\infty} a(p^2 n) q^n + p^{k-1} \sum_{n=0}^{\infty} \left(\frac{(-1)^k n}{p} \right) a(n) q^n + p^{2k-1} \sum_{n=0}^{\infty} a(n/p^2) q^n.$$

For $p = 2$ we have

$$T_{2^2} f = \sum_{n=0}^{\infty} a(4n) q^n.$$

Here $a(n/p^2)$ is defined to be zero if n/p^2 is not an integer, and $(\frac{n}{p})$ is the Legendre symbol.

We shall also need the following.

Lemma 3.4. *If n is a positive integer and p is an odd prime, then*

$$r_4(n) = 8 \sum_{d \mid n, 4 \nmid d} d \tag{3.1}$$

$$= \begin{cases} 8 \displaystyle\prod_{p \text{ odd}} \frac{p^{\alpha_p+1} - 1}{p - 1} & \text{if } n \text{ is odd} \\ 24 \displaystyle\prod_{p \text{ odd}} \frac{p^{\alpha_p+1} - 1}{p - 1} & \text{if } n \text{ is even} \end{cases} \tag{3.2}$$

$$r_4(p) = 8(p+1) \tag{3.3}$$

$$\frac{r_4(mn)}{8} = \frac{r_4(n)}{8} \frac{r_4(m)}{8} \tag{3.4}$$

$$r_8(n) = 16(-1)^n \sum_{d \mid n} (-1)^d d^3 \tag{3.5}$$

$$r_8(p) = 16(p^3 + 1). \tag{3.6}$$

Proof: Equations (3.1) and (3.5) are essentially due to Jacobi [16, p. 160, Eq. (8) and p. 170, Eq. (8)]. See [5] for simple proofs. Equations (3.3) and (3.6) follow right away from (3.1) and (3.5), respectively. Equation (3.2) follows from (3.1), and seems to have been first written down by Ramanujan [23, p. 305, Eqs. (387) and (388)]. Equation (3.4) follows from (3.2). □

Lemma 3.5. *Let p be an odd prime, and let $a(n)$, $b(n)$, $c(n)$ and $d(n)$ be defined by*

$$\frac{z^3x}{16} = \sum_{n=0}^{\infty} a(n)q^n$$

$$-\frac{z^3(1-x)}{4} = \sum_{n=0}^{\infty} b(n)q^n$$

$$\frac{z^4(1-x^2)}{16} = \sum_{n=0}^{\infty} c(n)q^n$$

$$-\frac{z^4(1-x)^2}{16} = \sum_{n=0}^{\infty} d(n)q^n.$$

Then

$$\begin{aligned} &a(0) = 0, && b(0) = -\frac{1}{4}, && c(0) = \frac{1}{16}, && d(0) = -\frac{1}{16},\\ &a(p) = p^2 + (-1)^{(p-1)/2}, && b(p) = 1 + (-1)^{(p-1)/2}p^2, && c(p) = p^3 + 1, && d(p) = p^3 + 1; \end{aligned}$$

and each of $a(n)$, $b(n)$, $c(n)$ and $d(n)$ are multiplicative.

Proof: From Entry 17(ii) of Chapter 17 of [3] or [22], we have

$$\frac{z^3x}{16} = \sum_{k=1}^{\infty} \frac{k^2q^k}{1+q^{2k}}.$$

Therefore $a(0) = 0$, $a(1) = 1$, and for $n \geq 2$ we have

$$\begin{aligned} a(n) &= \sum_{k(2j+1)=n} (-1)^j k^2 \\ &= 2^{2\lambda} \prod_{p\equiv 1 \pmod 4} \frac{p^{2\lambda_p+2}-1}{p^2-1} \prod_{p\equiv 3 \pmod 4} \frac{p^{2\lambda_p+2}+(-1)^{\lambda_p}}{p^2+1}. \end{aligned}$$

Multiplicativity and the value of $a(p)$ follow immediately from this.

The results for $b(n)$, $c(n)$ and $d(n)$ follow similarly from Entries 17(vii), 14(ii) and 14(v), respectively, of Chapter 17 of [3] or [22]. □

Lemma 3.6.

$$z^6(2x-1) = -1 + 8\sum_{k=1}^{\infty}\frac{(-1)^{k-1}k^5q^k}{1+q^k} + 512\sum_{k=1}^{\infty}\frac{k^5q^{2k}}{1-q^{4k}}$$
$$z^8(1+14x+x^2)^2 = 1 + 480\sum_{k=1}^{\infty}\frac{k^7q^k}{1-q^k}$$

Proof: The first part follows by combining Entries 14(iii) and 15(vi) of Chapter 17 of [3] or [22]. The second part follows by combining Entry 12(ii) of Chapter 15 of [2] or [22] with Entry 13(iii) of Chapter 17 of [3] or [22]. □

Lemma 3.7. *The q-series expansion of $z^4x(2-x)$ contains only odd powers of q, and the q-series expansion of $z^8(2192-4384x+6606x^2-4414x^3+2192x^4)$ contains only even powers.*

Proof: Observe that under the change of variables

$$z \to z(1-x)^{1/2}, \quad x \to \frac{x}{x-1},$$

$z^4x(2-x)$ changes sign, while $z^8(2192-4384x+6606x^2-4414x^3+2192x^4)$ is invariant. The truth of the lemma therefore follows by the Change of Sign Principle [3, p. 126] or [22, Ch. 17, Entry 14(xii)]. □

4. The value of $r_{2k+1}(p^2)$

In this section, we give a simple proof of Stieltjes' conjectured formula for the value of $r_5(p^2)$ for odd primes p. Our proof is different from the one given by Hurwitz [13, 15, pp. 5–7]. Similar results for the values of $r_7(p^2)$ and $r_9(p^2)$ will also be given.

Theorem 4.1. *Let p be an odd prime. Then*

$$r_5(p^2) = 10(p^3-p+1)$$
$$r_7(p^2) = 14\big(p^5-(-1)^{(p-1)/2}p^2+1\big)$$
$$r_9(p^2) = \frac{274}{17}(p^7+1) - 18p^3 + \frac{32}{17}\theta(p)$$

Proof: Since p is odd, $p^2 \equiv 1 \pmod 8$. It follows that the only integer solutions $(x_1, x_2, x_3, x_4, x_5)$ of

$$p^2 = x_1^2+x_2^2+x_3^2+x_4^2+x_5^2$$

occur when one of the x_i is odd and the other four are even. Therefore

$$\begin{aligned} r_5(p^2) &= 10\sum_{i\text{ odd}} r_{4,0}(p^2-i^2)\\ &= 10\sum_{i\text{ odd}} r_4((p-i)(p+i)/4)\end{aligned}$$

$$= 10 \sum_{j=0}^{(p-1)/2} r_4(j(p-j))$$
$$= 5 \sum_{j=0}^{p} r_4(j(p-j))$$
$$= 10 + 5 \sum_{j=1}^{p-1} r_4(j(p-j)).$$

Since $(j, p-j) = 1$, Lemma 3.4 implies

$$\begin{aligned} r_5(p^2) &= 10 + \frac{5}{8} \sum_{j=1}^{p-1} r_4(j) r_4(p-j) \\ &= 10 - \frac{5}{4} r_4(p) + \frac{5}{8} \sum_{j=0}^{p} r_4(j) r_4(p-j) \\ &= 10 - \frac{5}{4} r_4(p) + \frac{5}{8} r_8(p) \\ &= 10 - 10(p+1) + 10(p^3+1) \\ &= 10(p^3 - p + 1). \end{aligned}$$

This proves the first part of the theorem.

The remaining parts are similar apart from some extra details. In order to compute $r_7(p^2)$, let us write

$$p^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2 + x_7^2.$$

Since p is odd, this implies that either one or five of the x_i are odd, and the others are all even. Accordingly,

$$\begin{aligned} r_7(p^2) &= 14 \sum_{i \text{ odd}} r_{6,0}(p^2 - i^2) + \frac{14}{5} \sum_{i \text{ odd}} r_{2,4}(p^2 - i^2) \\ &= \sum_{i \text{ odd}} \left[q^{p^2 - i^2}\right] \left\{ 14\phi(q^4)^6 + \frac{14}{5} \times \binom{6}{4} \times 16 q^4 \phi(q^4)^2 \psi(q^8)^4 \right\} \\ &= 2 \sum_{i \text{ odd}} \left[q^{(p^2 - i^2)/4}\right] \left\{ \binom{7}{1} \phi(q)^6 + \binom{7}{5} \times 16 q \phi(q)^2 \psi(q^2)^4 \right\} \\ &= \sum_{j=0}^{p} \left[q^{j(p-j)}\right] z^3 \left\{ \binom{7}{1} + \binom{7}{5} x \right\} \\ &= \sum_{j=0}^{p} \left[q^{j(p-j)}\right] z^3 (7 + 21x). \end{aligned}$$

In general we have

$$r_{2k+1}(p^2) = \sum_{j=0}^{p} \left[q^{j(p-j)}\right] z^k \left(\sum_{i=0}^{\lfloor k/2 \rfloor} \binom{2k+1}{4i+1} x^i \right). \tag{4.1}$$

Now

$$z^3(7+21x) = 448\left(\frac{z^3x}{16}\right) + 28\left(\frac{z^3(1-x)}{4}\right) = 448\sum_{j=0}^{\infty} a(j)q^j - 28\sum_{j=0}^{\infty} b(j)q^j.$$

Using Lemma 3.5, this gives

$$\begin{aligned}
r_7(p^2) &= 448\sum_{j=0}^{p} a(j(p-j)) - 28\sum_{j=0}^{p} b(j(p-j)) \\
&= 896a(0) - 56b(0) + 448\sum_{j=1}^{p-1} a(j(p-j)) - 28\sum_{j=1}^{p-1} b(j(p-j)) \\
&= 896a(0) - 56b(0) + 448\sum_{j=1}^{p-1} a(j)a(p-j) - 28\sum_{j=1}^{p-1} b(j)b(p-j) \\
&= 896a(0) - 56b(0) - 896a(0)a(p) + 56b(0)b(p) \\
&\quad + 448\sum_{j=0}^{p} a(j)a(p-j) - 28\sum_{0}^{p} b(j)b(p-j) \\
&= 896a(0) - 56b(0) - 896a(0)a(p) + 56b(0)b(p) \\
&\quad + 448[q^p]\frac{z^6x^2}{256} - 28[q^p]\frac{z^6(1-x)^2}{16} \\
&= -14(-1)^{(p-1)/2}p^2 + \frac{7}{4}[q^p]z^6(2x-1).
\end{aligned}$$

From Lemma 3.6 we have

$$z^6(2x-1) = -1 + 8\sum_{k=1}^{\infty} \frac{(-1)^{k-1}k^5q^k}{1+q^k} + 512\sum_{k=1}^{\infty} \frac{k^5q^{2k}}{1-q^{4k}},$$

and it is readily checked that for any odd prime p, the coefficient of q^p on the right hand side is $8p^5+8$. Substituting this into the previous expression for $r_7(p^2)$ gives

$$r_7(p^2) = 14(-1)^{(p+1)/2}p^2 + \frac{7}{4}(8p^5+8) = 14\left(p^5 - (-1)^{(p-1)/2}p^2 + 1\right).$$

This completes the proof of the second part of the theorem.

Lastly, starting with (4.1) with $k=4$ we have

$$r_9(p^2) = \sum_{j=0}^{p} \left[q^{j(p-j)}\right] z^4(9+126x+x^2).$$

Now

$$z^4(9+126x+x^2) = -1024\frac{z^4(1-x^2)}{16} + 1168\frac{z^4(1-x)^2}{16} + 136z^4x(2-x).$$

Using Lemmas 3.5 and 3.7 we obtain

$$\begin{aligned}
r_9(p^2) &= -1024\sum_{j=0}^{p} c(j(p-j)) - 1168\sum_{j=0}^{p} d(j(p-j)) + 0 \\
&= -2048c(0) - 2336d(0) - 1024\sum_{j=1}^{p-1} c(j(p-j)) - 1168\sum_{j=1}^{p-1} d(j(p-j)) \\
&= -2048c(0) - 2336d(0) - 1024\sum_{j=1}^{p-1} c(j)c(p-j) - 1168\sum_{j=1}^{p-1} d(j)d(p-j) \\
&= -2048c(0) - 2336d(0) + 2048c(0)c(p) + 2336d(0)d(p) \\
&\quad - 1024[q^p]\frac{z^8(1-x^2)^2}{256} - 1168[q^p]\frac{z^8(1-x)^4}{256} \\
&= -18p^3 - [q^p]\frac{z^8}{16}(137 - 292x + 310x^2 - 292x^3 + 137x^4).
\end{aligned}$$

Now

$$\begin{aligned}
&-\frac{z^8}{16}(137 - 292x + 310x^2 - 292x^3 + 137x^4) \\
&\quad = \frac{137}{4080}z^8(x^2+14x+1)^2 + \frac{2}{17}z^8x(1-x) - z^8p(x)
\end{aligned}$$

where

$$p(x) = \frac{1}{255}[2192 - 4384x + 6606x^2 - 4414x^3 + 2192x^4].$$

By Lemmas 3.6 and 3.7 together with Eq. (2.1) we further obtain

$$\begin{aligned}
r_9(p^2) &= -18p^3 + \frac{137}{4080}[q^p]z^8(x^2+14x+1)^2 + \frac{2}{17}[q^p]z^8x(1-x) + 0 \\
&= -18p^3 + \frac{274}{17}(p^7+1) + \frac{32}{17}\theta(p).
\end{aligned}$$

This completes the proof of the theorem. $\square$

5. Sums of five and seven squares

Lemma 5.1. *Let p be an odd prime. Then*

$$r_5(p^2n) = \left[p^3 - p\left(\frac{n}{p}\right) + 1\right]r_5(n) - p^3r_5(n/p^2)$$
$$r_7(p^2n) = \left[p^5 - p^2\left(\frac{-n}{p}\right) + 1\right]r_7(n) - p^5r_7(n/p^2).$$

Proof: Lemmas 3.1–3.3 imply

$$T_{p^2}(\phi(q)^5) = c_1\phi(q)^5 + c_2\phi(q)(16q\psi(q^2)^4)$$
$$T_{p^2}(\phi(q)^5) = \sum_{n=0}^{\infty} r_5(p^2n)q^n + p\sum_{n=0}^{\infty}\left(\frac{n}{p}\right) r_5(n)q^n + p^3\sum_{n=0}^{\infty} r_5(n/p^2)q^n$$

for some constants c_1 and c_2. Comparing the constant term and coefficient of q gives

$$p^3 + 1 = c_1,$$
$$r_5(p^2) + pr_5(1) = c_1r_5(1) + 16c_2.$$

Now $r_5(1) = 10$ and the value of $r_5(p^2)$ is given by Theorem 4.1. These imply

$$16c_2 = r_5(p^2) - r_5(1)\left(p^3 - p + 1\right) = 10(p^3 - p + 1) - 10(p^3 - p + 1) = 0.$$

Therefore

$$\sum_{n=0}^{\infty} r_5(p^2n)q^n + p\sum_{n=0}^{\infty}\left(\frac{n}{p}\right) r_5(n)q^n + p^3\sum_{n=0}^{\infty} r_5(n/p^2)q^n = (1+p^3)\sum_{n=0}^{\infty} r_5(n)q^n.$$

Comparing the coeffients of q^n on both sides gives

$$r_5(p^2n) = \left[p^3 - p\left(\frac{n}{p}\right) + 1\right] r_5(n) - p^3r_5(n/p^2),$$

as required. This completes the proof of the first part.

Next, Lemmas 3.1–3.3 imply that

$$T_{p^2}(\phi(q)^7) = d_1\phi(q)^7 + d_2\phi(q)^3(16q\psi(q^2)^4)$$
$$T_{p^2}(\phi(q)^7) = \sum_{n=0}^{\infty} r_7(p^2n)q^n + p^2\sum_{n=0}^{\infty}\left(\frac{-n}{p}\right) r_7(n)q^n + p^5\sum_{n=0}^{\infty} r_7(n/p^2)q^n$$

for some constants d_1 and d_2. Comparing the constant term and coefficient of q gives

$$p^5 + 1 = d_1,$$
$$r_7(p^2) + p^2\left(\frac{-1}{p}\right) r_7(1) = d_1r_7(1) + 16d_2.$$

Now $r_7(1) = 14$ and the value of $r_7(p^2)$ is given by Theorem 4.1. Consequently we find

$$16d_2 = r_7(p^2) - r_7(1)\left(p^5 + (-1)^{(p+1)/2}p^2 + 1\right) = 0.$$

Therefore

$$\sum_{n=0}^{\infty} r_7(p^2n)q^n + p^2\sum_{n=0}^{\infty}\left(\frac{-n}{p}\right) r_7(n)q^n + p^5\sum_{n=0}^{\infty} r_7(n/p^2)q^n = (1+p^5)\sum_{n=0}^{\infty} r_7(n)q^n.$$

Comparing the coeffients of q^n on both sides gives

$$r_7(p^2n) = \left[p^5 - p^2\left(\frac{-n}{p}\right) + 1\right] r_7(n) - p^5 r_7(n/p^2),$$

as required. □

Remark 5.2. The fact that $c_2 = 0$ implies that for the space $M_{5/2}(\tilde{\Gamma}_0(4))$, $\phi(q)^5$ is an eigenfunction of the Hecke operator T_{p^2} with corresponding eigenvalue $p^3 + 1$. Similarly, $d_2 = 0$ implies that for the space $M_{7/2}(\tilde{\Gamma}_0(4))$, $\phi(q)^7$ is an eigenfunction of T_{p^2} with corresponding eigenvalue $p^5 + 1$.

Theorem 5.3. *If p is an odd prime and $p^2 \nmid n$ then*

$$r_5(p^{2\lambda}n) = \left[\frac{p^{3\lambda+3} - 1}{p^3 - 1} - p\left(\frac{n}{p}\right)\frac{p^{3\lambda} - 1}{p^3 - 1}\right] r_5(n)$$
$$r_7(p^{2\lambda}n) = \left[\frac{p^{5\lambda+5} - 1}{p^5 - 1} - p^2\left(\frac{-n}{p}\right)\frac{p^{5\lambda} - 1}{p^5 - 1}\right] r_7(n).$$

Proof: Use Lemma 5.1 and induction on λ. □

Analysis for the prime 2 is more complicated. We will need

Lemma 5.4.

$$r_5(4n) = \begin{cases} 9r_5(n) & \text{if } n \equiv 1 \pmod 8 \\ 5r_5(n) & \text{if } n \equiv 2 \text{ or } 3 \pmod 4 \\ 47r_5(n)/7 & \text{if } n \equiv 5 \pmod 8 \end{cases}$$
$$r_7(4n) = \begin{cases} 41r_7(n) & \text{if } n \equiv 1 \text{ or } 2 \pmod 4 \\ 33r_7(n) & \text{if } n \equiv 3 \pmod 8 \\ 1157r_7(n)/37 & \text{if } n \equiv 7 \pmod 8 \end{cases}$$
$$r_5(16n) = 9r_5(4n) - 8r_5(n)$$
$$r_7(16n) = 33r_7(4n) - 32r_7(n).$$

Proof: The first two parts have been proved by Barrucand and Hirschhorn [1]; the special case in which n is square-free was proved earlier by Cohen [4]. We will prove the last two parts. By Lemmas 3.1–3.3 we have

$$T_4(\phi(q)^5) = c_1\phi(q)^5 + c_2\phi(q)(16q\psi(q^2)^4),$$

for some constants c_1 and c_2, and

$$T_4(\phi(q)^5) = \sum_{n=0}^{\infty} r_5(4n)q^n.$$

Equating the two equations and comparing coefficients yields $c_1 = 1, c_2 = 80$. Next, applying T_4 again gives

$$T_4\left(\sum_{n=0}^{\infty} r_5(4n)q^n\right) = \sum_{n=0}^{\infty} r_5(16n)q^n,$$

and

$$T_4\left(\sum_{n=0}^{\infty} r_5(4n)q^n\right) = d_1\phi(q)^5 + d_2\phi(q)(16q\psi(q^2)^4),$$

for some constants d_1 and d_2. Equating the last two equations and comparing coefficients gives $d_1 = 1$, $d_2 = 720$. Thus we have the pair of equations

$$\begin{aligned}\sum_{n=0}^{\infty} r_5(4n)q^n &= \sum_{n=0}^{\infty} r_5(n)q^n + 80\phi(q)(16q\psi(q^2)^4),\\ \sum_{n=0}^{\infty} r_5(16n)q^n &= \sum_{n=0}^{\infty} r_5(n)q^n + 720\phi(q)(16q\psi(q^2)^4).\end{aligned}$$

Eliminating the $\phi(q)(16q\psi(q^2)^4)$ terms gives

$$\sum_{n=0}^{\infty} r_5(16n)q^n = 9\sum_{n=0}^{\infty} r_5(4n)q^n - 8\sum_{n=0}^{\infty} r_5(n)q^n,$$

as required.

For seven squares, apply the Hecke operator T_4 to $\phi(q)^7$, and use Lemmas 3.1–3.3 to get

$$T_4(\phi(q)^7) = c_1\phi(q)^7 + c_2\phi(q)^3(16q\psi(q^2)^4),$$

for some constants c_1 and c_2, and

$$T_4(\phi(q)^7) = \sum_{n=0}^{\infty} r_7(4n)q^n.$$

Equating the two equations and comparing coefficients yields $c_1 = 1$, $c_2 = 560$. Next, applying T_4 again gives

$$T_4\left(\sum_{n=0}^{\infty} r_7(4n)q^n\right) = \sum_{n=0}^{\infty} r_7(16n)q^n,$$

and

$$T_4\left(\sum_{n=0}^{\infty} r_7(4n)q^n\right) = d_1\phi(q)^7 + d_2\phi(q)^3(16q\psi(q^2)^4),$$

for some constants d_1 and d_2. Equating the last two equations and comparing coefficients gives $d_1 = 1$, $d_2 = 18480$. Thus we have the pair of equations

$$\sum_{n=0}^{\infty} r_7(4n)q^n = \sum_{n=0}^{\infty} r_7(n)q^n + 560\phi(q)^3(16q\psi(q^2)^4),$$
$$\sum_{n=0}^{\infty} r_7(16n)q^n = \sum_{n=0}^{\infty} r_7(n)q^n + 18480\phi(q)^3(16q\psi(q^2)^4).$$

Eliminating the $\phi(q)^3(16q\psi(q^2)^4)$ terms gives

$$\sum_{n=0}^{\infty} r_7(16n)q^n = 33\sum_{n=0}^{\infty} r_7(4n)q^n - 32\sum_{n=0}^{\infty} r_7(n)q^n. \qquad \square$$

Stieltjes [25], [26], [27, pp. 329–331, 360–361] stated, without proof, results equivalent to

Theorem 5.5. *If n is not divisible by 4 then*

$$r_5(2^{2\lambda}n) = \left[\frac{2^{3\lambda+3}-1}{2^3-1} - \epsilon_5(n)\frac{2^{3\lambda}-1}{2^3-1}\right] r_5(n)$$
$$r_7(2^{2\lambda}n) = \left[\frac{2^{5\lambda+5}-1}{2^5-1} - \epsilon_7(n)\frac{2^{5\lambda}-1}{2^5-1}\right] r_7(n),$$

where

$$\epsilon_5(n) = \begin{cases} 0 & \text{if } n \equiv 1 \pmod 8 \\ 4 & \text{if } n \equiv 2 \text{ or } 3 \pmod 4 \\ 16/7 & \text{if } n \equiv 5 \pmod 8, \end{cases}$$
$$\epsilon_7(n) = \begin{cases} 0 & \text{if } n \equiv 3 \pmod 8 \\ -8 & \text{if } n \equiv 1 \text{ or } 2 \pmod 4 \\ 64/37 & \text{if } n \equiv 7 \pmod 8. \end{cases}$$

Proof: These results follow by induction using the previous lemma. $\square$

Combining Theorems 5.3 and 5.5 we get

Theorem 5.6.

$$r_5(n) = r_5(n')\left[\frac{2^{3\lfloor\lambda/2\rfloor+3}-1}{2^3-1} - \epsilon_5(n')\,\frac{2^{3\lfloor\lambda/2\rfloor}-1}{2^3-1}\right]$$
$$\times \prod_p \left[\frac{p^{3\lfloor\lambda_p/2\rfloor+3}-1}{p^3-1} - p\left(\frac{n'}{p}\right)\frac{p^{3\lfloor\lambda_p/2\rfloor}-1}{p^3-1}\right]$$

$$r_7(n) = r_7(n')\left[\frac{2^{5\lfloor\lambda/2\rfloor+5}-1}{2^5-1} - \epsilon_7(n')\frac{2^{5\lfloor\lambda/2\rfloor}-1}{2^5-1}\right] \times \prod_p \left[\frac{p^{5\lfloor\lambda_p/2\rfloor+5}-1}{p^5-1} - p^2\left(\frac{-n'}{p}\right)\frac{p^{5\lfloor\lambda_p/2\rfloor}-1}{p^5-1}\right],$$

where the functions ϵ_5 and ϵ_7 are the same as in Theorem 5.5.

Corollary 5.7.

$$r_5(n^2) = 10\left[\frac{2^{3\lambda+3}-1}{2^3-1}\right]\prod_p\left[\frac{p^{3\lambda_p+3}-1}{p^3-1} - p\frac{p^{3\lambda_p}-1}{p^3-1}\right]$$

$$r_7(n^2) = 14\left[\frac{2^{5\lambda+5}-1}{2^5-1} + 8\frac{2^{5\lambda}-1}{2^5-1}\right]\prod_p\left[\frac{p^{5\lambda_p+5}-1}{p^5-1} - (-1)^{(p-1)/2}p^2\frac{p^{5\lambda_p}-1}{p^5-1}\right].$$

Corollary 5.8. *The sequences $r_5(n^2)/10$ and $r_7(n^2)/14$ are multiplicative.*

6. Sums of nine squares

The analysis in this section is similar to that in the previous section. However $\phi(q)^9$ is not an eigenfunction of the Hecke operator T_{p^2}, so there are some extra details.

6.1. The prime 2

Theorem 6.1.

$$\begin{aligned} T_4\phi_{9,0} &= \phi_{9,0} + 126\phi_{5,4} + 9\phi_{1,8} \\ T_4\phi_{5,4} &= \qquad\quad 120\phi_{5,4} + 8\phi_{1,8} \\ T_4\phi_{1,8} &= \qquad\quad 128\phi_{5,4}. \end{aligned}$$

That is,

$$\begin{aligned} \phi_{9,0}(4n) &= \phi_{9,0}(n) + 126\phi_{5,4}(n) + 9\phi_{1,8}(n) \\ \phi_{5,4}(4n) &= \qquad\qquad 120\phi_{5,4}(n) + 8\phi_{1,8}(n) \\ \phi_{1,8}(4n) &= \qquad\qquad 128\phi_{5,4}(n). \end{aligned}$$

Proof: Let us consider $\phi_{9,0}$ first. By Lemmas 3.1 and 3.2, we have

$$T_4\phi_{9,0} = \alpha_1\phi_{9,0} + \alpha_2\phi_{5,4} + \alpha_3\phi_{1,8},$$

for some constants α_1, α_2, α_3. By Lemma 3.3, we have

$$T_4\phi_{9,0} = \sum_{n=0}^{\infty}\phi_{9,0}(4n)q^n.$$

Therefore

$$\sum_{n=0}^{\infty} \phi_{9,0}(4n)q^n = \alpha_1 \sum_{n=0}^{\infty} \phi_{9,0}(n)q^n + \alpha_2 \sum_{n=0}^{\infty} \phi_{5,4}(n)q^n + \alpha_3 \sum_{n=0}^{\infty} \phi_{1,8}(n)q^n.$$

Equating coefficients of 1, q and q^2 gives

$$\begin{aligned} \alpha_1 &= 1 \\ 18\alpha_1 + 16\alpha_2 &= 2034 \\ 144\alpha_1 + 160\alpha_2 + 256\alpha_3 &= 22608 \end{aligned}$$

and so

$$\alpha_1 = 1, \quad \alpha_2 = 126, \quad \alpha_3 = 9.$$

This proves the first part of the theorem. The results for $T_4\phi_{5,4}$ and $T_4\phi_{1,8}$ follow similarly. □

Corollary 6.2. *The eigenfunctions and eigenvalues of T_4 are*

$$\begin{aligned} w_1 &:= 127\phi_{9,0} - 142\phi_{5,4} + 7\phi_{1,8}, & \lambda_1 &= 1, \\ w_2 &:= 16\phi_{5,4} + \phi_{1,8}, & \lambda_2 &= 128, \\ w_3 &:= \phi_{5,4} - \phi_{1,8}, & \lambda_3 &= -8. \end{aligned}$$

Let us write

$$w_j = w_j(q) = \sum_{n=0}^{\infty} w_j(n)q^n$$

so that $w_j(n)$ is the coefficient of q^n in the series expansion of $w_j(q)$, $j = 1,\ 2,\ 3$.

Theorem 6.3.

$$\begin{aligned} w_3(n) &= 0 & &\textit{if } n = 4^\lambda(8k+5) \textit{ for some nonnegative integers } \lambda \textit{ and } k \\ 16w_1(n) + 15w_2(n) &= 0 & &\textit{if } n \equiv 2 \textit{ or } 3 \pmod 4 \\ 128w_1(n) - 7w_2(n) &= 0 & &\textit{if } n \equiv 1 \pmod 8 \\ 2176w_1(n) + 135w_2(n) &= 0 & &\textit{if } n \equiv 5 \pmod 8. \end{aligned}$$

A proof of Theorem 6.3 has been given by Barrucand and Hirschhorn [1].

As a consequence of the previous results we have

Theorem 6.4.

$$r_9(4^\lambda(8k+5)) = \frac{2^{7\lambda+7}-1}{2^7-1} r_9(8k+5).$$

Proof:

$$\begin{aligned}
&r_9(4^\lambda(8k+5))\\
&\quad = \phi_{9,0}(4^\lambda(8k+5))\\
&\quad = \frac{1}{17\times 127}(17w_1(4^\lambda(8k+5)) + 135w_2(4^\lambda(8k+5)) + 254w_3(4^\lambda(8k+5)))\\
&\quad = \frac{1}{17\times 127}(17w_1(8k+5) + 135\times 128^\lambda w_2(8k+5) + 254\times(-8)^\lambda w_3(8k+5)).
\end{aligned}$$

Applying Theorem 6.3 gives

$$\begin{aligned}
r_9(4^\lambda(8k+5)) &= \frac{1}{17\times 127}(17w_1(8k+5) - 2176\times 128^\lambda w_1(8k+5) + 0)\\
&= \frac{-1}{127}(128^{\lambda+1}-1)w_1(8k+5).
\end{aligned}$$

Taking $\lambda = 0$ in this gives

$$r_9(8k+5) = -w_1(8k+5)$$

and so

$$r_9(4^\lambda(8k+5)) = \frac{1}{127}(128^{\lambda+1}-1)r_9(8k+5) = \frac{2^{7\lambda+7}-1}{2^7-1}r_9(8k+5),$$

as required. □

6.2. The case p an odd prime

Theorem 6.5. *Let*

$$\alpha = p^7+1, \quad \beta = \theta(p),$$

Then

$$\begin{pmatrix} T_{p^2}\phi_{9,0}\\ T_{p^2}\phi_{5,4}\\ T_{p^2}\phi_{1,8}\end{pmatrix} = \frac{1}{17}\begin{pmatrix} 17\alpha & -2\alpha+2\beta & 2\alpha-2\beta\\ 0 & 16\alpha+\beta & \alpha-\beta\\ 0 & 16\alpha-16\beta & \alpha+16\beta\end{pmatrix}\begin{pmatrix}\phi_{9,0}\\ \phi_{5,4}\\ \phi_{1,8}\end{pmatrix}.$$

Proof: By Lemmas 3.2 and 3.3 we have

$$\begin{aligned}
T_{p^2}\phi_{9,0} &= \sum_{n=0}^{\infty}\phi_{9,0}(p^2n)q^n + p^3\sum_{n=0}^{\infty}\left(\frac{n}{p}\right)\phi_{9,0}(n)q^n + p^7\sum_{n=0}^{\infty}\phi_{9,0}(n/p^2)q^n\\
&= c_1\phi_{9,0}(q) + c_2\phi_{5,4}(q) + c_3\phi_{1,8}(q), && (6.1)\\
T_{p^2}\phi_{5,4} &= \sum_{n=0}^{\infty}\phi_{5,4}(p^2n)q^n + p^3\sum_{n=0}^{\infty}\left(\frac{n}{p}\right)\phi_{5,4}(n)q^n + p^7\sum_{n=0}^{\infty}\phi_{5,4}(n/p^2)q^n\\
&= d_1\phi_{9,0}(q) + d_2\phi_{5,4}(q) + d_3\phi_{1,8}(q), && (6.2)
\end{aligned}$$

$$
\begin{aligned}
T_{p^2}\phi_{1,8} &= \sum_{n=0}^{\infty}\phi_{1,8}(p^2n)q^n + p^3\sum_{n=0}^{\infty}\left(\frac{n}{p}\right)\phi_{1,8}(n)q^n + p^7\sum_{n=0}^{\infty}\phi_{1,8}(n/p^2)q^n \\
&= e_1\phi_{9,0}(q) + e_2\phi_{5,4}(q) + e_3\phi_{1,8}(q),
\end{aligned} \tag{6.3}
$$

for some constants c_1, c_2, c_3, d_1, d_2, d_3, e_1, e_2, e_3. Equating the constant terms in each of the three equations above gives

$$
\begin{aligned}
c_1 &= p^7 + 1 \\
d_1 &= 0 \\
e_1 &= 0.
\end{aligned}
$$

Equating the coefficients of q in (6.1)–(6.3) gives

$$
\begin{aligned}
18c_1 + 16c_2 &= \phi_{9,0}(p^2) + 18p^3 \\
18d_1 + 16d_2 &= \phi_{5,4}(p^2) + 16p^3 \\
18e_1 + 16e_2 &= \phi_{1,8}(p^2).
\end{aligned}
$$

From Theorem 4.1 we have

$$
\phi_{9,0}(p^2) = \frac{274}{17}(p^7+1) - 18p^3 + \frac{32}{17}\theta(p). \tag{6.4}
$$

By the same methods it can be shown that

$$
\phi_{5,4}(p^2) = \frac{256}{17}(p^7+1) - 16p^3 + \frac{16}{17}\theta(p) \tag{6.5}
$$

$$
\phi_{1,8}(p^2) = \frac{256}{17}(p^7+1) - \frac{256}{17}\theta(p)). \tag{6.6}
$$

It follows that

$$
\begin{aligned}
c_2 &= -\frac{2}{17}(p^7+1) + \frac{2}{17}\theta(p) \\
d_2 &= \frac{16}{17}(p^7+1) + \frac{1}{17}\theta(p) \\
e_2 &= \frac{16}{17}(p^7+1) - \frac{16}{17}\theta(p).
\end{aligned}
$$

Next, equating coefficients of q^4 in (6.1)–(6.3) gives

$$
\begin{aligned}
\phi_{9,0}(4p^2) + 2034p^3 &= 2034c_1 + 1920c_2 + 2048c_3 \\
\phi_{5,4}(4p^2) + 1920p^3 &= 2034d_1 + 1920d_2 + 2048d_3 \\
\phi_{1,8}(4p^2) + 2048p^3 &= 2034e_1 + 1920e_2 + 2048e_3.
\end{aligned}
$$

By Theorem 6.1 and Eqs. (6.4)–(6.6) we have

$$
\begin{aligned}
\phi_{9,0}(4p^2) &= \phi_{9,0}(p^2) + 126\phi_{5,4}(p^2) + 9\phi_{1,8}(p^2) \\
&= \frac{34834}{17}(p^7+1) - 2034p^3 - \frac{256}{17}\theta(p).
\end{aligned}
$$

Similarly,

$$\begin{aligned}\phi_{5,4}(4p^2) &= \frac{32768}{17}(p^7+1) - 1920p^3 - \frac{128}{17}\theta(p)\\ \phi_{1,8}(4p^2) &= \frac{32768}{17}(p^7+1) - 2048p^3 + \frac{2048}{17}\theta(p).\end{aligned}$$

It follows that

$$\begin{aligned}c_3 &= \frac{2}{17}(p^7+1) - \frac{2}{17}\theta(p)\\ d_3 &= \frac{1}{17}(p^7+1) - \frac{1}{17}\theta(p)\\ e_3 &= \frac{1}{17}(p^7+1) + \frac{16}{17}\theta(p).\end{aligned}$$

This completes the proof. □

Corollary 6.6. *The eigenfunctions and eigenvalues of T_{p^2} are*

$$\begin{aligned}v_1 &:= \phi_{9,0} - 2\phi_{5,4}, & \lambda_1 &= \alpha\\ v_2 &:= 16\phi_{5,4} + \phi_{1,8}, & \lambda_2 &= \alpha\\ v_3 &:= \phi_{5,4} - \phi_{1,8}, & \lambda_3 &= \beta.\end{aligned}$$

That is,

$$\begin{aligned}T_{p^2}v_1 &= (p^7+1)v_1\\ T_{p^2}v_2 &= (p^7+1)v_2\\ T_{p^2}v_3 &= \theta(p)v_3.\end{aligned}$$

Remark 6.7. Observe that $v_2 = w_2$ and $v_3 = w_3$, where w_2 and w_3 are the eigenfunctions for T_4 in Corollary 6.2.

Theorem 6.8. *Let us write*

$$v_j = v_j(q) = \sum_{n=0}^{\infty} v_j(n)q^n$$

so that $v_j(n)$ is the coefficient of q^n in the series expansion of $v_j(q)$, $j = 1, 2, 3$. If $p^2 \nmid n$, then

$$\begin{aligned}v_1(p^{2\mu}n) &= \left[\frac{p^{7\mu+7}-1}{p^7-1} - p^3\left(\frac{n}{p}\right)\frac{p^{7\mu}-1}{p^7-1}\right]v_1(n)\\ v_2(p^{2\mu}n) &= \left[\frac{p^{7\mu+7}-1}{p^7-1} - p^3\left(\frac{n}{p}\right)\frac{p^{7\mu}-1}{p^7-1}\right]v_2(n)\\ v_3(p^{2\mu}n) &= \left[c_1(n)r_1^{\mu} + c_2(n)r_2^{\mu}\right]v_3(n),\end{aligned}$$

where

$$\begin{aligned}
r_1 &= \frac{\theta(p) + \sqrt{\Delta(p)}}{2} \\
r_2 &= \frac{\theta(p) - \sqrt{\Delta(p)}}{2} \\
c_1(n) &= \frac{1}{2} + \frac{\theta(p) - 2p^3\left(\frac{n}{p}\right)}{2\sqrt{\Delta(p)}} \\
c_2(n) &= \frac{1}{2} - \frac{\theta(p) - 2p^3\left(\frac{n}{p}\right)}{2\sqrt{\Delta(p)}} \\
\Delta(p) &= \theta(p)^2 - 4p^7.
\end{aligned}$$

Proof: Using Lemma 3.3, together with Corollary 6.6, we have

$$v_1(p^2n) + p^3\left(\frac{n}{p}\right)v_1(n) + p^7v_1(n/p^2) = (p^7+1)v_1(n) \tag{6.7}$$

$$v_2(p^2n) + p^3\left(\frac{n}{p}\right)v_2(n) + p^7v_2(n/p^2) = (p^7+1)v_2(n) \tag{6.8}$$

$$v_3(p^2n) + p^3\left(\frac{n}{p}\right)v_3(n) + p^7v_3(n/p^2) = \theta(p)v_3(n). \tag{6.9}$$

Iterating (6.7)–(6.9) gives the result. □

Remark 6.9. Ramanujan [21, Eq. (163)] stated that $\Delta(p) < 0$ for all primes p.

6.3. *Sums of nine squares*

Theorem 6.10. *Let $n = 2^\lambda \prod_p p^{\lambda_p}$ be the prime factorisation of n and let n' be the squarefree part of n. If n' is of the form $8k+5$ then*

$$r_9(n) = r_9(n')\frac{2^{7\lfloor\lambda/2\rfloor+7}-1}{2^7-1}\prod_p\left[\frac{p^{7\lfloor\lambda_p/2\rfloor+7}-1}{p^7-1} - p^3\left(\frac{n'}{p}\right)\frac{p^{7\lfloor\lambda_p/2\rfloor}-1}{p^7-1}\right].$$

Proof: Let

$$\begin{aligned}
g_{p,\mu,m} &= \frac{p^{7\mu+7}-1}{p^7-1} - p^3\left(\frac{m}{p}\right)\frac{p^{7\mu}-1}{p^7-1} \\
h_{p,\mu,m} &= c_1(m)r_1^\mu + c_2(m)r_2^\mu,
\end{aligned}$$

where c_1, c_2, r_1, r_2 are as for Theorem 6.8. Let n^o be the greatest divisor of n which is not divisible by an odd square. If $p^2 \nmid m$ then

$$\begin{aligned}
r_9(p^{2\mu}m) &= \frac{1}{17}[17v_1(p^{2\mu}m) + 2v_2(p^{2\mu}m) + 2v_3(p^{2\mu}m)] \\
&= \frac{1}{17}[17g_{p,\mu,m}v_1(m) + 2g_{p,\mu,m}v_2(m) + 2h_{p,\mu,m}v_3(m)]
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{17}[17g_{p,\mu,m}v_1(m) + 2g_{p,\mu,m}v_2(m) + 2g_{p,\mu,m}v_3(m)] \\
&\quad + \frac{2}{17}[h_{p,\mu,m} - g_{p,\mu,m}]v_3(m) \\
&= g_{p,\mu,m}r_9(m) + \frac{2}{17}[h_{p,\mu,m} - g_{p,\mu,m}]v_3(m).
\end{aligned}
$$

By induction on j, it follows that if $p_1, p_2, \ldots, p_j$ are distinct odd primes, and $p_i^2 \nmid m$ for $1 \le i \le j$, then

$$
\begin{aligned}
&r_9\left(p_1^{2\mu_1} p_2^{2\mu_2} \ldots p_j^{2\mu_j} m\right) \\
&\quad = \left(\prod_{i=1}^{j} g_{p_i,\mu_i,m}\right) r_9(m) + \frac{2}{17}\left[\left(\prod_{i=1}^{j} h_{p_i,\mu_i,m}\right) - \left(\prod_{i=1}^{j} g_{p_i,\mu_i,m}\right)\right] v_3(m).
\end{aligned}
$$

Consequently, if either $p^{2\mu_p} \| n$ or $p^{2\mu_p+1} \| n$, then

$$
\begin{aligned}
r_9(n) &= r_9(n^o) \prod_p \left[\frac{p^{7\mu_p+7} - 1}{p^7 - 1} - p^3 \left(\frac{n^o}{p}\right) \frac{p^{7\mu_p} - 1}{p^7 - 1}\right] \\
&\quad + \frac{2}{17} v_3(n^o) \left[\prod_p h_{p,\mu_p,n^o} - \prod_p \left(\frac{p^{7\mu_p+7} - 1}{p^7 - 1} - p^3 \left(\frac{n^o}{p}\right) \frac{p^{7\mu_p} - 1}{p^7 - 1}\right)\right].
\end{aligned}
$$

If $n^o = 4^\mu(8k + 5)$ for some nonnegative integers μ and k, then since $v_3(q) = w_3(q)$, the first part of Theorem 6.3 implies $v_3(n^o) = 0$. In this case we have

$$
r_9(n) = r_9(n^o) \prod_p \left[\frac{p^{7\mu_p+7} - 1}{p^7 - 1} - p^3 \left(\frac{n^o}{p}\right) \frac{p^{7\mu_p} - 7}{p^7 - 1}\right].
$$

This, together with Theorem 6.4 completes the proof. □

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Bosonic Formulas for $\widehat{\mathfrak{sl}}_2$ Coinvariants

M. JIMBO jimbomic@ms.u-tokyo.ac.jp
Graduate School of Mathematical Sciences, The University of Tokyo, Tokyo 153-8914, Japan

T. MIWA tetsuji@kusm.kyoto-u.ac.jp
Division of Mathematics, Graduate School of Science, Kyoto University, Kyoto 606-8502, Japan

E. MUKHIN mukhin@math.iupui.edu
Department of Mathematics, Indiana University-Purdue University, Indianapolis, Indiana 46202

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Abstract. We derive bosonic-type formulas for the characters of $\widehat{\mathfrak{sl}}_2$ coinvariants.

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1. Introduction

Let us fix a natural number k which we call level. Let L_l be the integrable $\widehat{\mathfrak{sl}}_2$ module of level k and of weight l, generated by a highest weight vector v_l such that

$$h_0 v_l = l v_l, \quad f_0^{l+1} v_l = e_1^{k-l+1} v_l = f_i v_l = e_{i+1} v_l = h_i v_l = 0, \quad i \in \mathbb{Z}_{<0},$$

where e_i, f_i, h_i are the standard generators of $\widehat{\mathfrak{sl}}_2$. Then the space of coinvariants $L^e_{k,l}$ is the quotient space

$$L^e_l = L_l/(e_i L_l, \ i \in \mathbb{Z}_{\geq 0}).$$

This quotient is naturally doubly graded by actions of h_0 and the degree operator d. In this paper we present bosonic-type formulas for the corresponding character

$$\chi^e_l(q, z) = \mathrm{Tr}_{L^e_l}\, q^d z^{h_0}.$$

The spaces of coinvariants L^e_l were studied in the series of papers [2–4] (the space L^e_l is denoted there by $L^{(0,\infty)}_{k,l}$), where several other spaces with the same character χ_l are given. One such space, a space of rigged configurations, produced a fermionic-type formula for $\chi^e_l(q, z)$, see Theorems 3.5.2 and 3.5.3 in [3]. Another description via "combinatorial paths" led to a difference equation for $\chi^e_l(q, z)$. This difference equation was the main theme in all proofs of the papers [2–4]. It also plays the central role in this paper. Let us recall the construction.

For $0 \le i \le l \le k$, denote by $\mathcal{C}_l[i]$ the set of pairs $(\mathbf{a};\mathbf{b}) = (a_0, a_1 \dots ; b_0, b_1, \dots)$, where a_i, b_i are non-negative integers such that only finitely many are different from zero, and

$$a_0 = i, \quad a_r + b_{r+1} + a_{r+1} \le k, \qquad b_r + a_r + b_{r+1} \le k, \quad \sum_{s=m}^{n} b_s \le k + \sum_{s=m+1}^{n-2} a_s, \quad (1.1)$$

where $r \ge 0$, $-1 \le m < n \le \infty$. Here we set $b_\infty = k$, $b_{-1} = l$, $b_0 = a_{-1} = a_\infty = 0$. Then by Corollary 5.4.10 of [2],

$$\chi_l(q, z) = \sum_{i=0}^{l} \chi_{i,l}(q, z, z^{-1}), \quad \chi_{i,l}(q, z_1, z_2) := \sum_{(\mathbf{a},\mathbf{b}) \in \mathcal{C}_l[i]} q^{\sum j(a_j + b_j)} z_1^{\sum b_j} z_2^{\sum a_j + b_j}.$$

By Proposition 3.3.1 in [2], we have a difference equation of the form

$$\chi_{i,l}(q, z_1, z_2) = \sum_{i',l'} M_{i,l}^{i',l'}(q, z_1, z_2)\chi_{i',l'}(q, z_1, qz_2), \quad \chi_{i,l}(q, z_1, 0) = \delta_{i,0}\delta_{l,0},$$

where the matrix M is given by (2.1) below.

The set $\mathcal{C}_l[i]$ can be thought of as the set of integer points in a polytope in an infinite-dimensional space cut out by the hyperplanes (1.1). Then one may expect an existence of a bosonic-type formula for $\chi_{i,l}$ in the spirit of [6], written as a sum over vertices of the characters of the corresponding cones. Such a bosonic formula was studied in a simpler situation in [5]. However, our polytope is rather complicated and a direct approach does not look promising.

Instead we use the difference equation to reduce our infinite-dimensional problem to a problem in two dimensions. The non-zero entries of the matrix M are written as $M_{i,l}^{i',l'} = m_1^i m_2^l m_3^{i'} m_4^{l'}$, where the m_j are monomials of the form

$$q^{\alpha_j} z_1^{\beta_j} z_2^{\gamma_j}, \quad \alpha_j, \beta_j, \gamma_j \in \mathbb{Z}. \quad (1.2)$$

Moreover, for a generic (i, l), the set of (i', l') such that $M_{i,l}^{i',l'} \ne 0$ is a pentagon (see Fig. 1). Given a monomial n_1, n_2 of the form (1.2), these properties allow us to rewrite $\sum_{i',l'} M_{i,l}^{i',l'} n_1^{i'} n_2^{l'}$ as a sum of 5 rational functions corresponding to the vertices of the pentagon. We denote these five terms by $A(n_1^i n_2^l), \dots, E(n_1^i n_2^l)$. Then we can formally write

$$\chi_{i,l} = \lim_{N\to\infty} (A + B + C + D + E)^N (\delta_{i,0}\delta_{l,0}). \quad (1.3)$$

Expanding the RHS we obtain an expression for the character as a sum over all possible non-commutative monomials in the operators $A, \dots, E$.

This naıve expansion suffers from difficulties for two reasons. First, the number of terms grows exponentially as $N \to \infty$. Second, there are terms which contain zero denominators.

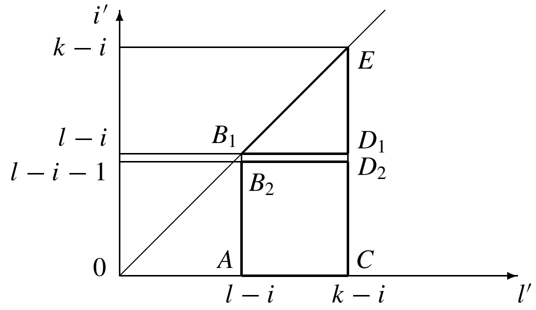

Figure 1. Region of summation in (2.2) and (2.4).

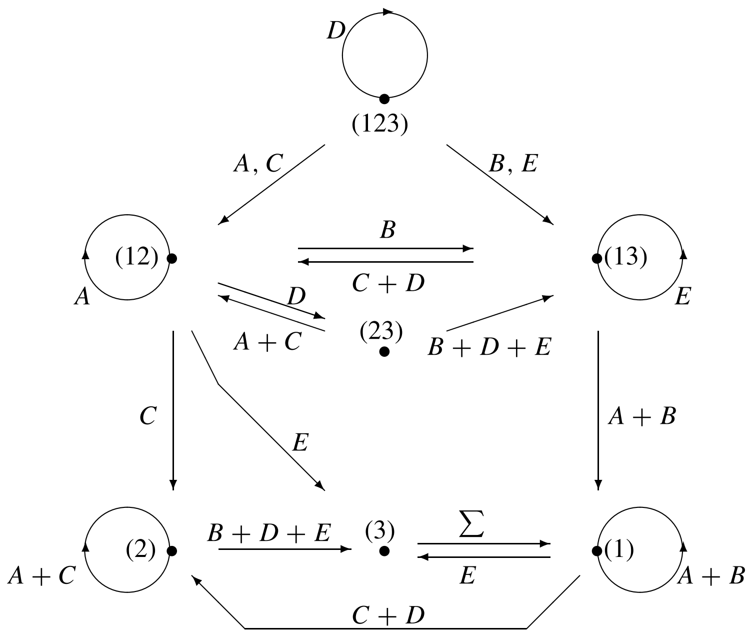

Figure 2. Summation graph (here $\sum = A + B + C + D + E$).

The aim of the present paper is to find a formula for $\chi_{i,l}$ as a sum over a subset of monomials, such that the number of terms grows polynomially and that all terms are well defined. Informally the existence of such a formula means that a huge cancellation takes place. The result is given in Theorem 6.1. In the preceding sections we explain the origin of this formula. For that purpose, an appropriate language is provided by what we call summation graph; see Fig. 2. Monomials in the operators A, B, C, D, E are identified with directed paths in this graph, and the cancellation pattern can be visualized. Terms corresponding to the same path (of infinite length) can be summed up with the use of Jackson's ${}_6\Phi_5$ formula (see (7.1) below). Then, all terms which cancel do so in pairs. Therefore, we need to perform no additional summation to observe the desired cancellation. We emphasize that the actual proof of Theorem 6.1 is done by a direct computation and is logically independent of these considerations.

Our final answer for $\chi_{i,l}$ given in Section 7 is a sum of 18 families of meromorphic functions. The functions in each family are parameterized by 3 non-negative integers. Each of them has a factorized form

$$m_0 m_1^i m_2^l m_3^k \frac{\prod_r (1-f_r)}{\prod_s (1-g_s)},$$

where m_j, f_r, g_s are monomials as in (1.2). Note that the structure of this formula does not depend on i, l in contrast to the fermionic formulas.

We expect that one can write in a similar fashion bosonic formulas for solutions of certain class of difference equations. Another example of such equations and the corresponding formulas are given in [1].

2. The difference equation

Fix a natural number $k \in \mathbb{Z}_{>0}$. Define a matrix $M(q, z_1, z_2) = (M_{i,l}^{i',l'})$ where $0 \le i \le l \le k$, $0 \le i' \le l' \le k$, by the formulas

$$M_{i,l}^{i',l'} = \begin{cases} (qz_1z_2)^{l'-i'} z_2^i & \text{if } l-i \le i' \le l' \le k-i; \\ (qz_1z_2)^{l'-l+i} z_2^i & \text{if } i' < l-i \le l' \le k-i; \\ 0 & \text{otherwise.} \end{cases} \tag{2.1}$$

It is a square matrix of size $(k+1)(k+2)/2$. We denote $M(q, z_1, 0)$ simply by $M(0)$. All entries of the matrix $M(0)$ are equal to 0 or 1.

For example, for $k = 1$, we have

$$M = \begin{pmatrix} 1 & qz_1z_2 & 1 \\ 0 & 1 & 1 \\ z_2 & 0 & 0 \end{pmatrix}, \quad M(0) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

where the components are arranged in the order $(i, l) = (0, 0), (0, 1), (1, 1)$. Consider the system of difference equations

$$\chi(q, z_1, z_2) = M(q, z_1, z_2)\chi(q, z_1, qz_2) \tag{2.2}$$

for a vector $\chi(q, z_1, z_2) = (\chi_{i,l})$ where $0 \le i \le l \le k$, and $\chi_{i,l}$ is a formal power series in the variables q, z_2, whose coefficients are formal Laurent series in z_1^{-1}. We impose the following initial condition

$$\chi_{i,l}(q, z_1, 0) = \delta_{i,0}\delta_{l,0}. \tag{2.3}$$

Lemma 2.1. *The system* (2.2), (2.3) *has a unique solution in* $\mathbb{C}[[q, z_2]]((z_1^{-1}))^m$ *where* $m = (k+1)(k+2)/2$.

Proof: Let us write $\chi(q, z_1, z_2) = \sum_{j=0}^{\infty} f_j(q, z_1) z_2^j$, where $f_j(q, z_1) \in \mathbb{C}[[q]]((z_1^{-1}))^m$. By (2.3), $f_0(q, z_1)_{i,l} = (\delta_{i,0}\delta_{l,0})$. Comparing the coefficients of z^n in (2.2), we obtain

$$f_n(q, z_1) = q^n M(0) f_n(q, z_1) + \cdots,$$

where the dots denote terms which depend on $f_j(q, z_1)$ with $j < n$. There exists the inverse matrix $(\mathrm{Id} - q^n M(0))^{-1}$ whose coefficients are power series in q alone, and therefore the vector $f_n(q, z_1)$ is uniquely determined via $f_j(q, z_1)$ with $j < n$. □

For $N \in \mathbb{Z}_{\geq 0}$, define vector valued functions $\chi^{(N)}(q, z_1, z_2) = (\chi_{i,l}^{(N)})$ $(0 \leq i \leq l \leq k)$ recursively by

$$\begin{aligned} \chi^{(N+1)}(q, z_1, z_2) &= M(q, z_1, z_2)\chi^{(N)}(q, z_1, qz_2), \\ \chi_{i,l}^{(0)}(q, z_1, z_2) &= \delta_{i,0}\delta_{l,0}. \end{aligned} \tag{2.4}$$

The function $\chi_{i,l}^{(N)}$ is a polynomial in q, z_1, z_2 with non-negative integer coefficients.

Lemma 2.2. *As* N *tends to infinity, the limit of* $\chi_{i,l}^{(N)}(q, z_1, z_2)$ *exists in* $\mathbb{C}[[q]][z_1, z_2]$ *and is equal to* $\chi_{i,l}(q, z_1, z_2)$.

Proof: Note that

$$M(0)\chi^{(0)}(q, z_1, z_2) = \chi^{(0)}(q, z_1, z_2),$$

and that for $m \geq n$, $M(q, z_1, q^m z_2) \equiv M(0) \mod q^n$. Therefore, for $N \geq n$ we have,

$$\begin{aligned} \chi^{(N)}(q, z_1, z_2) &= M(q, z_1, z_2)M(q, z_1, qz_2)\ldots M(q, z_1, q^{N-1}z_2)\chi^{(0)}(q, z_1, q^N z_2) \\ &\equiv M(q, z_1, z_2)M(q, z_1, qz_2)\ldots M(q, z_1, q^{n-1}z_2)M(0)^{N-n}\chi^{(0)}(q, z_1, z_2) \\ &\equiv M(q, z_1, z_2)M(q, z_1, qz_2)\ldots M(q, z_1, q^{n-1}z_2)\chi^{(0)}(q, z_1, z_2) \bmod q^n. \end{aligned}$$

Therefore the limit of $\chi_{i,l}^{(N)}(q, z_1, z_2)$ exists as N tends to infinity. Then it is equal to $\chi_{i,l}(q, z_1, z_2)$ by the uniqueness part of Lemma 2.1. □

Corollary 2.3. *The components of* $\chi(q, z_1, z_2)$ *are power series in* q *with coefficients in* $\mathbb{Z}_{\geq 0}[z_1, z_2]$.

The main purpose of this paper is to derive a bosonic-type formula for $\chi(q, z_1, z_2)$.

3. Bosonic formula for a triangle and a rectangle

The sum in the difference Eqs. (2.2) and (2.4) is taken over the union of a triangle B_1D_1E and a rectangle AB_2D_2C shown in Fig. 1. We divide the pentagon into two regions (the triangle B_1D_1E and the rectangle AB_2D_2C) since the formulas we apply will be different for these two.

The following two lemmas give bosonic type expressions for a sum over integer points inside a triangle and a rectangle. These lemmas are easily proved by a direct computation.

Lemma 3.1. *Let a, b be integers such that $a \leq b$. Then*

$$\sum_{a\leq n\leq m\leq b} x^m y^n = \frac{(xy)^a}{(1-x)(1-xy)} + \frac{x^b y^a}{(1-x^{-1})(1-y)} + \frac{(xy)^b}{(1-y^{-1})(1-(xy)^{-1})}. \tag{3.1}$$

Lemma 3.2. *Let a, b, c, d be integers such that $a \leq b+1$ and $c \leq d+1$. Then*

$$\sum_{\substack{a\leq m\leq b \\ c\leq n\leq d}} x^m y^n = \frac{x^a y^c}{(1-x)(1-y)} + \frac{x^a y^d}{(1-x)(1-y^{-1})} + \frac{x^b y^c}{(1-x^{-1})(1-y)} + \frac{x^b y^d}{(1-x^{-1})(1-y^{-1})}. \tag{3.2}$$

Note that Lemmas 3.1 and 3.2 state an equality of rational functions. Therefore, the equality still holds if we choose to expand the RHS into power series in any consistent way. There are four choices for the expansion of (3.2), depending on whether x, y are taken in the neighborhood of 0 or ∞. For (3.1) there are 6 ways to expand since there is an additional choice of xy being in the neighborhood of 0 or ∞.

The Lemmas 3.1 and 3.2 are simple examples of writing a character of a convex polytope as a sum over vertices of characters of the corresponding cones. The second lemma in fact follows from an even more simple example, namely from the expression for a 1-dimensional segment:

$$\sum_{a\leq m\leq b} x^m = \frac{x^a}{1-x} + \frac{x^b}{1-x^{-1}}.$$

We remark that the Weyl formula for the characters of irreducible finite dimensional modules over semisimple Lie algebras have a similar structure.

4. The extremal operators

Let P, Q, R be monomials of the form $q^\alpha z_1^\beta z_2^\gamma$ with $\alpha, \beta, \gamma \in \mathbb{Z}_{\geq 0}$. We denote by $[P, Q, R]$ the $(k+1)(k+2)/2$-dimensional vector whose (i, l)-th component is given by

$$[P, Q, R]_{i,l} = \left[P\left(1 : \frac{Q}{P} : \frac{R}{P}\right)\right]_{i,l} = P^{k-l} Q^{l-i} R^i.$$

In this notation we have

$$\begin{aligned}
&\chi^{(0)}_{i,l}(q, z_1, z_2) = \delta_{i,0}\delta_{l,0} = [1, 0, 0]_{i,l},\\
&\chi^{(1)}_{i,l}(q, z_1, z_2) = z_2^i\delta_{l,i} = [1, 0, z_2]_{i,l},\\
&\chi^{(2)}_{i,l}(q, z_1, z_2) = \sum_{j=0}^{k-l} q^{l-i+j} z_2^{l+j} = \frac{[1, qz_2, z_2]_{i,l}}{1 - qz_2} + \frac{[qz_2, z_2, z_2]_{i,l}}{1 - (qz_2)^{-1}}.
\end{aligned} \tag{4.1}$$

Consider the linear span V of vectors

$$f[P, Q, R], \quad f \in \mathbb{C}[[q, z_2]]\big(\big(z_1^{-1}\big)\big). \tag{4.2}$$

Normally we will write an element of V simply as v, without exhibiting the dependence on q, z_1, z_2 explicitly. We say v is *simple* if it has the form (4.2). We call f and $[P, Q, R]$ the *scalar part* and the *vector part*, respectively. Since $f[aP, aQ, aR] = fa^k[P, Q, R]$ holds for a monomial a, the scalar part and the vector part are determined up to this freedom.

For a vector-valued function $g(q, z_1, z_2)$, let

$$(Sg)(q, z_1, z_2) = g(q, z_1, qz_2) \tag{4.3}$$

denote the q-shift operator in z_2. We introduce the *extremal operators* A, B, C, D, E by the formulas

$$\begin{aligned}
A(f[P, Q, R]) &= S\left(\frac{f[P, Q, q^{-1}z_2P]}{(1 - z_1z_2Q/P)(1 - R/Q)}\right),\\
B(f[P, Q, R]) &= S\left(\frac{f(1 - Q/P)[P, R, q^{-1}z_2P]}{(1 - z_1z_2Q/P)(1 - Q/R)(1 - R/P)}\right),\\
C(f[P, Q, R]) &= S\left(\frac{f[z_1z_2Q, Q, q^{-1}z_2P]}{(1 - (z_1z_2)^{-1}P/Q)(1 - R/Q)}\right),\\
D(f[P, Q, R]) &= S\left(\frac{(1 - (z_1z_2)^{-1})f[z_1z_2Q, R, q^{-1}z_2P]}{(1 - (z_1z_2)^{-1}P/Q)(1 - (z_1z_2)^{-1}R/Q)(1 - Q/R)}\right),\\
E(f[P, Q, R]) &= S\left(\frac{f[R, R, q^{-1}z_2P]}{(1 - z_1z_2Q/R)(1 - P/R)}\right),
\end{aligned}$$

extended by linearity. The meaning of the RHS is as follows. For a monomial $X = q^\alpha z_1^\beta z_2^\gamma$ ($\alpha, \gamma \in \mathbb{Z}_{\geq 0}, \beta \in \mathbb{Z}$) such that $\alpha + \gamma > 0$ or $\alpha = \gamma = 0, \beta < 0$, $1/(1 - X^{\pm 1})$ means its expansion in non-negative powers of X. We define an operator $G(=A, B, C, D, E)$ on $f[P, Q, R]$ when the denominators appearing in the RHS are all of this form. Otherwise we do not define $G(f[P, Q, R])$.

The main point of introducing the operators A, B, C, D, E is:

Lemma 4.1. *Suppose operators A, B, C, D, E are defined on a simple vector v, and let $M(q, z_1, z_2)$ be the matrix* (2.1). *Then we have*

$$M(q, z_1, z_2)v(q, z_1, qz_2) = ((A + B + C + D + E)v)(q, z_1, z_2).$$

Proof: It suffices to consider the case $v = [P, Q, R]$. Written out explicitly, the (i, l)-component of the LHS reads

$$\sum_{l-i\le i'\le l'\le k-i} (qz_1z_2)^{l'-i'} z_2^i\, S(P)^{k-l'} S(Q)^{l'-i'} S(R)^{i'} + \sum_{i'<l-i\le l'\le k-i} (qz_1z_2)^{l'-l+i} z_2^i S(P)^{k-l'} S(Q)^{l'-i'} S(R)^{i'}.$$

We apply Lemmas 3.1 and 3.2 to obtain seven terms corresponding to vertices in Fig. 1. Namely

$$M(q, z_1, z_2)v(q, z_1, qz_2) = ((B_1 + E + D_1 + A + B_2 + D_2 + C)\, v)(q, z_1, z_2), \quad (4.4)$$

where

$$B_1([P, Q, R]) = S\left(\frac{[P, R, q^{-1}z_2P]}{(1 - z_1z_2Q/P)(1 - R/P)}\right),$$
$$B_2([P, Q, R]) = S\left(\frac{-[P, R, q^{-1}z_2P]}{(1 - z_1z_2Q/P)(1 - R/Q)}\right),$$
$$D_1([P, Q, R]) = S\left(\frac{[z_1z_2Q, R, q^{-1}z_2P]}{(1 - (z_1z_2)^{-1}P/Q)(1 - (z_1z_2)^{-1}R/Q)}\right),$$
$$D_2([P, Q, R]) = S\left(\frac{-[z_1z_2Q, R, q^{-1}z_2P]}{(1 - (z_1z_2)^{-1}P/Q)(1 - R/Q)}\right).$$

The lemma follows from the identities $B = B_1 + B_2$, $D = D_1 + D_2$. □

If the operators A, B, C, D, E were always defined, the lemma could be applied repeatedly to write the character as

$$\chi^{(N)}(q, z_1, z_2) = (A + B + C + D + E)^{N-1}[1, 0, z_2],$$
$$\chi(q, z_1, z_2) = (A + B + C + D + E)^{\infty}[1, 0, z_2]. \quad (4.5)$$

Unfortunately it is not the case. For example,

$$CBCAE[P, Q, R] = f(P, Q, R)\big[q^3z_1z_2^2\tilde{R}, q^2z_2\tilde{R}, q^3z_1z_2^2\tilde{R}\big], \quad \tilde{R} = S^5(R).$$

Therefore the operators $BCBCAE$ and $ECBCAE$ are not defined because 0 is produced in the denominator. (Note however that the sum $(B + E)CBCAE$ is well defined.) This example shows that there is no subspace W which contains $[1, 0, 0]$ and is stable under A, B, C, D, E. Such a difficulty does not arise in the case treated in [1].

Nevertheless it is possible to express $\chi(q, z_1, z_2)$ as a sum over a subset of non-commutative monomials in A, B, C, D, E, with all terms being defined in the sense above. In Theorem 6.1 below we write down the formula and show directly that it is the unique solution of the difference Eq. (2.2). Informally speaking, it means that the rest of the terms in (4.5) cancel with each other, including the non-defined ones. In the next two sections we analyze the mechanism of the cancellation and explain how the formula was found. This part is meant to motivate Theorem 6.1, although it is logically unnecessary.

5. Summation graph

In this section we prepare some language for Section 6.

A *monomial* $\mathcal{M}$ is an ordered composition of operators $G_1G_2\dots G_n$, where $G_i \in \{A, B, C, D, E\}$. We call n the degree of $\mathcal{M}$.

Now, our goal is to find different monomials which, when acted upon a simple vector $v \in V$, give rise to the same vector part. For example, the vector parts of $BC([P, Q, R])$ and $BD([P, Q, R])$ are both equal to $S^2([z_1z_2Q, q^{-1}z_2P, z_1z_2^2q^{-2}Q])$. Therefore the sum $B(C + D)$ applied to a simple vector is again a simple vector. Note that even though some monomials may not be defined, the action of a monomial on the vector part is always defined.

Introduce the maps $\sigma_G : \{1, 2, 3\} \to \{1, 2, 3\}$, where $G \in \{A, B, C, D, E\}$, by the formulas

$$\bar{\sigma}_A = (1, 2, 1),\quad \bar{\sigma}_B = (1, 3, 1),\quad \bar{\sigma}_C = (2, 2, 1),\quad \bar{\sigma}_D = (2, 3, 1),\quad \bar{\sigma}_E = (3, 3, 1),$$

where $\bar{\sigma}_G = (\sigma_G(1), \sigma_G(2), \sigma_G(3))$.

We define the map $\sigma_{\mathcal{M}}$ for all monomials $\mathcal{M}$ by the product rule

$$\sigma_{\mathcal{M}_1\mathcal{M}_2} = \sigma_{\mathcal{M}_2}\sigma_{\mathcal{M}_1}.$$

In other words σ is an anti-homomorphism of the semi-group of non-commutative monomials in A, B, C, D, E to the semi-group of endomorphisms of $\{1, 2, 3\}$.

The maps $\sigma_{\mathcal{M}}$ describe how $\mathcal{M}$ permutes P, Q, R in the vector part. Namely,

$$\mathcal{M}([P_1, P_2, P_3]) = S^{\deg\mathcal{M}}\big(f\big[u_1P_{\sigma(1)}, u_2P_{\sigma(2)}, u_3P_{\sigma(3)}\big]\big),$$

where u_1, u_2, u_3 denote factors independent of P_1, P_2, P_3.

Introduce a directed graph Γ, which we call *summation graph* (see Fig. 2). The vertices of Γ are non-empty subsets I of $\{1, 2, 3\}$. Each vertex emits 5 arrows, labeled A, B, C, D, E. The arrow with label G which starts from a vertex I ends at the vertex $\sigma_G(I)$.

Note that the summation graph Γ has three *floors* consisting of subsets I of given cardinality. We picture vertices corresponding to subsets of larger cardinality above those corresponding to subsets of smaller cardinality. Then no arrow goes "up".

In our picture, arrows with the same beginning and end are represented by one arrow with several labels. For arrows G_1, G_2 with a common source $I \subset \{1, 2, 3\}$, we write $G_1 + G_2$ if and only if the i-th component of the vector parts of $G_1([P, Q, R])$ and $G_2([P, Q, R])$ coincide for all $i \in I$. For example, the vector parts of $A([P, Q, R])$ and $C([P, Q, R])$ coincide in the second and the third components. Therefore we write $A + C$ above the arrows coming from (2) and (2, 3), but do not do so for (1, 2, 3).

We identify monomials in A, B, C, D, E with (oriented) paths in Γ starting at (1, 2, 3), by reading a monomial $\mathcal{M}$ from left to right and choosing the corresponding arrows accordingly.

For example, the monomial ACB corresponds to the path

$$(1,2,3)\overset{A}{\to}(1,2)\overset{C}{\to}(2)\overset{B}{\to}(3).$$

The vector part of $\mathcal{M}([P_1, P_2, P_3])$ corresponding to a path ending at a vertex I depends only on P_i with $i \in I$. Therefore we have the following obvious lemma.

Lemma 5.1. *Let $\mathcal{M}$ be a monomial. Suppose that the corresponding path in the summation graph ends at a source of an arrow $G_1 + \cdots + G_s$, where G_i are distinct elements of $\{A, B, C, D, E\}$. Then the operator $\mathcal{M}(G_1 + \cdots + G_s)$ maps simple vectors $v \in V$ to simple vectors if it is defined.*

We will use such summations in Section 6 to combine several terms in (4.5) in one. It turns out that the sum of the corresponding scalar parts in our computations is completely factorized to "linear" factors. We warn the reader that this is not necessarily the case in general (e.g. in the bottom floor for finite N).

6. Cancellation in the case $N = \infty$

From now on we concentrate on the case $N = \infty$. The purpose of this section is to describe the structure of the resulting formula and the way how it was obtained. In the argument below, we assume that all compositions of operators are defined. After finding the final formula we turn to proving it by direct means.

In formula (4.5), the RHS is a sum of monomials in A, B, C, D, E of infinite degree applied to $[1, 0, z_2]$. Our first step is to choose another vector in place of $[1, 0, z_2]$, so that it suffices to deal only with monomials of finite degree.

Let $\mathcal{M}$ be an arbitrary monomial of infinite degree and $\mathcal{N}$ of finite degree. We observe that $\mathcal{M}G\mathcal{N}([1, 0, z_2]) = 0$ holds for $G = C, D, E$.

The reason is as follows. We claim, that if $G\mathcal{N}([1, 0, z_2]) \neq 0$, then all components of its vector part contain a factor z_2. Indeed, if $\mathcal{N} = 1$ then $C([1, 0, z_2]) = D([1, 0, z_2]) = 0$, and the vector part of $E([1, 0, z_2])$ is $[qz_2, qz_2, z_2]$. If $\mathcal{N} \neq 1$, then the only component which does not depend on z_2 in $\mathcal{N}([1, 0, z_2])$ may be the first one, and our claim follows. Then $[z_2 P, z_2 Q, z_2 R] = z_2^k [P, Q, R]$ has an overall factor z_2^k. It vanishes when $\mathcal{M}$ is applied, because of an infinite shift S^∞.

The formula (4.5) would then take the form

$$\chi(q, z_1, z_2) = \sum_{\deg \mathcal{M} < \infty} \mathcal{M}(A + B)^\infty([1, 0, z_2]). \tag{6.1}$$

In the language of paths in the summation graph Γ, the formula (6.1) means that we sum over all paths of finite length which start at $(1, 2, 3)$ and end at (1).

We compute $(A + B)^\infty([1, 0, z_2])$ using the identity

$$(A + B)^n = A^n + \sum_{j=0}^{n-1} A^j B (A + B)^{n-j-1}.$$

We have explicitly

$$A^n B(A+B)^m[(1,0,z_2)] = \frac{[1, q^{n+1}z_2, z_2]}{(1-q^{n+1}z_2)\prod_{\substack{j=-n\\ j\neq 0}}^{m-1}(1-q^j)\prod_{\substack{j=n+2\\ j\neq 2n+2}}^{2n+m+1}\left(1-q^j z_1 z_2^2\right)},$$

and $A([1,0,z_2]) = 0$. Taking the limit $n\to\infty$ we obtain the following definition.

Introduce the vector $v_\infty \in V$ by the formulas

$$v_\infty = \sum_{n=1}^{\infty} f_n v_n, \tag{6.2}$$

where

$$v_n = [1, q^n z_2, z_2], \quad f_n = \frac{(-1)^{n-1}q^{n(n-1)/2}\left(1-q^{2n}z_1z_2^2\right)}{(q)_\infty\left(q^{n+1}z_1z_2^2\right)_\infty(q)_{n-1}(1-q^n z_2)}. \tag{6.3}$$

We have an identity $(A+B)v_\infty = v_\infty$ (see Lemma 7.1 below).

Now, we describe the mechanism of the cancellation and the structure of the final formula.

First of all we infer, on the basis of examples, that paths which pass through vertices (2) or (3) cancel out. In other words the surviving monomials correspond to paths in the summation graph which never go to the bottom level until the tail $(A+B)^\infty$ is reached. We have no direct proof of this statement.

After that reduction, we still have several cycles which paths can wrap around in the middle floor. Let us set

$$L = (C+D)D(B+D+E), \qquad \bar{L} = D(B+D+E)(C+D).$$

We have several operator identities which explain a part of the cancellation. For example, we have

$$BE = 0,$$

which implies in particular that if E cycle appears after L then the corresponding term is zero. We have also the identities valid for any $m\in\mathbb{Z}_{\geq 0}$,

$$X\bar{L}^m D(A+C) = 0, \quad X\bar{L}^m B = 0 \quad (X = A, C),$$
$$YL^m(C+D)D(A+C) = 0, \quad YL^m(C+D)B = 0 \quad (Y = B, E),$$

which imply that the cycle $D(A+C)$, $(C+D)B$ give zero contributions. (We stress that we do not use any of these operator identities for our proofs.) So, we are left with the cycles A (at the vertex $(1,2)$), E (at the vertex $(1,3)$) and L (starting at the vertex $(1,3)$).

Finally, there is an ordering of the cycles in the following sense. We call a path in the summation graph Γ *good* if the following 3 conditions are fulfilled:

- it ends at the vertex $(1, 3)$,
- it does not contain both A and E cycles,
- it enters neither A nor E cycles after passing through L cycle.

A path which is not good is called *bad*. We infer that in the formula (6.1) bad paths cancel out and only good paths survive.

These considerations lead us to the following theorem.

Theorem 6.1. *We have the following identity*

$$\chi(q, z_1, z_2) = \sum_{n,m,s=0}^{\infty} (D^n A^m BL^s v_\infty + D^n CA^m BL^s v_\infty + D^n E^{m+1} L^s v_\infty \\ + D^n CA^m D(B + D + E)L^s v_\infty + D^n A^{m+1} D(B + D + E)L^s v_\infty), \quad (6.4)$$

where v_∞ *is defined in* (6.2) *and* (6.3). *All terms in the RHS are defined.*

Proof: For the proof we use the explicit formulas for each term in the RHS given in Section 7. By a direct computation we verify that they are all defined, and that each operator A, B, C, D, E are also defined on them.

Substitution $z_2 = 0$ to the RHS clearly gives 0 (because of the nontrivial vector part) unless $l = i = 0$, when we get 1 (coming from the term Bv_∞). By the uniqueness part of Lemma 2.1, it is enough to show that multiplication by $A + B + C + D + E$ does not change the RHS.

There are two things to show. First, we have to show that multiplication by $A + B + C + D + E$ reproduces all the terms, and second, that the extra terms appearing cancel out.

The first statement, informally speaking, can be reformulated as follows. If a monomial $G\mathcal{M}$ with $G \in \{A, B, C, D, E\}$ is good, then monomial $\mathcal{M}$ itself is also good. We verify it by case checking. The nontrivial cases are $BL^s v_\infty$, $EL^s v_\infty$, $CD^2 L^s v_\infty$ and $AD^2 L^s v_\infty$ with $s \in \mathbb{Z}_{>0}$. We have to show that $L^s v_\infty$ and $D^2 L^s v_\infty$ occur in the RHS. Let us consider the case of $L^s v_\infty$, the other case is similar.

Expanding L into a sum of monomials, we see that all monomials in $L^s v_\infty$ are good, except for the monomial $D^{3s} v_\infty$, which we replace by the sum of good monomials $\sum_{n=0}^{\infty} D^{3s} A^n B v_\infty$, using the identity (see Lemma 7.1)

$$(1 - A)^{-1} B v_\infty = \sum_{n=0}^{\infty} A^n B v_\infty = v_\infty.$$

For the second statement, we claim that cancellations always happen in pairs. We list the necessary equalities which are checked using the explicit formulas given in Section 7.

In all equalities below, m, n, s stand for arbitrary non-negative integers. First we have cancellations of bad monomials of type $A\mathcal{M}$, where $\mathcal{M}$ is a good monomial.

$$
\begin{aligned}
(AD^{3n+3}A^mBL^s + AD^{3n+2}CA^mBL^s)v_\infty &= 0,\\
(AD^{3n+1}A^{m+1}BL^s + AD^{3n+1}CA^mBL^s)v_\infty &= 0,\\
(AD^{3n+2}A^mBL^{s+1} + AD^{3n+2}A^{m+1}D(B+D+E)L^s)v_\infty &= 0,\\
(AD^{3n}CA^mBL^s + AD^{3n}CA^mD(B+D+E)L^s)v_\infty &= 0,\\
(AD^{3n}E^{2m+1}L^s + AD^{3n}E^{2m+2}L^s)v_\infty &= 0,\\
(AD^{3n+1}E^{2m+3}L^s + AD^{3n+2}E^{2m+2}L^s)v_\infty &= 0,\\
(AD^{3n+2}E^{2m+1}L^s + AD^{3n+1}E^{2m+2}L^s)v_\infty &= 0,\\
(AD^{3n+3}A^{m+1}D(B+D+E)L^s + AD^{3n+2}CA^{m+1}D(B+D+E)L^s)v_\infty &= 0,\\
(AD^{3n+1}A^{m+1}D(B+D+E)L^s + AD^{3n+1}CA^mD(B+D+E)L^s)v_\infty &= 0.
\end{aligned}
$$

Then another set of 9 equalities are obtained from the above 9 by replacing the first factor A by C. They describe cancellations of bad monomials of type $C\mathcal{M}$, where $\mathcal{M}$ is a good monomial.

In the case $B\mathcal{M}$, where $\mathcal{M}$ is a good monomial, we have the following equalities.

$$
\begin{aligned}
(BD^{3n}A^mBL^{s+1} + BD^{3n}A^{m+1}D(B+D+E)L^s)v_\infty &= 0,\\
(BD^{3n+1}A^mBL^s + BD^{3n}CA^mBL^s)v_\infty &= 0,\\
(BD^{3n+2}A^{m+1}BL^s + BD^{3n+2}CA^mBL^s)v_\infty &= 0,\\
(BD^{3n+1}CA^mBL^s + BD^{3n+1}CA^mD(B+D+E)L^s)v_\infty &= 0,\\
(BD^{3n+3}E^{2m+1}L^s + BD^{3n+2}E^{2m+2}L^s)v_\infty &= 0,\\
(BD^{3n+1}E^{2m+1}L^s + BD^{3n+1}E^{2m+2}L^s)v_\infty &= 0,\\
(BD^{3n+2}E^{2m+3}L^s + BD^{3n+3}E^{2m+2}L^s)v_\infty &= 0,\\
(BD^{3n+1}A^{m+1}D(B+D+E)L^s + BD^{3n}CA^{m+1}D(B+D+E)L^s)v_\infty &= 0,\\
(BD^{3n+2}A^{m+1}D(B+D+E)L^s + BD^{3n+2}CA^mD(B+D+E)L^s)v_\infty &= 0.
\end{aligned}
$$

The cancellations of bad monomials of type $E\mathcal{M}$, where $\mathcal{M}$ is a good monomial are given by replacing first factor B in the above formulas by E.

Note that if $\mathcal{M}$ is good then $D\mathcal{M}$ is also good. $\square$

Note that each term in the RHS of Theorem 6.1 is a product of a monomial in q, z_1, z_2 with "linear" factors of the form $(1-q^{s+1})^{\pm 1}$, $(1-q^sz_1^{-1})^{\pm 1}$, $(1-q^sz_2)^{\pm 1}$, $(1-q^sz_1z_2)^{\pm 1}$, $(1-q^sz_1z_2^2)^{\pm 1}$ with $s \geq 0$.

Also note, that in order to prove the cancellation (or in order to write closed formulas) one has to break 5 families in the RHS of (6.4) into 18 families. Then each function on the RHS is a power series in q whose coefficients are formal Laurent series in z_1^{-1} over the ring of polynomials $\mathbb{Z}[z_2]$. We know by Corollary 2.3 that terms containing z_1^{-1} disappear after the summation.

Finally, we remark that some cancellations of the above happen between monomials of different degrees, for example $CEv_\infty + CE^2 v_\infty = 0$. However, we could rewrite this equality as a sum of monomials of the same degree: $CE(A + B)v_\infty + CE^2 v_\infty = 0$.

7. The explicit formula

We start with an identity which plays a crucial role in our computations:

$$\frac{(q)_\infty}{(qx)_\infty (qy)_\infty} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} q^{n(n-1)/2}(1 - q^{2n}xy)}{(q)_{n-1}(q^{n+1}xy)_\infty (1 - q^n x)(1 - q^n y)}. \tag{7.1}$$

In fact this is a special case of Jackson's ${}_6\Phi_5$ formula which states (see for example [7] p.102, formula (3.4.2.3)):

$$\sum_{n=0}^{\infty} \frac{(1 - q^{2n}a)(a)_n(b)_n(c)_n(d)_n(qa/bcd)^n}{(1 - a)(q)_n(qa/b)_n(qa/c)_n(qa/d)_n} = \frac{(qa)_\infty (qa/cd)_\infty (qa/bd)_\infty (qa/bc)_\infty}{(qa/b)_\infty (qa/c)_\infty (qa/d)_\infty (qa/bcd)_\infty}.$$

Identity (7.1) is obtained by setting $a = xy$, $b = x$, $c = y$, $d = \infty$.

Here is an example.

Lemma 7.1. *We have $(A+B)v_\infty = v_\infty$. Moreover, $A^n f_1 v_1 = f_{n+1} v_{n+1}$ and $Bv_\infty = f_1 v_1$.*

Proof: The identity $A^n f_1 v_1 = f_{n+1} v_{n+1}$ is obvious. The identity $Bv_\infty = f_1 v_1$ is equivalent to (7.1) with $x = q^2 z_1 z_2^2$ and $y = 1$. □

Note that v_∞ is a sum of infinitely many simple vectors with vector parts $[1, q^n z_2, z_2]$ and the first component of these vector parts is always 1. Therefore, if $\mathcal{M}$ is a monomial corresponding to a path which ends at vertex (1) in the summation graph, then $\mathcal{M}[1, q^n z_2, z_2]$ does not depend on n, and $\mathcal{M}v_\infty$ is a simple vector. In particular, all formulas below are simple vectors. The summation of scalar parts can be always explicitly performed using (7.1).

Now we finish with explicit formulas for the terms in the RHS of the formula in Theorem 6.1. These formulas are obtained by an explicit computation, induction on m, n, s and using formula (7.1).

We define quadratic forms α, β, γ and δ by

$$\begin{aligned}
\alpha_{n,m,s} &= 3(n+s)^2 + 2ms,\\
\beta_{n,m,s} &= 3(n+s)^2 + m^2 + 4ms + 3mn,\\
\gamma_{n,m,s} &= \frac{21}{2}n^2 + \frac{1}{2}m^2 + \frac{13}{2}s^2 + 3nm + 15ns + 5ms,\\
\delta_{n,m,s} &= \frac{11}{2}n^2 + \frac{3}{2}m^2 + 5s^2 + 5nm + 10ns + 6ms.
\end{aligned}$$

Then we have the following formulas.

$$D^{3n-2}A^mBL^sv_\infty=(-1)^{n+m+s}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{17}{2}n-\frac12m-\frac72s+2}$$
$$\times\frac{(qz_1z_2)_{2n-1}(q^{3n+m+s}z_1z_2)_s\big(1-q^{6n+2m+2s-2}z_1z_2^2\big)}{(q)_\infty(q)_{2n-2}(q)_{2n+m+2s-1}(q^{3n+m+s-1}z_2)_{n+s}(q^{2n+m+2s}z_1z_2)_n\big(q^{4n+m+2s-1}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}+m}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n-m-2s}(z_1z_2)^{-1}:q^{1-4n-m-2s}(z_1z_2)^{-1})\big],\quad n\geq1;m,s\geq0,$$
$$D^{3n-1}A^mBL^sv_\infty=(-1)^{n+m+s}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac72n-\frac12m-\frac12s}$$
$$\times\frac{(qz_1z_2)_{2n-1}(q^{3n+m+s+1}z_1z_2)_s\big(1-q^{6n+2m+2s}z_1z_2^2\big)}{(q)_\infty(q)_{2n-1}(q)_{2n+m+2s}(q^{3n+m+s}z_2)_{n+s}(q^{2n+m+2s+1}z_1z_2)_n\big(q^{4n+m+2s+1}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n}(z_1z_2)^{-1}:q^{2n+m+2s}z_2)\big],\quad n\geq1,m\geq0,s\geq0,$$
$$D^{3n}A^{m-1}BL^sv_\infty=(-1)^{n+m+s+1}\big(z_1z_2^3\big)^{n+s}q^{\gamma_{n,m,s}+\frac12n-\frac12m-\frac12s}$$
$$\times\frac{(qz_1z_2)_{2n}(q^{3n+m+s+1}z_1z_2)_s\big(1-q^{6n+2m+2s}z_1z_2^2\big)}{(q)_\infty(q)_{2n}(q)_{2n+m+2s-1}(q^{3n+m+s}z_2)_{n+s+1}(q^{2n+m+2s+1}z_1z_2)_n\big(q^{4n+m+2s+1}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}}\big(z_1z_2^2\big)^{n+s}(1:q^{4n+m+2s}z_2:q^{2n}z_2)\big],\quad n\geq0,m\geq1,s\geq0,$$
$$D^{3n-2}CA^{m-1}BL^sv_\infty=(-1)^{n+m+s}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{13}{2}n-\frac12m-\frac72s+1}$$
$$\times\frac{(qz_1z_2)_{2n-1}(q^{3n+m+s}z_1z_2)_s\big(1-q^{6n+2m+2s-2}z_1z_2^2\big)}{(q)_\infty(q)_{2n-1}(q)_{2n+m+2s-1}(q^{3n+m+s-1}z_2)_{n+s}(q^{2n+m+2s}z_1z_2)_n\big(q^{4n+m+2s}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}+m}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n-m-2s}(z_1z_2)^{-1}:q^{2n+m+2s}z_2)\big],\quad n\geq1,m\geq1,s\geq0,$$
$$D^{3n-1}CA^{m-1}BL^sv_\infty=(-1)^{n+m+s}\big(z_1z_2^3\big)^{n+s}q^{\gamma_{n,m,s}+\frac12n+\frac12m+\frac32s}$$
$$\times\frac{(qz_1z_2)_{2n}(q^{3n+m+s+1}z_1z_2)_s\big(1-q^{6n+2m+2s}z_1z_2^2\big)}{(q)_\infty(q)_{2n-1}(q)_{2n+m+2s}(q^{3n+m+s}z_2)_{n+s+1}(q^{2n+m+2s+1}z_1z_2)_n\big(q^{4n+m+2s+1}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}}\big(z_1z_2^2\big)^{n+s}(1:q^{4n+m+2s}z_2:q^{2n+m+2s}z_2)\big],\quad n\geq1,m\geq1,s\geq0,$$
$$D^{3n-3}CA^mBL^sv_\infty=(-1)^{n+m+s+1}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{17}{2}n-\frac12m-\frac72s+2}(q)_\infty^{-1}$$
$$\times\frac{(qz_1z_2)_{2n-2}(q^{3n+m+s}z_1z_2)_s\big(1-q^{6n+2m+2s-2}z_1z_2^2\big)}{(q)_{2n-2}(q)_{2n+m+2s-1}(q^{3n+m+s-1}z_2)_{n+s}(q^{2n+m+2s+1}z_1z_2)_{n-1}\big(q^{4n+m+2s-1}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{m,n,s}+m}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n+1}(z_1z_2)^{-1}:q^{2-4n-m-2s}(z_1z_2)^{-1})\big],\quad n\geq1,m\geq0,s\geq0,$$
$$D^{3n-1}A^{m+1}D(B+D+E)L^{s-1}v_\infty=(-1)^{n+m+s}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{11}{2}n-\frac32m-\frac52s}$$
$$\times\frac{(qz_1z_2)_{2n-1}(q^{3n+m+s+1}z_1z_2)_{s-1}\big(1-q^{6n+2m+2s}z_1z_2^2\big)}{(q)_\infty(q)_{2n-1}(q)_{2n+m+2s-1}(q^{3n+m+s}z_2)_{n+s}(q^{2n+m+2s+1}z_1z_2)_{n-1}\big(q^{4n+m+2s}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n}(z_1z_2)^{-1}:q^{-4n-m-2s}(z_1z_2)^{-1})\big],\quad n\geq1,m\geq0,s\geq1,$$

$$D^{3n-2}A^m D(B+D+E)L^s v_\infty = (-1)^{n+m+s+1}\big(z_1z_2^3\big)^{n+s} q^{\gamma_{n,m,s}-\frac{5}{2}n+\frac{3}{2}m+\frac{1}{2}s+1}$$
$$\times \frac{(qz_1z_2)_{2n-1}(q^{3n+m+s}z_1z_2)_s\big(1-q^{6n+2m+2s-2}z_1z_2^2\big)}{(q)_\infty(q)_{2n-2}(q)_{2n+m+2s}(q^{3n+m+s-1}z_2)_{n+s+1}(q^{2n+m+2s+1}z_1z_2)_{n-1}\big(q^{4n+m+2s}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}+m}\big(z_1z_2^2\big)^{n+s}(1:q^{4n+m+2s-1}z_2:q^{2n+m+2s}z_2)\big],\quad n\ge 1, m\ge 1, s\ge 0,$$
$$D^{3n}A^m D(B+D+E)L^{s-1}v_\infty = (-1)^{n+m+s+1}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{7}{2}n-\frac{3}{2}m-\frac{5}{2}s}$$
$$\times \frac{(qz_1z_2)_{2n}(q^{3n+m+s+1}z_1z_2)_{s-1}\big(1-q^{6n+2m+2s}z_1z_2^2\big)}{(q)_\infty(q)_{2n}(q)_{2n+m+2s-1}(q^{3n+m+s}z_2)_{n+s}(q^{2n+m+2s}z_1z_2)_n\big(q^{4n+m+2s+1}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n-m-2s}(z_1z_2)^{-1}:q^{2n}z_2)\big],\quad n\ge 0, m\ge 1, s\ge 1,$$
$$D^{3n-2}CA^{m-1}D(B+D+E)L^s v_\infty = (-1)^{n+m+s}\big(z_1z_2^3\big)^{n+s}q^{\gamma_{n,m,s}-\frac{5}{2}n+\frac{1}{2}m-\frac{3}{2}s}(q)_\infty^{-1}$$
$$\times \frac{(qz_1z_2)_{2n-1}(q^{3n+m+s}z_1z_2)_s\big(1-q^{6n+2m+2s-2}z_1z_2^2\big)}{(q)_{2n-1}(q)_{2n+m+2s-1}(q^{3n+m+s-1}z_2)_{n+s+1}(q^{2n+m+2s+1}z_1z_2)_{n-1}\big(q^{4n+m+2s}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}+m}\big(z_1z_2^2\big)^{n+s}(1:q^{4n+m+2s-1}z_2:q^{2n-1}z_2)\big],\quad n\ge 1, m\ge 1, s\ge 0,$$
$$D^{3n-1}CA^m D(B+D+E)L^{s-1}v_\infty = (-1)^{n+m+s+1}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{11}{2}n-\frac{3}{2}m-\frac{5}{2}s}$$
$$\times \frac{(qz_1z_2)_{2n}(q^{3n+m+s+1}z_1z_2)_{s-1}\big(1-q^{6n+2m+2s}z_1z_2^2\big)}{(q)_\infty(q)_{2n-1}(q)_{2n+m+2s-1}(q^{3n+m+s}z_2)_{n+s}(q^{2n+m+2s}z_1z_2)_n\big(q^{4n+m+2s}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha_{n,m,s}}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n-m-2s}(z_1z_2)^{-1}:q^{-4n-m-2s}(z_1z_2)^{-1})\big],\quad n\ge 1, m\ge 0, s\ge 1,$$
$$D^{3n-3}CA^m D(B+D+E)L^s v_\infty = (-1)^{n+m+s+1}\big(z_1z_2^3\big)^{n+s}z_2^{-1}q^{\gamma_{n,m,s}-\frac{13}{2}n+\frac{1}{2}m-\frac{3}{2}s+2}$$
$$\times \frac{(qz_1z_2)_{2n-2}(q^{3n+m+s}z_1z_2)_s\big(1-q^{6n+2m+2s-2}z_1z_2^2\big)}{(q)_\infty(q)_{2n-2}(q)_{2n+m+2s}(q^{3n+m+s-1}z_2)_{n+s}(q^{2n+m+2s+1}z_1z_2)_{n-1}\big(q^{4n+m+2s}z_1z_2^2\big)_\infty}$$
$$\times\big[q^{\alpha+n,m,s+m}\big(z_1z_2^2\big)^{n+s}(1:q^{-2n+1}(z_1z_2)^{-1}:q^{2n+m+2s}z_2)\big],\quad n\ge 1, m\ge 0, s\ge 0,$$
$$D^{3n-2}E^{2m+1}L^s v_\infty = (-1)^{n+m+1}\big(z_1z_2^2\big)^{n+s}z_2^m q^{\delta_{n,m,s}-\frac{3}{2}n-\frac{1}{2}m}(q)_s^{-1}$$
$$\times \frac{(q^{-2s}z_1)_s(qz_1z_2)_{n-1}}{(q)_{n-1}(q)_{2n+m+2s-1}(q^{1-n-m-2s}z_1)_{n+m+2s-1}(q^{n+m+2s+1}z_2)_\infty(q^{2n+m+2s}z_1z_2)_\infty}$$
$$\times\big[q^{\beta_{n,m,s}}\big(z_1z_2^2\big)^{n+s}z_2^m\big(1:q^{-2n-m-2s}(z_1z_2)^{-1}:q^{-n}z_1^{-1}\big)\big],\quad n\ge 1, m\ge 0, s\ge 0,$$
$$D^{3n-1}E^{2m+1}L^s v_\infty = (-1)^{n+m}\big(z_1z_2^2\big)^{n+s}z_2^m q^{\delta_{n,m,s}+\frac{1}{2}n+\frac{1}{2}m+\frac{1}{2}s}$$
$$\times \frac{(q^{-2s}z_1)_s(qz_1z_2)_n}{(q)_s(q)_{n-1}(q)_{2n+m+2s}(q^{-n-m-2s}z_1)_{n+m+2s}(q^{n+m+2s+1}z_2)_\infty(q^{2n+m+2s+1}z_1z_2)_\infty}$$
$$\times\big[q^{\beta_{n,m,s}}\big(z_1z_2^2\big)^{n+s}z_2^m\big(1:q^{n+m+2s}z_1^{-1}:q^{2n+m+2s}z_2\big)\big],\quad n\ge 1, m\ge 0, s\ge 0,$$

$$D^{3n}E^{2m-1}L^s v_\infty = (-1)^{n+m}\left(z_1 z_2^2\right)^{n+s} z_2^m q^{\delta_{n,m,s}-\frac{1}{2}n-\frac{1}{2}m}$$
$$\times \frac{(q^{-2s}z_1)_s(qz_1z_2)_n}{(q)_s(q)_n(q)_{2n+m+2s-1}(q^{-n-m-2s+1}z_1)_{n+m+2s-1}(q^{n+m+2s}z_2)_\infty(q^{2n+m+2s+1}z_1z_2)_\infty}$$
$$\times \left[q^{\beta_{n,m,s}}\left(z_1z_2^2\right)^{n+s} z_2^m(1:q^n:q^{-n-m-2s})\right], \quad n\geq 0, m\geq 1, s\geq 0,$$
$$D^{3n-2}E^{2m+2}L^s v_\infty = (-1)^{n+m}\left(z_1 z_2^2\right)^{n+s} z_2^m q^{\delta_{n,m,s}-\frac{3}{2}n-\frac{1}{2}m}$$
$$\times \frac{(q^{-2s}z_1)_s(qz_1z_2)_{n-1}}{(q)_s(q)_{n-1}(q)_{2n+m+2s-1}(q^{-n-m-2s}z_1)_{n+m+2s}(q^{n+m+2s+1}z_2)_\infty(q^{2n+m+2s+1}z_1z_2)_\infty}$$
$$\times \left[q^{\beta_{n,m,s}}\left(z_1z_2^2\right)^{n+s} z_2^m\left(1:q^{n+m+2s}z_1^{-1}:q^{-n}z_1^{-1}\right)\right], \quad n\geq 1, m\geq 0, s\geq 0,$$
$$D^{3n-1}E^{2m}L^s v_\infty = (-1)^{n+m}\left(z_1 z_2^2\right)^{n+s} z_2^m q^{\delta_{n,m,s}-\frac{3}{2}n-\frac{1}{2}m}$$
$$\times \frac{(q^{-2s}z_1)_s(qz_1z_2)_n}{(q)_s(q)_{n-1}(q)_{2n+m+2s-1}(q^{-n-m-2s+1}z_1)_{n+m+2s-1}(q^{2n+m+2s}z_1z_2)_\infty(q^{n+m+2s}z_2)_\infty}$$
$$\times \left[q^{\beta_{n,m,s}}\left(z_1z_2^2\right)^{n+s} z_2^m(1:q^{-2n-m-2s}(z_1z_2)^{-1}:q^{-n-m-2s})\right], \quad n\geq 1, m\geq 1, s\geq 0,$$
$$D^{3n}E^{2m}L^s v_\infty = (-1)^{n+m}\left(z_1 z_2^2\right)^{n+s} z_2^m q^{\delta_{n,m,s}+\frac{3}{2}n+\frac{1}{2}m+2s}$$
$$\times \frac{(q^{-2s}z_1)_s(qz_1z_2)_n}{(q)_s(q)_n(q)_{2n+m+2s}(q^{-n-m-2s+1}z_1)_{n+m+2s-1}(q^{n+m+2s+1}z_2)_\infty(q^{2n+m+2s+1}z_1z_2)_\infty}$$
$$\times \left[q^{\beta_{n,m,s}}\left(z_1z_2^2\right)^{n+s} z_2^m(1:q^n:q^{2n+m+2s}z_2)\right], \quad n\geq 0, m\geq 1, s\geq 0.$$

Our convention for expansions of RHS of these formulas into power series can be now written in the form:

$$|q|\ll 1, \qquad \left|z_1^{-1}\right|\ll 1, \qquad |z_2|\ll 1, \qquad |z_1z_2|\sim 1.$$

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